

The RH counterfactual in the original tau-based Split-Zero programme

Adversarial reconstruction and complete new calculations

13 September 2026

The original absolute base, supported zero, theta quotient, full arithmetic jets, coefficient purity, relative tensor fibres, source boundaries and four arithmetic volume endpoints are retained throughout.

This edition gives proved new restrictions on an actual hypothetical off-critical packet. It establishes neither a contradiction to that counterfactual nor an RH disproof. The complete source audit and inherited proof dependencies accompany the editable manuscript.

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Chapter 1

Reconstruction of the actual argument

What has been reconstructed, and what has been decided

This manuscript reconstructs the actual programme in the shared conversation Branch · Compile Navier Stokes Literature, together with the two supplied continuation instructions. Its question is the counterfactual to RH inside that programme: take a genuine off-critical zero of its original arithmetic function, and carry its complete cohomological packet through the constructed maps. The question is not answered by assigning arbitrary roots to an auxiliary polynomial.

The recovered source does construct split support, an absolute base, coefficient Frobenius, coefficient purity, arithmetic cohomology over that base, its right adjoint, actual theta representatives, tensor and symmetric pushforwards, conormal maps, exponential periods, and finite holonomy comparisons. These constructions are used below. The source's final auxiliary counterexample does not establish that RH is false, or that the whole programme fails. This continuation also establishes neither an RH contradiction nor an off-critical zeta zero. Its new results are exact comparisons, explicit retained kernels, and quantitative restrictions on an actual putative zero. Each is proved in the subsequent chapters.

Complete source coverage and its limits

The saved public response contains 5,350 mapping nodes. Following every parent from the declared current node to the root produces exactly the same 5,350 nodes in exactly the published linear order. There are 5,349 message records, of which 67 are user messages and 148 are readable assistant prose messages. All 215 readable user/assistant records were read in five complete contiguous assignments. The complete user history was also read by the integrating auditor. The original node identifiers, message hashes, verbatim bodies, assignment coverage and independent reviews are included in the source repository. A second public capture has exactly the same 5,349 message identifiers and body hashes. Differences in transient HTML metadata do not change any message body.

The share explicitly redacts tool payloads and does not embed the bodies of every historical attachment. Those absent contents are not counted as read. Original local proof files are retained separately and are cited as such. The reconstruction uses the actual visible messages and inspected proof files; it does not infer hidden mathematics from a redacted tool call.

The source begins with Navier–Stokes and thermodynamic calculations. It does not begin with the eventual tau-base construction. In particular the early function denoted $e(y)$ and early time or dual-trace coordinates denoted τ are not retrospectively identified with the later supported zero and absolute base. The user's negative-polynomial detour is expressly withdrawn at U0010. The later user instruction at U0043 demands the tau base itself, and U0046 accepts the corrected arithmetic construction of A1523. These corrections determine the programme continued here.

Source order and the surviving mathematical content

Source	Constructed object	Exact role in this continuation
U0001–15	Heat/NS test maps and the arithmetic seed	The full early audit retains its integral identities and constants. The test-field realization is kinematic with its specified force; it does not prove that an arbitrary fixed Navier–Stokes evolution supplies arithmetic positivity.
U0016–28	Split support, origin fibres, arithmetic amplitude, mixed-support synchronization, coefficient Frobenius	The packet chapter uses the actual localization and its integral coefficient action, and constructs a trace retract retaining the arithmetic eigenclass.
U0029–40	Original theta quotient, source labels, raw and balanced Mellin comparison, finite full jets	BK, DP and PL compute the previously retained balanced kernel and every smaller-label finite defect on these spaces.
U0041–45	Absolute base $\mathbf{b}_\tau$, marked chart, A_τ , restriction and K_τ	The companion tau chapter computes restriction’s complete homotopy fibre and its adjoint, including the nonzero arithmetic quotient at supported zero.
U0046–54	Continuous theta retraction, tensor and symmetric source, relative sum algebra, Fisher and conormal maps	The scaling chapter calculates the original cocycle and boundary pairing. The conormal chapter retains all local factors and computes the intrinsic symmetric trace, including negative directions within one exact sum fibre.
U0055–66	Original cyclic interpolation, volume windows, exponential periods, residue, periodization, all-phase recovery	The holonomy and volume chapters calculate what the original off-line packet forces into the boundary and compare the proved actual upper and lower estimates.
U0067, A2611	Auxiliary exponential/purity and Gamma-reference counterexamples	These refute specified reduced implications. Their polynomials are not asserted to divide 2ξ ; their arithmetic correction is retained, rather than set to zero.

Every row has a complete source-pinned audit in `audit/`; the table is a navigation device, not a substitute for those proofs. The original 821-page source cut, including its editable proof dependencies, is retained in `original_programme/`. The later original holonomy and audit messages occur in the complete saved transcript. None of the new results is attributed to a historical source that only supplied its starting objects.

The original absolute base and the arithmetic scalar square

For a commutative ring R , retain the disjoint union

$$G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}, \quad e = 0_R^\bullet.$$

Addition has identity τ , with $r^\bullet \oplus s^\bullet = (r + s)^\bullet$; multiplication has identity $1_R^\bullet$, has absorbing element τ , and satisfies $r^\bullet s^\bullet = (rs)^\bullet$. The laws follow directly from the ring laws when all inputs are supported; cases containing τ use its stated identity or absorption law. In particular $e \neq \tau$, $e \oplus e = e$, and $er^\bullet = e$ for every supported scalar.

Let $\mathbb{B} = \{0, 1\}$ have addition OR and multiplication AND. The amplitude and support observations are the unital semiring maps

$$p_R : G(R) \longrightarrow R, \quad p_R(\tau) = 0, \quad p_R(r^\bullet) = r, \quad \chi_R : G(R) \longrightarrow \mathbb{B}, \quad \chi_R(\tau) = 0, \quad \chi_R(r^\bullet) = 1. \quad (\text{R.1})$$

To verify additivity of χ_R , two supported summands have a supported sum even if their amplitudes cancel. Their OR is still one. The cases with an absent input follow from the additive identity. Multiplicativity is the corresponding support conjunction. The pair (p_R, χ_R) is injective: support zero identifies τ uniquely, whereas support one leaves the amplitude to identify $r^\bullet$ uniquely. Thus the zero amplitude at supported zero retains an observable absent from the ordinary ring quotient.

For a unital ring map $f : R \rightarrow S$, the map $G(f)$ fixes absence and sends $r^\bullet$ to $f(r)^\bullet$. The operation rules prove it is a unital semiring map, and $p_S G(f) = f p_R$ on both sorts of input. In particular the actual scalar square is

$$\begin{aligned} G(\mathbb{Z}) @>> G(\iota) >>> G(\mathbb{C}) \\ @V p_{\mathbb{Z}} V V @V p_{\mathbb{C}} V \\ \mathbb{Z} @>> \iota >>> \mathbb{C}. \end{aligned} \quad (\text{R.2})$$

The lower-left arithmetic target is the full infinite ring $\mathbb{Z}$. No finite characteristic or Boolean quotient replaces it in (R.2).

The original pointed-monoid blueprint base has elements $\{\tau, 1\}$, absorbing τ , and the empty-sum relation $\emptyset \equiv \tau$. It does not impose $1 + 1 = 1$. Its original structural map sends these elements to absence and the unit in $G(R)$. This defines the chart's structure over $\mathfrak{b}_\tau$. The support map in (R.1), whose codomain does impose Boolean addition, is a further map, not that structural map. The stated operations prove the two maps are compatible; their different additive relations are not identified.

For a complex linear map $f : E \rightarrow F$ at an admitted support label λ , its split lift sends (λ, v) to $(\lambda, f v)$ and external τ to external τ . It is additive because f is, and commutes with supported scalar multiplication for the same reason. If $f v = 0$, its value is $(\lambda, 0)$. This proves the support rule used by every linear complex, quotient and kernel below. Between labels, the same statement uses the explicitly specified linear transport; the PL chapter calculates a transport that kills a nonzero relation class to supported zero, without deleting its source label.

The actual cohomological route of a hypothetical zero

The full-space definitions of $V, \mathcal{B}, \Theta, D, \mathcal{M}$ are given in BK.1 and proved against the original source in the companion chapters. In particular

$$g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \quad \mathcal{M}\Theta\phi = gH_\phi.$$

The exact arithmetic object is $Q = \mathcal{B}/\Theta V$, with its original locally convex quotient topology. The later global theta retraction proves this quotient Hausdorff in that topology; it does not substitute an L^2 completion for Q .

On the original four-point chart $+, - < \eta < \sigma$, the arithmetic diagram is

$$\mathcal{T} = (V, V, \mathcal{B}, 0; \Theta, \Theta\mathcal{F}, 0).$$

The two-leg complex in degrees 0, 1 is

$$A_\tau(\mathcal{T}) = [V \oplus V \xrightarrow{(\phi, \psi) \mapsto \Theta(\phi - \widehat{\psi})} \mathcal{B}]. \quad (\text{R.3})$$

Its full H^1 is Q . The derived restriction comparison of the diagram is $A_\tau(\mathcal{T}) \rightarrow \mathcal{T}_\sigma[0] = 0$; its homotopy fibre is (R.3) itself. The companion tau proof computes the projective resolution, both

representatives of restriction, their homotopy, the fibre, its right adjoint, and the induced residue map. Thus nonzero classes sent to supported zero at sigma remain in the actual relative arithmetic cohomology. There is no inference here that the zero stalk makes Q vanish.

The counterfactual studied throughout the calculations is the existence of an actual nontrivial zero

$$\rho = \frac{1}{2} + \delta + i\gamma, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 0,$$

with its full positive order m . The original conjugation and functional equation give the complete reflection-conjugation quartet of the same order. Choose that full packet and retain

$$h(s) = \prod_{\epsilon, \eta \in \{-1, 1\}} (s - \frac{1}{2} - \epsilon\delta - i\eta\gamma)^m, \quad v_h = g/h, \quad v_h = j_h(v_h). \quad (\text{R.4})$$

Here division is entire because these are stipulated actual zeros, and v_h is a unit because their complete orders were used. Its entire Taylor class, including every nonconstant coefficient, is retained. This is the temporary counterfactual input to deductions, not evidence that any such zero exists.

The original arithmetic jet sequence, proved in the retained source and extended to every nonzero polynomial in BK, is

$$0 \longrightarrow Q[h(D)] \longrightarrow E_h \xrightarrow{\times j_h g} E_h \longrightarrow Q/h(D)Q \longrightarrow 0, \quad E_h = \mathbb{C}[s]/(h). \quad (\text{R.5})$$

For the actual complete packet, $j_h g = 0$. Its finite torsion class is therefore a full copy of E_h , and the Mellin observation on that copy includes the invertible factor v_h rather than dropping it. The original representative map uses the inverse unit to obtain $J_h \sigma_h = 1$. The Duhamel chapter gives its source representatives, its nonsplit source extension and its full scaling action explicitly.

At each local zero its action under $U_a F(x) = F(x/a)$, $a > 0$, is

$$a^\rho \sum_{j=0}^{m-1} \frac{(\log a)^j}{j!} N_\rho^j. \quad (\text{R.6})$$

This is not replaced by its reduced eigenvalue. BK and DP prove that the full-source balanced Mellin kernel cannot contain, cancel or hide this finite primary block. PL proves that proper source labels have additional finite kernel classes, even at parameters that are not zeros of g , and calculates their exact transition to the full source. This is why the source labels cannot be suppressed when using (R.5).

The mixed-support chapter next carries (R.4)–(R.6) through the actual coefficient synchronization. Its trace retract proves that the original arithmetic eigenclass remains nonzero. The complex-linear lift of the integral coefficient Frobenius adds roots of unity to the spectrum, and has a fixed coefficient line on which the arithmetic eigenvalue is unchanged. All tensor amplifications in this statement are over $\mathbb{C}$, as in the original external product; tensoring the local top nilpotent over E_h instead would give a different map and could annihilate it.

The tensor programme also contains more than its cyclic sum variable. The full conormal and symmetric chapters preserve the relative occupation algebra, its local lengths, original cochain signs, residue/Jacobian comparison and source representatives. In particular two full symmetric-square occupation factors can both have sum eigenvalue 1 while the difference of their idempotents has intrinsic arithmetic trace $-2m^2$. The detailed proof identifies the actual mixed derivative jet that detects this trace. Purity of a sum eigenvalue alone therefore cannot erase these retained relative directions.

Finally the actual holonomy comparison retains the common source generated by v_h and its tensor convolution. Its canonical section has a computed relation-valued action defect. The holonomy chapter proves a lower bound for its full original boundary energy at every phase and every admitted finite degree, and a spectral-distance lower bound for the associated Laplacian boundary. The concluding volume calculation compares those restrictions with the proved scalar arithmetic upper estimate. That comparison, rather than an auxiliary Gamma example, determines the remaining numerical range obtained by this continuation.

What the purity construction does, and the precise remaining calculation

The original finite coefficient algebra has proved weight-zero coefficient purity. The later finite-field exponential family has its proved weight-one cohomology under its stated good-reduction hypotheses; its external k -fold product retains degree k and the corresponding weight. The original source computes specialization and period maps relating its full arithmetic packet to those families. These are actual constructions. The supplied audits retain their periods, connections, cohomological degrees and kernels, rather than calling the families unrelated to the arithmetic source.

The operational numerical use of the Deligne-style amplification in this programme is to control the original arithmetic action or its equivalent original source volume defect uniformly as tensor degree grows. A period congruence transports both the metric and its action defect. It does not set the defect to zero. For the following formula let E be the retained finite tensor coefficient space (or a specified invariant constituent), let $A_k : E \rightarrow E$ be the restriction of the sum operator, and let $R : E \rightarrow L^2((\mathbb{R}_{>0})^k, dx_1 \cdots dx_k)$ be the original source section. Its columns are in $\mathcal{B}^{\otimes k}$, so $D^{(k)} = \sum_i (-x_i \partial_{x_i})$ acts on them. Put $G = R^* R$ and $\mathcal{B} = D^{(k)} R - R A_k$ as maps on this exact E . The symbol $\mathcal{B}$ in this formula denotes the boundary map; the function space with the same inherited symbol is distinguished by its domain and codomain. The exact boundary formula, proved on the original $dx_1 \cdots dx_k$ source in the following chapters, is

$$A_k^* G + G A_k - kG = -(R^* \mathcal{B} + \mathcal{B}^* R), \quad \mathcal{B} = D^{(k)} R - R A_k. \quad (\text{R.7})$$

The source relation producing $\mathcal{B}$ is explicitly retained. Neither its zero cohomology class nor its zero observation at sigma sets either cross-pairing in (R.7) to zero. This continuation computes the contribution and gives a strictly positive counterfactual lower bound on its original energy.

For the actual four endpoints take $k \geq 2$ and the original cyclic sum quotient of length $q = (1 + k(m - 1))(k + 1)^2$. In that quotient let $V_{k,N}^\nu = \det G_{k,N}^\nu > 0$, $N \geq q - 1$, be the determinant of the canonical source Gram for the specified measure $\nu \in \{\text{ar}, \Gamma\}$, as defined in the holonomy and volume chapters. Define $\mathcal{B}_{h,k}^\nu = \log(V_{k,q-1}^\nu / V_{k,2q-1}^\nu) + \log(V_{k,q}^\nu / V_{k,2q}^\nu)$. Substituting the positive volume ratios at all four endpoints gives the exact retained comparison

$$\mathcal{B}_{h,k}^{\text{ar}} = \mathcal{B}_{h,k}^\Gamma + \log \frac{T_{k,q-1} T_{k,q}}{T_{k,2q-1} T_{k,2q}}, \quad T_{k,N} = V_{k,N}^{\text{ar}} / V_{k,N}^\Gamma. \quad (\text{R.8})$$

The final auxiliary Gamma counterexample retains the reference volumes but does not calculate the actual signed correction in (R.8). The new concluding calculation bounds the actual four-window quantity on both sides, retaining this exact signed term. The coarse scalar upper and lower bound functions are compatible; the refined common-source interval retains unevaluated quantities, as detailed in the final combined restriction. No actual off-line zero is certified by the coarse compatibility, and no RH contradiction follows from those bounds.

Consequently the remaining issue after the mathematics below is the stronger original arithmetic control of (R.7), or of the joint signed term in (R.8), that would close the amplification argument. This is a precise limit of the results obtained, not an assumption used to prove them. The manuscript supplies the new exact maps and inequalities in full and retains the complete original sources for further calculation.

Chapter 2

The absolute-base restriction fibre and its adjoint

1. Objects, provenance, and cochain conventions

The complete successor *A global retraction of the original theta complex over the tau base*, retained as `Tau_Global_Comparison_2026-09-12/RESEARCH_NOTE.md`, was also read completely. Its SHA256 is 496f3ec752e3e67ddafaae7423feb300b1faa789ca6907872f8d539bdf6e1859. Equations (6.12a)–(6.12i) below reproduce its constructive retraction and evaluate the previously general closure kernel for this exact source.

This calculation continues equations (18)–(48) of *Arithmetic cohomology over the tau-base*. The complete source `NOTE.md` has SHA256 d03e71ad18f187bd89880cb8ca6fc9c5d6f260b3de8e8e5702c27db97e7b63c3. Its complete retained dependency sources/`Split_Support_Actual_Comparison__CONTINUATION.tex` has SHA256 98e8aaa63b1f7ee4661c043cd985ca274646b749f36d23d2ec46cbc4b0737dc5. Both were read completely for this calculation. The category, original theta source, masks, analytic topology, multiplicities, and actions below are those sources’ objects.

Let $E = \mathbb{C}$, let P have order $+$ $< \eta < \sigma$ and $-$ $< \eta < \sigma$, and let $\mathcal{A} = \text{Fun}(P, \text{Vect}_E)$. A diagram $F \in \mathcal{A}$ is specified by

$$U = F_+, \quad V_- = F_-, \quad M = F_\eta, \quad S = F_\sigma, \quad r_+ : U \rightarrow M, \quad r_- : V_- \rightarrow M, \quad t : M \rightarrow S. \quad (1.1)$$

Its remaining two arrows are the composites tr_+ and tr_- . The symbol V_- here denotes a general stalk; V in Section 5 denotes the original even Schwartz source.

Use cohomological complexes. For a chain map $f : C \rightarrow D$, define

$$\text{Cone}(f)^n = D^n \oplus C^{n+1}, \quad d(y, x) = (d_D y + f x, -d_C x), \quad d_{C[k]} = (-1)^k d_C. \quad (1.2)$$

After ordering its summands as $C^n \oplus D^{n-1}$, the homotopy fibre is

$$\text{Fib}(f) = \text{Cone}(f)[-1], \quad d(c, y) = (d_C c, -d_D y - f c). \quad (1.3)$$

For internal Hom use

$$d_{\text{Hom}} h = d_W h - (-1)^{|h|} h d_C. \quad (1.4)$$

These conventions, including the minus sign in the second component of (1.3), remain fixed throughout.

2. The original pushforward and the complete restriction fibre

Define $P_x(y) = E$ for $x \leq y$, and zero otherwise, with identity arrows between nonzero stalks. A natural map $P_x \rightarrow F$ is uniquely specified by the image of $1 \in P_x(x)$, so $\text{Hom}(P_x, F) = F_x$. Evaluation is exact, hence P_x is projective. The sequence

$$0 \longrightarrow P_\eta \xrightarrow{v \mapsto (v, -v)} P_+ \oplus P_- \xrightarrow{(u, v) \mapsto u+v} \mathbf{E} \longrightarrow 0 \quad (2.1)$$

is exact: at $+$ and $-$ its last arrow is identity; at η, σ its kernel is the displayed anti-diagonal; all restriction squares commute. Consequently a representative of the source functor is

$$C(F) = A_\tau(F) = [U \oplus V_- \xrightarrow{\delta} M], \quad \delta(u, v) = r_+u - r_-v, \quad (2.2)$$

in degrees 0, 1. With (1.4), applying internal Hom to (2.1) initially gives differential $-\delta$; the explicit isomorphism to (2.2) is identity in degree zero and minus identity in degree one. This is a change between two stated cochain presentations, not a modification of the arithmetic differential in (2.2).

The two representatives of derived restriction are

$$f_+(u, v) = tr_+u, \quad f_-(u, v) = tr_-v, \quad C(F) \longrightarrow S[0]. \quad (2.3)$$

They are chain maps because the target has only degree zero. Their difference is $t\delta$. Thus $h^1 = t : M \rightarrow S$ satisfies $f_+ - f_- = d_S h + h d_C$. In particular both represent the same natural derived morphism.

Using f_+ , the complete fibre is

$$\boxed{B_\sigma(F) = [U \oplus V_- \xrightarrow{b} M \oplus S], \quad b(u, v) = (r_+u - r_-v, -tr_+u).} \quad (2.4)$$

The cone itself has degrees $-1, 0$ and differential

$$U \oplus V_- \longrightarrow S \oplus M, \quad (u, v) \mapsto (tr_+u, -r_+u + r_-v). \quad (2.5)$$

No stalk or quotient has been dropped in either presentation.

Put

$$I = \text{im } r_+ \cap \text{im } r_- \subset M. \quad (2.6)$$

Then the complete cohomology of (2.4) is

$$H^0 B_\sigma(F) = \{(u, v) : r_+u = r_-v \in \ker t\}, \quad (2.7)$$

$$H^1 B_\sigma(F) = \frac{M \oplus S}{\{(r_+u - r_-v, -tr_+u) : u \in U, v \in V_-\}}, \quad (2.8)$$

and all its other cohomology groups vanish. There are exact sequences

$$0 \longrightarrow \ker r_+ \oplus \ker r_- \longrightarrow H^0 B_\sigma(F) \xrightarrow{(u, v) \mapsto r_+u} I \cap \ker t \longrightarrow 0, \quad (2.9)$$

$$0 \longrightarrow S/t(I) \xrightarrow{[s] \mapsto [(0, s)]} H^1 B_\sigma(F) \xrightarrow{[(m, s)] \mapsto [m]} M/(\text{im } r_+ + \text{im } r_-) \longrightarrow 0. \quad (2.10)$$

For (2.9), every element of the displayed intersection has a preimage under each leg, and the kernel is exactly the indicated direct sum. For (2.10), the last map is well defined and surjective. A class in its kernel has $m = \delta(u, v)$; subtracting $b(u, v)$ replaces it by $(0, s + tr_+u)$. The class of $(0, s)$ vanishes exactly when there exist u, v with $r_+u = r_-v$ and $s = -tr_+u$, which is precisely $s \in t(I)$. This proves the sequence, including the sign in its presentation. These are exact sequences of specified objects, without a chosen splitting.

Equivalently the restriction long exact sequence is

$$0 \rightarrow H^0 B_\sigma(F) \rightarrow \ker \delta \xrightarrow{f_+} S \xrightarrow{s \mapsto [(0, -s)]} H^1 B_\sigma(F) \rightarrow \operatorname{coker} \delta \rightarrow 0. \quad (2.11)$$

Its signs agree with (1.2): when the fibre-to-source arrow is the positive projection, the rotated distinguished triangle has boundary arrow minus the inclusion of S in $\operatorname{Cone}(f) = B_\sigma(F)[1]$. This accounts for the minus in (2.11); the independently specified exact presentation (2.10) uses the positive inclusion. The only nonzero possible cone cohomology is $H^{-1} \operatorname{Cone}(f) = H^0 B_\sigma(F)$ and $H^0 \operatorname{Cone}(f) = H^1 B_\sigma(F)$.

The two representative fibres are related by the chain isomorphism

$$B_{\sigma,+}(F) \rightarrow B_{\sigma,-}(F), \quad \Phi^0 = 1, \quad \Phi^1(m, s) = (m, s + tm), \quad (2.12)$$

whose inverse subtracts tm . Indeed $-f_+ + t\delta = -f_-$. Thus the fibre calculation does not depend on a privileged choice of the plus leg.

3. Naturality and the right-adjoint comparison

A morphism $a : F \rightarrow F'$ consists of maps $a_+, a_-, a_\eta, a_\sigma$ commuting with r_+, r_-, t . The fibre map is exactly

$$B_\sigma(a)^0 = a_+ \oplus a_-, \quad B_\sigma(a)^1 = a_\eta \oplus a_\sigma. \quad (3.1)$$

For example its second differential component is $-a_\sigma t r_+ u = -t' r'_+ a_+ u$; the first follows from the two other commutative squares. Formula (3.1) also commutes with (2.12), because $a_\sigma t = t' a_\eta$. This proves the naturality of every kernel, quotient, exact sequence, and representative comparison above.

For a vector space W , let $I_x(W)(y) = W$ for $y \leq x$, and zero otherwise. A natural map $F \rightarrow I_x(W)$ is uniquely determined by its component $F_x \rightarrow W$; the other nonzero components are precomposition with $F_y \rightarrow F_x$. Hence

$$\operatorname{Hom}_{\mathcal{A}}(F, I_x(W)) = \operatorname{Hom}_E(F_x, W). \quad (3.2)$$

Every vector space is injective over E : a linear map on a subspace extends after extending a basis. Therefore the right side is exact contravariantly in F , and $I_x(W)$ is injective.

Let

$$K_\tau(W) = [I_\eta(W) \xrightarrow{(+\operatorname{res}, -\operatorname{res})} I_+(W) \oplus I_-(W)] \quad (3.3)$$

have degrees $-1, 0$. It is a bounded injective complex. Applying (3.2) gives

$$R \operatorname{Hom}_{\mathcal{A}}(F, K_\tau(W)) = [\operatorname{Hom}(M, W) \xrightarrow{\ell \mapsto (\ell r_+, -\ell r_-)} \operatorname{Hom}(U, W) \oplus \operatorname{Hom}(V_-, W)]. \quad (3.4)$$

The internal Hom of (2.2) into $W[0]$ has exactly this differential: the element ℓ has degree -1 , so (1.4) gives $d\ell = +\ell\delta$. In particular the signs in the original single-factor adjoint and residue arrow require no repair.

For completeness the adjunction holds for bounded complexes, with all totalization signs specified. Define

$$C(F)^n = (F_+^n \oplus F_-^n) \oplus F_\eta^{n-1}, \quad d(b, m) = (d_F b, \delta b - d_F m), \quad (3.5)$$

and $K_\tau(W) = K_\tau(E) \otimes W$, with tensor differential. A degree n map $F \rightarrow K_\tau(W)$ consists of $b_p : F_+^p \oplus F_-^p \rightarrow W^{p+n}$ and $a_p : F_\eta^p \rightarrow W^{p+n+1}$. The isomorphism to $\operatorname{Hom}^n(C(F), W)$ keeps b_p and takes

$$\ell_p = (-1)^{n+1} a_p. \quad (3.6)$$

On the source differential these components become

$$a'_p = -d_W a_p - (-1)^n a_{p+1} d_F, \quad b'_p = d_W b_p + a_p \delta - (-1)^n b_{p+1} d_F. \quad (3.7)$$

On the target they become

$$\ell'_p = d_W \ell_p + (-1)^n \ell_{p+1} d_F, \quad b'_p = d_W b_p - (-1)^n (b_{p+1} d_F + \ell_p \delta). \quad (3.8)$$

Substitution of (3.6) gives equality in both components. Bounded injective targets compute derived Hom, and (2.1), with the degree n multiplier $(-1)^n$ on its eta component, gives (3.5) from its projective resolution. This proves the bounded derived adjunction as an actual natural chain isomorphism.

The stalk complexes of (3.3) are $[W \xrightarrow{1} W]$ at $+$, $[W \xrightarrow{-1} W]$ at $-$, $W[1]$ at η , and zero at σ . The stalkwise kernel diagram is $S_\eta W$, supported only at η . Its inclusion in degree -1 is a natural quasi-isomorphism

$$S_\eta W[1] \longrightarrow K_\tau(W). \quad (3.9)$$

Since every point of P lies below σ , $I_\sigma(W)$ is the constant diagram W . It represents the dual of the sigma stalk. The actual adjoint of restriction is

$$\alpha_+ : I_\sigma(W) \longrightarrow K_\tau(W), \quad (3.10)$$

given in degree zero by restriction into $I_+(W)$, and zero into $I_-(W)$. The alternative α_- restricts into $I_-(W)$. Their difference is $d_K h$, where $h : I_\sigma(W) \rightarrow I_\eta(W)$ is restriction in degree -1 . Applying Hom to (3.10) sends $w : S \rightarrow W$ to $(wtr_+, 0)$, which is exactly precomposition with f_+ . This identifies the morphism, rather than only comparing its endpoints.

4. The dual of the restriction fibre and its nonsplit extension

Define the bounded injective complex

$$R_\tau(W) = [I_\eta(W) \oplus I_\sigma(W) \xrightarrow{d_R} I_+(W) \oplus I_-(W)], \quad (4.1)$$

in degrees $-1, 0$, where

$$d_R(\ell, w) = (\ell \text{res}_+ - w \text{res}_{\sigma,+}, -\ell \text{res}_-). \quad (4.2)$$

Here restriction symbols denote the natural maps of injective diagrams. Formula (3.2) makes its Hom differential

$$(\ell, w) \mapsto (\ell r_+ - wtr_+, -\ell r_-). \quad (4.3)$$

This is precisely precomposition with the two components $(\delta, -f_+)$ of (2.4), with the positive sign supplied by degree -1 in (1.4). Therefore

$$\boxed{R \text{Hom}_{\mathcal{A}}(F, R_\tau(W)) \cong R \text{Hom}_E(B_\sigma(F), W).} \quad (4.4)$$

Equivalently $R_\tau(W) = \text{Cone}(-\alpha_+)$. Multiplying its $I_\sigma(W)$ cone summand by -1 identifies this presentation with $\text{Cone}(\alpha_+)$; (4.2) fixes which presentation is being used in (4.4).

At $+$, the differential is $(\ell, w) \mapsto \ell - w$, with kernel (w, w) . At $-$, it is $(\ell, w) \mapsto -\ell$, with kernel $(0, w)$. At η it has domain $W \oplus W$ and zero target, and at σ it has domain W and zero target. All degree-zero cokernels vanish. Consequently, if $D(W)$ has stalks

$$D(W)_+ = W, \quad D(W)_- = W, \quad D(W)_\eta = W \oplus W, \quad D(W)_\sigma = W, \quad (4.5)$$

and arrows

$$w \mapsto (w, w) \quad (+ \rightarrow \eta), \quad w \mapsto (0, w) \quad (- \rightarrow \eta), \quad (u, w) \mapsto w \quad (\eta \rightarrow \sigma), \quad (4.6)$$

the kernel inclusion is a natural quasi-isomorphism $D(W)[1] \rightarrow R_\tau(W)$.

There is an exact sequence of diagrams

$$0 \longrightarrow S_\eta W \xrightarrow{u \mapsto (u,0)} D(W) \xrightarrow{\text{second coordinate}} I_\sigma(W) \longrightarrow 0. \quad (4.7)$$

At $+$, $-$, σ the last map is identity under (4.5); at η its kernel is the first coordinate. Naturality follows directly from (4.6). For $W \neq 0$, the extension has no natural splitting. Indeed a splitting is identity at the plus and minus stalks, since the quotient there is identity. Commutation along $+$ $\rightarrow \eta$ forces its η component to be $w \mapsto (w, w)$, whereas commutation along $- \rightarrow \eta$ forces $w \mapsto (0, w)$. They disagree on every nonzero $w \in W$. This proves nonsplitting explicitly.

In the original case $F_\sigma = 0$, every term $\text{Hom}(F, I_\sigma(W)) = \text{Hom}(0, W)$ is zero. Thus the inclusion $K_\tau(W) \rightarrow R_\tau(W)$ induces an actual equality of the represented Hom complexes, as well as an isomorphism in the derived category. The nonzero original global arithmetic classes are therefore all in the relative fibre, and their original adjoint is precisely the restriction of (4.4) to these objects.

5. Calculation on the original theta source and every mask

Retain the exact original spaces and maps

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \hat{\phi}(0) = 0\}, \quad \hat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx, \quad (5.1)$$

$$\mathcal{B} = \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |(-x\partial_x)^k F(x)| < \infty \text{ for all } b \in \mathbb{Z}, k \geq 0\}, \quad (5.2)$$

$$\Theta\phi(x) = \sum_{n \in \mathbb{Z} \setminus \{0\}} \phi(nx), \quad JF(x) = x^{-1}F(1/x), \quad J\Theta = \Theta\mathcal{F}. \quad (5.3)$$

The last identity is the retained Poisson identity. Explicitly, the full Poisson equation is $\sum_n \phi(nx) = x^{-1} \sum_n \hat{\phi}(n/x)$; the two zero terms vanish by (5.1), giving (5.3). At infinity every differentiated summand is bounded by a constant times $(|n|x)^{-N}$ for any chosen $N > 1$; summing $2 \sum_{n \geq 1} n^{-N}$ proves rapid decay. Applying (5.3) proves arbitrary power decay at zero, with all derivatives, so $\Theta : V \rightarrow \mathcal{B}$. The Fourier transform preserves (5.1) because it exchanges the value and integral conditions, and its square is reflection, which is identity on even functions.

The map Θ is injective. For $\Re s = c > 1$, absolute convergence gives

$$\mathcal{M}\Theta\phi(s) = 2\zeta(s)\mathcal{M}\phi(s), \quad \mathcal{M}F(s) = \int_0^\infty F(x)x^s \frac{dx}{x}. \quad (5.4)$$

For absolute convergence, the integral of $|\phi(nx)|x^{c-1}$ is $n^{-c} \int_0^\infty |\phi(u)|u^{c-1}du$, and both the integral and $\sum n^{-c}$ are finite. The Euler product is nonzero for $c > 1$, as follows from absolute convergence of its logarithm. If $\Theta\phi = 0$, (5.4) makes $\mathcal{M}\phi(c+it) = 0$ for every real t . This is the Fourier transform, with its frequency sign explicitly reversed, of the integrable function $y \mapsto e^{cy}\phi(e^y)$. Fourier uniqueness gives zero almost everywhere and continuity gives zero everywhere. Evenness then gives $\phi = 0$ on the full real line.

The joint sheaf is

$$\mathcal{T} = (V, V, \mathcal{B}, 0; \Theta, \Theta\mathcal{F}, 0), \quad C(\mathcal{T}) = [V \oplus V \xrightarrow{\Theta(\phi-\hat{\psi})} \mathcal{B}]. \quad (5.5)$$

Its restriction map is zero on both chain representatives, so

$$B_\sigma(\mathcal{T}) = C(\mathcal{T}), \quad \text{Cone}(\text{res}_\sigma) = C(\mathcal{T})[1]. \quad (5.6)$$

Injectivity just proved gives

$$H^0 B_\sigma(\mathcal{T}) = \{(\widehat{\psi}, \psi) : \psi \in V\}, \quad H^1 B_\sigma(\mathcal{T}) = Q := \mathcal{B}/\Theta V. \quad (5.7)$$

The quotient here is the algebraic quotient with its original quotient locally convex topology. The later global retraction source proves that this image is closed; its exact retraction and the resulting zero closure kernel are included after (6.12) below.

Let $L = \mathcal{P}(\{+, -\})$. For each nonempty mask $A \in L$, retain V at each admitted leg and the zero coefficient space at a missing leg, retain $\mathcal{B}$ at eta and zero at sigma, and retain the restrictions in (5.5). The empty mask has the zero diagram. Every nonempty A has restriction fibre exactly $C(\mathcal{T}_A)$; its differential image is ΘV , since both Θ and $\Theta \mathcal{F}$ have that image. A one-leg differential is injective, with its literal plus or minus sign retained. It follows that

$$H^0 B_\sigma(\mathcal{T}_A) = \begin{cases} V, & A = \{+, -\}, \\ 0, & |A| = 1, \end{cases} \quad H^1 B_\sigma(\mathcal{T}_A) = Q \quad (A \neq \emptyset). \quad (5.8)$$

For mask inclusions the eta map is identity and leg maps are coordinate inclusions, hence the induced H^1 map is exactly identity on Q . This explicitly proves compatibility with every original mask transport.

Reconstruction gives

$$\widetilde{H}^1 B_\sigma = \{\tau\} \sqcup \bigsqcup_{\emptyset \neq A \subseteq \{+, -\}} \{A\} \times Q. \quad (5.9)$$

The scalar $e = 0_E^\bullet \in G(E)$ sends (A, q) to $(A, 0)$. The scalar τ sends it to the absent element. Linear restriction to the zero coefficient at sigma sends q to zero, but its supported reconstruction is $(A, q) \mapsto (A, 0)$, preserving A . No class of (5.9) becomes absent merely because its coefficient restricts to zero. The same statement applies to (2.4), (3.4), and (4.4) fibrewise, with dual label arrows given by actual precomposition and hence indexed by L^{op} .

6. Finite jets, their full kernel, and the retained topology

Put $g = 2\xi$, where

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s). \quad (6.1)$$

The source's full-plane multiplier identity is

$$\mathcal{M}\Theta\phi = gH_\phi, \quad H_\phi = \frac{\mathcal{M}\phi}{\frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)} \in \mathcal{O}(\mathbb{C}). \quad (6.2)$$

Here is the full continuation used in this identity. Write the Taylor expansion of the even Schwartz function at zero as $\phi(x) = \sum_{j=0}^N a_{2j}x^{2j} + O(x^{2N+2})$. Subtract this finite sum in its Mellin integral over $(0, 1)$. The remainder is holomorphic for $\Re s > -2N - 2$, while the subtracted terms have integrals $a_{2j}/(s + 2j)$. The integral over $(1, \infty)$ is entire by Schwartz decay. Since N is arbitrary, $\mathcal{M}\phi$ has meromorphic continuation with only possible simple poles at the nonpositive even integers. The reciprocal $1/\Gamma(s/2)$ has a simple zero at each such integer and cancels each possible pole. At $s = 0$, the condition $a_0 = \phi(0) = 0$ removes the Mellin pole, and $1/\Gamma(s/2)$ cancels the explicit factor s in the denominator of H_ϕ . At $s = 1$, evenness and the integral condition give $\mathcal{M}\phi(1) = \frac{1}{2} \int_{\mathbb{R}} \phi = 0$, cancelling $s - 1$. No other denominator zero occurs, since the gamma function has no zeros. Thus the displayed H_ϕ is entire. Formula (5.4) proves (6.2) first on $\Re s > 1$, and analytic continuation proves it everywhere, with the original factor $g = 2\xi$ retained.

Let Z be a finite set of actual zeros of g with their actual orders m_ρ , and write

$$z_\rho = s - \rho, \quad g(s) = u_\rho(z_\rho)z_\rho^{m_\rho}, \quad u_\rho(0) \neq 0, \quad A_Z = \bigoplus_{\rho \in Z} \mathcal{O}_\rho/(g). \quad (6.3)$$

The germ coordinate z_ρ , the full jet length, and the actual unit u_ρ are retained. The jet arrow is

$$J_Z : \mathcal{B} \longrightarrow A_Z, \quad (J_Z F)_\rho = \sum_{j=0}^{m_\rho-1} \frac{(\mathcal{M}F)^{(j)}(\rho)}{j!} z_\rho^j. \quad (6.4)$$

Equation (6.2) proves $J_Z \Theta = 0$, so it induces $q_Z : Q \rightarrow A_Z$.

Here is a full finite interpolation proof, with the original dx Riesz density preserved. Index jets by $\alpha = (\rho, j)$, and fix any $R > 0$ and $0 < R' < R$. Let

$$\eta_{R'}(y) = \begin{cases} \exp[-1/((R')^2 - y^2)], & |y| < R', \\ 0, & |y| \geq R', \end{cases} \quad r_\alpha(x) = x^{\bar{\rho}-1}(\log x)^j/j!. \quad (6.5)$$

Set

$$B_{\alpha\beta} = \frac{1}{j!k!} \int_{-R'}^{R'} \eta_{R'}(y) e^{(\rho+\bar{\sigma}-1)y} y^{j+k} dy, \quad \beta = (\sigma, k). \quad (6.6)$$

This is $\ell_\alpha(\eta_{R'}(\log x)r_\beta(x))$, where ℓ_α is the corresponding component of (6.4). It is Hermitian. For a column c , its quadratic form is

$$\bar{c}^\top B c = \int_{-R'}^{R'} \eta_{R'}(y) e^{-y} \left| \sum_\alpha \bar{c}_\alpha e^{\rho y} y^j / j! \right|^2 dy. \quad (6.7)$$

The exponential polynomials are independent: for a fixed exponent ρ , apply $\prod_{\sigma \neq \rho} (\partial_y - \sigma)^{m_\sigma}$ to a vanishing linear combination. It kills every other exponent block, and acts on the remaining block by $e^{\rho y}$ times $\prod_{\sigma \neq \rho} (\partial_y + \rho - \sigma)^{m_\sigma}$. Each factor on finite-degree polynomials is invertible, since it is a nonzero constant plus a nilpotent differentiation operator, with inverse given by the terminating geometric series. Hence that polynomial block is zero. This holds for every ρ , proving independence. Therefore (6.7) is strictly positive for $c \neq 0$, so B is invertible.

For $v \in A_Z$, the function

$$s_Z(v)(x) = \eta_{R'}(\log x) \sum_\beta (B^{-1}v)_\beta r_\beta(x) \quad (6.8)$$

is smooth, belongs to $\mathcal{B}$, has support in $[e^{-R'}, e^{R'}] \subset (e^{-R}, e^R)$, and satisfies $J_Z s_Z(v) = v$ by matrix multiplication. Thus (6.4) and q_Z are surjective, with an explicitly calculated finite-dimensional continuous section of (6.4).

For continuity, choose an integer $N > |\Re \rho|$ and put $p_N(F) = \max(\sup x^N |F(x)|, \sup x^{-N} |F(x)|)$. Splitting the integral at one gives the exact bound

$$|\ell_{\rho,j}(F)| \leq p_N(F) ((N + \Re \rho)^{-j-1} + (N - \Re \rho)^{-j-1}). \quad (6.9)$$

Indeed the two integrals of $|\log x|^j/j!$ against $x^{N+\Re \rho} dx/x$ and $x^{-N+\Re \rho} dx/x$ are the two displayed terms. This also proves that the Mellin jet derivatives used here are well-defined by dominated differentiation.

Define the actual retained kernel spaces, using $\mathcal{K}_Z$ to avoid changing the different space named $\mathcal{B}_Z$ in the source's finite arithmetic extension:

$$\mathcal{K}_Z = \ker J_Z, \quad Q_Z^0 = \mathcal{K}_Z / \Theta V. \quad (6.10)$$

They give exact sequences

$$0 \longrightarrow \mathcal{K}_Z \longrightarrow \mathcal{B} \xrightarrow{J_Z} A_Z \longrightarrow 0, \quad \boxed{0 \longrightarrow Q_Z^0 \longrightarrow Q \xrightarrow{q_Z} A_Z \longrightarrow 0}. \quad (6.11)$$

The first is split by (6.8); the second is split as a topological vector-space sequence by the induced section. No equivariance of this section is asserted: only the arrows (6.11) are canonical scaling morphisms.

To verify the topology assertion, the map $(F_0, v) \mapsto F_0 + s_Z(v)$ is a continuous linear bijection $\mathcal{K}_Z \oplus A_Z \rightarrow \mathcal{B}$, with continuous inverse $F \mapsto (F - s_Z J_Z F, J_Z F)$. Since $\Theta V \subset \mathcal{K}_Z$, passing to the quotient gives the claimed topological direct sum. By (6.9), $\mathcal{K}_Z$ is closed, so $\overline{\Theta V} \subset \mathcal{K}_Z$. The same proof gives

$$0 \rightarrow \mathcal{K}_Z / \overline{\Theta V} \rightarrow Q^H := \mathcal{B} / \overline{\Theta V} \rightarrow A_Z \rightarrow 0, \quad \ker(Q \rightarrow Q^H) = \overline{\Theta V} / \Theta V. \quad (6.12)$$

Thus every finite jet survives both specified quotients, with the closure kernel written explicitly. No new global topology has been substituted.

For this actual source the kernel in (6.12) is zero. We include the later source's constructive reason, retaining its auxiliary cutoffs and contraction constant. Let μ be the arithmetic Möbius function, and define on $x \neq 0$

$$\mathcal{U}F(x) = \frac{1}{2} \sum_{n \geq 1} \mu(n) F(n|x|). \quad (6.12a)$$

For $j \geq 0$, write $x^j \partial_x^j = P_j(D)$, where $P_j(T) = (-1)^j T(T+1) \cdots (T+j-1)$, and let $C_{N,j}(F)$ be the sum of $p_{N,k}(F) = \sup x^N |D^k F|$ weighted by the absolute coefficients of P_j . For each integer $N > 1$, termwise differentiation gives

$$|\partial_x^j \mathcal{U}F(x)| \leq \frac{1}{2} \zeta(N) C_{N,j}(F) x^{-N-j} \quad (x > 0). \quad (6.12b)$$

Uniform convergence away from zero justifies each derivative. On the original theta image the double sum is absolutely convergent, bounded by $\sum_{k \geq 1} d(k) |\phi(kx)|$, which converges by Schwartz decay and $d(k) \leq k$. The identity $\sum_{n|k} \mu(n) = 1$ for $k = 1$, zero otherwise, therefore proves

$$\mathcal{U}\Theta\phi = \phi, \quad \mathcal{U}J\Theta\phi = \hat{\phi} \quad (x \neq 0). \quad (6.12c)$$

Choose the original type of real even compactly supported smooth cutoffs $0 \leq \alpha, \beta \leq 1$, each equal to one near zero, and set on $L^2(\mathbb{R}, dx)$

$$T = M_\alpha \mathcal{F}^{-1} M_\beta \mathcal{F}, \quad c = \|T\|, \quad YF = (1 - \alpha) \mathcal{U}F + \alpha \mathcal{F}^{-1}((1 - \beta) \mathcal{U}JF). \quad (6.12d)$$

The kernel of T is $\alpha(x) \check{\beta}(x - y)$, with squared L^2 norm $\|\alpha\|_2^2 \|\beta\|_2^2$, so T is compact. Both its factors are contractions. If $c = 1$, compactness supplies a norm-attaining unit vector f ; equality in both contraction inequalities forces $\check{f}$ to have compact support where $\beta = 1$ and f to have compact support where $\alpha = 1$. Compact Fourier support makes the inverse Fourier integral entire, because the compactly supported L^2 transform is L^1 and allows all complex derivatives under its integral. This entire function vanishes on a real interval outside the support of f , hence vanishes identically, a contradiction. Therefore $c < 1$, and

$$(1 - T)^{-1} = \sum_{n \geq 0} T^n, \quad \|(1 - T)^{-1}\| \leq (1 - c)^{-1}. \quad (6.12e)$$

The cut-off exterior maps in Y are continuous into even Schwartz functions by (6.12b); their values are extended by zero where the cutoff vanishes near zero. The operator $T : L^2 \rightarrow \mathcal{S}$ is continuous: compact Fourier support bounds every derivative of $\mathcal{F}^{-1} M_\beta \mathcal{F} f$ by Cauchy–Schwarz, and multiplication by smooth compactly supported α controls every Schwartz seminorm. Thus $\Phi = (1 - T)^{-1} Y$ is Schwartz-valued and continuous: the equation $\Phi F = YF + T\Phi F$ bounds each Schwartz seminorm p by

$$p(\Phi F) \leq p(YF) + C_p (1 - c)^{-1} \|YF\|_2. \quad (6.12f)$$

It preserves evenness. With $\gamma_0 = e^{-\pi x^2}$ and $\gamma_2 = 2\pi x^2 e^{-\pi x^2}$, whose value/integral columns are $(1, 1)$, $(0, 1)$, define $P_V : \mathcal{S}(\mathbb{R})^{\text{even}} \rightarrow V$ by

$$P_V \phi = \phi - \phi(0)\gamma_0 - \left(\int_{\mathbb{R}} \phi - \phi(0) \right) \gamma_2, \quad \Lambda = P_V(1 - T)^{-1}Y. \quad (6.12g)$$

This continuous projection has image V and is identity on V . On $\Theta\phi$, (6.12c) gives $Y\Theta\phi = (1 - T)\phi$, so

$$\Lambda\Theta = 1_V, \quad K = 1_{\mathcal{B}} - \Theta\Lambda, \quad K^2 = K, \quad \ker K = \Theta V. \quad (6.12h)$$

The last equality follows in both directions by direct substitution. Since K is continuous, its kernel is closed. Therefore $\overline{\Theta V} = \Theta V$, $Q = Q^H$, and the closure kernel in (6.12) is exactly zero. The section

$$s : Q \rightarrow \mathcal{B}, \quad s[F] = KF \quad (6.12i)$$

is well-defined and continuous in the quotient topology, satisfies $qs = 1$ and $\Lambda s = 0$, and gives the topological decomposition $\mathcal{B} \cong V \oplus Q$ by $F \mapsto (\Lambda F, qF)$. Every cutoff and the factor $(1 - c)^{-1}$ remain in this construction. It establishes closedness of the theta image, and makes no assertion that the different kernel Q_Z^0 in (6.11) vanishes.

Let $\mathcal{T}_Z$ have the same source legs and sigma stalk as $\mathcal{T}$, eta stalk $\mathcal{K}_Z$, and the original restrictions. Then

$$0 \rightarrow \mathcal{T}_Z \rightarrow \mathcal{T} \xrightarrow{J_Z} S_{\eta}A_Z \rightarrow 0 \quad (6.13)$$

is exact at each stalk; its eta surjectivity is (6.8). Applying the explicit complexes gives a termwise exact sequence

$$0 \rightarrow C(\mathcal{T}_Z) \rightarrow C(\mathcal{T}) \rightarrow A_Z[-1] \rightarrow 0. \quad (6.14)$$

The kernel inclusion into the homotopy fibre of the last map is a quasi-isomorphism. This follows directly by quotienting that inclusion: the quotient fibre is the fibre of the identity of $A_Z[-1]$, whose two identical terms have an invertible differential and zero cohomology. Therefore the full cohomology of this jet-comparison fibre is

$$H^0 = V \quad \text{via } \psi \mapsto (\widehat{\psi}, \psi), \quad H^1 = Q_Z^0, \quad (6.15)$$

and nothing else. On a nonempty one-leg mask $H^0 = 0$ and $H^1 = Q_Z^0$; the same proof applies to each exact mask diagram. On $S_{\eta}A_Z$ itself, sigma restriction is zero, its restriction fibre is exactly $A_Z[-1]$, and its restriction cone is $A_Z[0]$.

For disjoint finite packets Z, W of actual zeros, simultaneous application of (6.8) to $Z \cup W$ with prescribed coordinates $(0, w)$ proves the further exact surjection

$$Q_Z^0 \longrightarrow A_W. \quad (6.16)$$

Consequently the kernel in (6.11) retains all these explicitly reachable omitted packets. It has not been set to zero.

7. Original scaling, generator, and dual actions

For $a > 0$, retain $U_a F(x) = F(x/a)$. Substitution $x = ay$ gives

$$\mathcal{M}U_a F(s) = a^s \mathcal{M}F(s), \quad \widehat{U_a \phi} = aU_{1/a} \widehat{\phi}. \quad (7.1)$$

These operators preserve the original test spaces: dilation rescales each defining Schwartz or $\mathcal{B}$ seminorm by finite constants, preserves evenness and the value-zero condition, and multiplies the integral by a . The source sheaf action is exactly

$$U_a^{\mathcal{T}} : (\phi, \psi, F) \mapsto (U_a \phi, aU_{1/a} \psi, U_a F). \quad (7.2)$$

The identities $\Theta U_a = U_a \Theta$ and (7.1) give

$$d(U_a \phi, a U_{1/a} \psi) = U_a \Theta(\phi - \widehat{\psi}). \quad (7.3)$$

Thus (7.2) is a diagram automorphism and (3.1) proves equivariance of every restriction-fibre map. On $H^0 = V$ in the parameter ψ of (5.7), its action is $\psi \mapsto a U_{1/a} \psi$; on Q it is $[F] \mapsto [U_a F]$. Every support mask is fixed.

The infinitesimal generator for $a = e^t$ at $t = 0$ is $D = -x \partial_x$ on the plus and eta stalks, and $1 - D$ on the minus stalk. Differentiating (7.1) gives the original relation $\mathcal{F} D = (1 - D) \mathcal{F}$. Integration by parts, with zero boundary terms because $F \in \mathcal{B}$, gives $\mathcal{M}(DF) = s \mathcal{M} F$.

On the full local jet algebra,

$$U_a|_\rho = a^\rho \exp((\log a) N_\rho) = a^\rho \sum_{k=0}^{m_\rho-1} \frac{(\log a)^k}{k!} N_\rho^k, \quad N_\rho(z_\rho^j) = z_\rho^{j+1}, \quad (7.4)$$

where $N_\rho(z_\rho^{m_\rho-1}) = 0$. The generator is $\rho I + N_\rho$. This follows by multiplying the jet by the exact germ $a^{\rho+z_\rho}$, so (6.4), (6.11), (6.13), and (6.14) intertwine it; in particular $\mathcal{K}_Z$ and Q_Z^0 are stable under scaling. Nilpotent directions have not been replaced by their trace.

Let $\mathcal{L}_1 = E$ with action $w \mapsto aw$, the original integral-moment line. On every dual object the action is the induced Hom action

$$(U_a \ell)(v) = a \ell(U_{1/a} v). \quad (7.5)$$

The formula is forced by the evaluation map $\ell \otimes v \mapsto \ell(v) \in \mathcal{L}_1$: applying (7.5) and U_a to both inputs multiplies the result by a . Since all maps in Sections 3 and 4 are defined by precomposition and restriction, this proves their equivariance with (7.5), including every contravariant mask transport.

The known continuous section (6.12i) has an exact, retained scaling defect. Denote the induced action on Q by $\overline{U}_a$, and define

$$k_a = \Lambda U_a s : Q \rightarrow V, \quad U_a s - s \overline{U}_a = \Theta k_a. \quad (7.6)$$

To prove the second equality, use $1_{\mathcal{B}} = sq + \Theta \Lambda$ on $U_a s$, and $q U_a s = \overline{U}_a$. The group identity then gives, without dropping a boundary term,

$$k_{ab} = U_a k_b + k_a \overline{U}_b, \quad k_1 = 0. \quad (7.7)$$

Indeed expand $U_{ab} s - s \overline{U}_{ab}$ by first applying the equation for b , then that for a , and cancel the injective Θ . These maps are continuous by their formulas. Thus the known topological splitting is accompanied by the full original source-valued action defect.

8. Full finite residue arrow and its chain degree

The symmetry of the original $g = 2\xi$ used here follows directly from the original Gaussian Poisson formula, with no assumption on its zeros. Put $\psi_0(t) = \sum_{n \geq 1} e^{-\pi n^2 t}$. Poisson gives $\psi_0(t) = t^{-1/2} \psi_0(1/t) + \frac{1}{2}(t^{-1/2} - 1)$. For $\Re s > 1$, termwise integration gives $\pi^{-s/2} \Gamma(s/2) \zeta(s) = \int_0^\infty \psi_0(t) t^{s/2} dt/t$. Splitting at one and substituting $t = 1/v$ in the reflected term therefore yields the entire continuation

$$g(s) = 1 + s(s-1) \int_1^\infty \psi_0(t) (t^{s/2} + t^{(1-s)/2}) \frac{dt}{t}. \quad (8.0)$$

The constant is exactly one: the remaining elementary integral is $1/(s-1) - 1/s = 1/[s(s-1)]$. The bound $\psi_0(t) \leq e^{-\pi t}/(1 - e^{-\pi t})$ for $t \geq 1$ proves locally uniform convergence with all s -derivatives.

The expression is invariant under $s \mapsto 1 - s$, and its real coefficients give conjugation symmetry. Consequently $g^\dagger = g$, and g is not the zero function (for example its continuation has $g(0) = 1$).

Assume Z is reflection-stable, meaning the finite chosen set satisfies $\iota Z = Z$ for $\iota\rho = 1 - \bar{\rho}$; this selects an actual finite divisor, without a hypothesis on the location of its points. The completed function has $g^\dagger = g$ for $f^\dagger(s) = \overline{f(1 - \bar{s})}$. In original coordinates,

$$f_\rho^\dagger(z_\rho) = \sum_j (-1)^j \overline{f_{\iota\rho, j}} z_\rho^j, \quad u_\rho(z) = (-1)^{m_\rho} \overline{u_{\iota\rho}(-\bar{z})}. \quad (8.1)$$

The residue pairing, with its raw orientation, is

$$R_Z(f, h) = \sum_{\rho \in Z} \text{Res}_\rho \frac{f^\dagger(s) h(s)}{g(s)} ds. \quad (8.2)$$

Changing any representative by a multiple of g adds a holomorphic differential, so this is well-defined. At ρ , if $u_\rho(z)^{-1} = \sum_{k \geq 0} c_{\rho, k} z^k$, its literal formula is

$$R_Z(f, h) = \sum_\rho \sum_{i, j=0}^{m_\rho-1} (-1)^i \overline{f_{\iota\rho, i}} h_{\rho, j} c_{\rho, m_\rho-1-i-j}, \quad c_{\rho, k} = 0 \ (k < 0). \quad (8.3)$$

The matrix $G_{ij} = c_{\rho, m_\rho-1-i-j}$ has determinant $(-1)^{m_\rho(m_\rho-1)/2} u_\rho(0)^{-m_\rho} \neq 0$: reversing the columns makes it triangular with diagonal $u_\rho(0)^{-1}$, and reversal has the displayed sign. Multiplication of the i -th row by $(-1)^i$, together with the invertible reflection-permutation of blocks, preserves invertibility. Hence R_Z is perfect, including all nilpotent jets and actual analytic units.

For a meromorphic function k , expansion in z_ρ gives

$$\text{Res}_\rho k^\dagger(s) ds = -\overline{\text{Res}_{\iota\rho} k(s) ds}. \quad (8.4)$$

Only its coefficient of z^{-1} contributes, and reflection multiplies that coefficient by $(-1)^{-1} = -1$. Apply (8.4) to $k = h^\dagger f/g$, sum over the permuted divisor, and use $g^\dagger = g$. The result is

$$R_Z(f, h) = -\overline{R_Z(h, f)}. \quad (8.5)$$

The raw pairing is therefore skew-Hermitian. No phase has been inserted.

The identity $(a^s)^\dagger = a^{1-s}$ gives

$$R_Z(U_a f, U_a h) = a R_Z(f, h). \quad (8.6)$$

Thus the following is an injective linear equivariant map from the conjugate vector space:

$$\boxed{\overline{A_Z} \longrightarrow H^{-1} R \text{Hom}_{\mathcal{A}}(\mathcal{T}, K_\tau(\mathcal{L}_1)), \quad \bar{f} \mapsto \ell_f, \quad \ell_f(F) = R_Z(f, J_Z F).} \quad (8.7)$$

Here ℓ_f is literally the degree -1 cochain on $\mathcal{B}$ in (3.4). It is closed, since (6.2) gives $\ell_f \Theta = 0$, and $\ell_f \Theta \mathcal{F} = 0$. There is no degree -2 term, so this cocycle represents zero exactly when it is the zero functional. Surjectivity of J_Z and perfectness of (8.2) then prove injectivity. Formula (8.6), evaluated on $J_Z U_{1/a} F$, proves $\ell_{U_a f}(F) = a \ell_f(U_{1/a} F)$, exactly (7.5). Bound (6.9) also proves that all these functionals are continuous on the original quotient topology, even though the adjunction (3.4) uses the full algebraic dual. Thus (8.7) exhibits a specified subspace of continuous functionals inside that full dual, by an actual injection.

The contraction $J_g : h \mapsto g' h$ retains its local form

$$g' = m_\rho u_\rho(z) z^{m_\rho-1} + u'_\rho(z) z^{m_\rho} \equiv m_\rho u_\rho(z) z^{m_\rho-1} \pmod{z^{m_\rho}}. \quad (8.8)$$

Using $g'/g = m_\rho/z + u'_\rho/u_\rho$ gives

$$R_Z(f, J_g h) = \sum_\rho m_\rho \overline{f(\iota\rho)} h(\rho) = \text{Tr}(M_{f^\dagger h} | A_Z). \quad (8.9)$$

The trace equality follows because multiplication by a germ in the ordered basis $1, z, \dots, z^{m-1}$ is triangular with its value repeated m times on the diagonal. The radical in either variable of (8.9) is exactly the subspace of all packets whose constant term is zero: the formula annihilates that subspace, and arbitrary constant terms in the reflected block detect every nonzero remaining value. The differentiated symmetry is $(g')^\dagger = -g'$, which together with (8.5) also confirms the Hermitian orientation of (8.9). No positivity of the raw residue form was used or asserted.

Every arrow in (8.7) is reconstructed at its original label A . A zero functional is $(A, 0)$, and the absent input alone is sent to τ . Precomposition with the mask inclusions commutes with the residue formula because H^1 transport in (5.8) is identity. This completes the labelled adjoint arrow with its degree, sign, topology, and action.

9. Tensor compatibility, including the dual Koszul sign

All coefficient tensors in this section are the original algebraic tensors over E ; no unproved completed tensor topology is substituted. For a product P^r , tensoring the stalkwise exact resolution (2.1) produces a finite projective resolution of the constant diagram. To see exactness without assuming a tensor theorem, at each stalk split an exact finite complex into its cohomology and elementary two-term identity complexes by choosing complements to boundaries and cycles. Tensoring an elementary identity complex with any complex is contractible, by its usual contracting homotopy tensored with identity. Only the tensor of the degree-zero constant cohomology remains. Each external product of the P_x 's represents evaluation at the corresponding product point and is projective. This proves the claimed resolution and yields the natural cochain identity

$$A_{\tau,r}(F_1 \boxtimes \dots \boxtimes F_r) \simeq C(F_1) \otimes \dots \otimes C(F_r), \quad (9.1)$$

with the full differential

$$d(x_1 \otimes \dots \otimes x_r) = \sum_{j=1}^r (-1)^{\sum_{k < j} |x_k|} x_1 \otimes \dots \otimes dx_j \otimes \dots \otimes x_r. \quad (9.2)$$

The standard cross product of cohomology classes is an isomorphism because the same split-complex argument kills all summands with a contractible factor. The proof uses a splitting to verify that the canonical cross product is bijective; it does not make that map dependent on the splitting.

Product restriction is the tensor $f_1 \otimes \dots \otimes f_r$ to $S_1 \otimes \dots \otimes S_r[0]$. Changes from plus to minus representatives are compatible: for $g_i = f_{i,-}$ and $f_i = f_{i,+}$, a homotopy is

$$H = \sum_{j=1}^r g_1 \otimes \dots \otimes g_{j-1} \otimes h_j \otimes f_{j+1} \otimes \dots \otimes f_r, \quad (9.3)$$

where tensor maps act with the Koszul sign and $h_j^1 = t_j$. The differential of each tensor summand is the same tensor with $d_{\text{Hom}} h_j = f_j - g_j$; every other factor is a degree-zero chain map. Summing telescopes to $\bigotimes f_i - \bigotimes g_i$, proving the homotopy identity.

For the original sheaves and all nonempty masks, every $S_i = 0$. Thus product restriction is zero and

$$B_{\sigma^r}(\mathcal{T}_{A_1} \boxtimes \dots \boxtimes \mathcal{T}_{A_r}) = \bigotimes_{i=1}^r C(\mathcal{T}_{A_i}) = \bigotimes_{i=1}^r B_{\sigma}(\mathcal{T}_{A_i}). \quad (9.4)$$

The complete cohomology is the graded tensor product of the two groups in (5.8). Explicitly its degree n is the direct sum over subsets $I \subseteq \{1, \dots, r\}$, $|I| = n$, with Q in factors indexed by I and $H^0 C(\mathcal{T}_{A_i})$ in the other factors. In particular

$$\boxed{H^r B_{\sigma^r}(\mathcal{T}^{\boxtimes r}) = Q^{\otimes r}.} \quad (9.5)$$

The numerator is $\mathcal{B}^{\otimes r}$, and its denominator is exactly

$$\sum_{j=1}^r \mathcal{B}^{\otimes(j-1)} \otimes \Theta V \otimes \mathcal{B}^{\otimes(r-j)}. \quad (9.6)$$

Signs in (9.2) change none of these subspaces but remain in the differential itself. The quotient map to $Q^{\otimes r}$ has kernel (9.6), by successive tensoring of the quotient presentation over the field.

For general diagrams one cannot replace the fibre of a tensor product by the tensor of fibres. The exact replacement can itself be computed. For composable chain maps $X \xrightarrow{f} Y \xrightarrow{g} Z$, put $B(f) = \text{Fib}(f)$ with (1.3). There are chain maps

$$u : B(f) \rightarrow B(gf), \quad (x, y) \mapsto (x, gy), \quad v : B(gf) \rightarrow B(g), \quad (x, z) \mapsto (fx, z). \quad (9.7)$$

Their composite has nullhomotopy $H(x, y) = (-y, 0)$, since $dH + Hd = (fx, gy)$. A surjective chain map

$$\text{Cone}(u) \rightarrow B(g), \quad (x, z; x', y') \mapsto (fx - y', z) \quad (9.8)$$

has kernel $z = 0, y' = fx$, with remaining coordinates $x \in X^n, x' \in X^{n+1}$ and differential $(x, x') \mapsto (dx + x', -dx')$. The degree -1 map $(x, x') \mapsto (0, x)$ contracts this kernel. Thus (9.8) is a quasi-isomorphism and proves the fibre triangle explicitly. Apply this with $f = f_1 \otimes 1_{C_2}$ and $g = 1_{S_1} \otimes f_2$. The resulting triangle is

$$B(f_1) \otimes C_2 \rightarrow B(f_1 \otimes f_2) \rightarrow S_1 \otimes B(f_2) \rightarrow (B(f_1) \otimes C_2)[1]. \quad (9.9)$$

The first fibre identification follows immediately from (1.3) and the tensor differential; in the second, S_1 is in degree zero. This proves the general tensor relation and explains precisely why it reduces to (9.4) for the original source.

The product right adjoint is

$$K_{\tau, r}(W) = K_{\tau}(E)^{\boxtimes r} \otimes W, \quad K_{\tau, r}(W) \simeq S_{(\eta, \dots, \eta)} W[r]. \quad (9.10)$$

Its terms are finite direct sums of $I_{(x_1, \dots, x_r)}(W)$, hence injective. Its only cohomology follows by the same stalkwise contraction proof used for (3.9) and (9.1). No assertion that an infinite-dimensional algebraic dual commutes with tensor products is used.

The adjunction on the product has a sign that must be stated. In total degree $-p$, multiply the coefficient functional arising from $K_{\tau}(E)^{\boxtimes r}$ by

$$\epsilon_p = (-1)^{p(p-1)/2} \quad (9.11)$$

to identify it with $\text{Hom}^{-p}(\bigotimes C(F_i), W)$. To check this, an index set I of p eta factors contributes to degree $-p$. Removing $j \in I$ in d_K has sign $(-1)^{|I \cap \{1, \dots, j-1\}|}$. The corresponding incoming cochain differential in (9.2) has the same sign, while its internal Hom differential has the additional factor $(-1)^{p+1}$ from (1.4). The equality $\epsilon_{p-1} = (-1)^{p+1} \epsilon_p$ proves commutation in every component. This verifies the sign on every product differential.

In top degree the identification is therefore

$$H^{-r} R \text{Hom}_{\mathcal{A}(P^r)}(\mathcal{T}^{\boxtimes r}, K_{\tau, r}(W)) \cong \text{Hom}_E(Q^{\otimes r}, W), \quad (9.12)$$

with the coefficient multiplier ϵ_r . In particular, for $r = 2$ it is -1 . If a product residue arrow is written on the right of (9.12) as the literal functional

$$([F_1] \otimes \dots \otimes [F_r]) \mapsto \prod_{i=1}^r R_Z(f_i, J_Z F_i), \quad (9.13)$$

its coefficient representative in the unaltered tensor complex $K_{\tau,r}$ is ϵ_r times that product. This is the cohomological Koszul sign, not a phase inserted into (8.2) and not a modification of its positivity properties.

Tensoring (6.11) gives the exact surjection $Q^{\otimes r} \rightarrow A_Z^{\otimes r}$, with kernel

$$\sum_{j=1}^r Q^{\otimes(j-1)} \otimes Q_Z^0 \otimes Q^{\otimes(r-j)}. \quad (9.14)$$

This follows by successively quotienting each factor; a vector-space complement verifies equality of the kernel, without imposing an equivariant complement. The fibre of the cochain jet map

$$C(\mathcal{T})^{\otimes r} \rightarrow A_Z^{\otimes r}[-r] \quad (9.15)$$

has the same cohomology as $C(\mathcal{T})^{\otimes r}$ in degrees below r , has (9.14) in degree r , and zero above r . Indeed the target has only degree r , its degree- r map is the proved surjection, and the explicit fibre long exact sequence then has exactly these kernels and cokernels.

The action in all tensor expressions is the tensor of the original (7.2) and (7.4). On a specified packet tuple $(\rho_1, \dots, \rho_r)$ it is

$$a^{\rho_1 + \dots + \rho_r} \exp \left((\log a) \sum_{j=1}^r 1 \otimes \dots \otimes N_{\rho_j} \otimes \dots \otimes 1 \right). \quad (9.16)$$

The summands commute, so their exponential is exactly the product of the terminating exponentials in (7.4). On the target line $\mathcal{L}_1^{\otimes r}$ the action is a^r ; (8.6) proves covariance of (9.13) by this exact factor. Formula (9.16) retains every nilpotent term, not only its eigenvalue. Support reconstruction is performed on each original admitted tensor fibre and its balancing relations: a zero tensor amplitude remains the supported zero of that fibre, whereas an absent tensor input is absorbing. The proof never identifies the coefficient tensor with a Cartesian product of label sets.

There is also the source's proved passage to completed projective tensor products, now obtained from the retained continuous contraction instead of an assumed exactness of completion. For the joint complex $C = C(\mathcal{T})$, let $H_0 = V[0] \oplus Q[-1]$, with zero differential, and put

$$p^0(\phi, \psi) = \psi, \quad p^1 = q, \quad i^0(\psi) = (\widehat{\psi}, \psi), \quad i^1 = s, \quad H^1(F) = (\Lambda F, 0). \quad (9.17)$$

Then $pi = 1$ and $1 - ip = dH + Hd$: in degree zero the latter equation is $(\phi, \psi) - (\widehat{\psi}, \psi) = (\phi - \widehat{\psi}, 0)$, obtained from $\Lambda\Theta = 1$; in degree one it is $1 - sq = \Theta\Lambda$. All the maps are continuous in the already specified topologies. If $P = ip$, the tensor homotopy

$$H_r = \sum_{j=1}^r P^{\otimes(j-1)} \otimes H \otimes 1^{\otimes(r-j)} \quad (9.18)$$

uses the sign $(-1)^{\sum_{k < j} |x_k|}$ on homogeneous inputs. The differential telescopes to $1 - P^{\otimes r}$, since $d_{\text{Hom}} H = 1 - P$, while $P, 1$ are chain maps of degree zero. Every finite tensor map is continuous for the projective tensor topology and therefore extends uniquely to its completion; the identity extends by density. The completed complex is thus continuously homotopy equivalent to $H_0^{\widehat{\otimes} \pi^r}$. Its degree n cohomology is the direct sum of ordered completed tensors with Q in n slots and V in the others, and in particular its top cohomology is $Q^{\widehat{\otimes} \pi^r}$. The comparison from (9.5) is the canonical completion map on these tensors. For an individual one-leg mask replace H_0 by $Q[-1]$, keep $p^1 = q, i^1 = s$, and take $H^1 = \Lambda$ on the plus leg or $H^1 = -\mathcal{F}\Lambda$ on the minus leg. On the latter the differential is $-\Theta\mathcal{F}$, so $Hd = \mathcal{F}\Lambda\Theta\mathcal{F} = 1_V$ and $dH = \Theta\Lambda = 1 - sq$; the plus leg has the same identities with its positive signs. Thus the same tensor proof applies to every mask. This specifies the exact completed continuation while retaining the algebraic tensor calculation and the action defect (7.6).

10. Exact consequences for continuation

The original global arithmetic object, its generic-boundary restriction, and its right adjoint are related by the explicit maps (2.3), (3.10), and (4.4), and the nonsplit extension (4.7). On the original theta sheaf, the restriction fibre is the original cochain complex itself, with every supported copy of Q retained. Finite jets give the exact kernel sequence (6.11), its sheaf and fibre realizations (6.13)–(6.15), and the injective continuous residue arrow into the original adjoint (8.7). Tensor products give (9.5), (9.14), and the full action (9.16), with the independently checked Koszul sign (9.11).

These are completed calculations on the original $\Theta, V, \mathcal{B}, \mathbf{A}_\tau, K_\tau$ and $g = 2\xi$. No vanishing of Q , of the finite-jet kernel Q_Z^0 , or of an arithmetic boundary contribution was used to obtain them. The calculations establish exact comparison morphisms and retain their kernels; they supply no upper estimate for the original tensor action beyond the identities proved here.

Chapter 3

Original scaling, the source extension, and the full boundary pairing

Finite dilation of the original arithmetic packet over the tau base

0. Scope and exact source correspondence

This calculation continues the original two-leg theta complex. The retained infinitesimal identity is (A11) of `arithmetic_input.tex`; the absolute-base pushforward, its two-leg action, and its tensor differential are (19), (21), (34), and (46) of *Arithmetic cohomology over the tau-base*. The full global retraction and its cocycle were already constructed in transcript turn **A1694**, node `1fa0ebbc-f569-40c1-b7ac-b523da19a5f0`, chain position 3576, and in *Tau_Global_Comparison*, equations (16), (19), (29)–(35), and (41). Those results are retained as prior work, not reported as newly discovered here.

The additional calculation is the explicit finite-packet dilation cocycle for every positive real dilation, its full Taylor-unit and nilpotent coefficients, convergent source-space integral, incomplete-gamma evaluation, exact finite/global change of representatives, two-leg homotopies, and tensor and boundary formulas. These formulas perform the group-level continuation of the existing infinitesimal extension. They do not assign positivity to a representative pairing.

The source bytes used are pinned at the end. Every integral on the positive half-line keeps the displayed measure. In particular the Hilbert pairing below uses dx , and its dilation mass is a , not 1. The notation $t = \log a$ is a real time parameter; the polynomial coordinate in $\mathbb{C}[X]/(h)$ is written X to avoid confusing it with that time. Thus $A = M_X$ here is exactly the source operator denoted M_t .

1. The original spaces, maps, and topology

Work over $\mathbb{C}$. Let

$$V = \left\{ \phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \quad \phi(0) = 0, \quad \int_{\mathbb{R}} \phi(x) dx = 0 \right\},$$
$$\mathcal{B} = \left\{ F \in C^\infty(\mathbb{R}_{>0}) : p_{b,k}(F) = \sup_{x>0} x^b |D^k F(x)| < \infty \quad (b \in \mathbb{Z}, k \geq 0) \right\}, \quad D = -x\partial_x.$$

Use the Schwartz seminorms

$$q_{N,k}(\phi) = \sup_{x \in \mathbb{R}} (1 + |x|)^N |\partial_x^k \phi(x)|, \quad N, k \geq 0.$$

The space V is a closed subspace of the complete Schwartz space: parity, value at zero, and integration are continuous. For example $|\int \phi| \leq q_{2,0}(\phi) \int_{\mathbb{R}} (1+|x|)^{-2} dx$. The space $\mathcal{B}$ is complete because a Cauchy sequence in all $p_{b,k}$ converges with all derivatives on compact subsets after $x = e^y$; the weighted uniform bounds pass to its limit. These observations justify limits of Riemann sums in the actual spaces below.

Retain

$$\hat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx, \quad \Theta \phi(x) = 2 \sum_{n \geq 1} \phi(nx), \quad JF(x) = x^{-1} F(1/x),$$

$$\mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}.$$

Schwartz estimates, followed by the original Poisson identity, give

$$\Theta : V \longrightarrow \mathcal{B}, \quad \Theta \hat{\phi} = J\Theta \phi, \quad D\Theta = \Theta D.$$

Here is the continuity argument. At $x \geq 1$, each $D^k \phi(nx)$ is bounded by a finite sum of Schwartz seminorms times $(nx)^{-M}$, where M can be any sufficiently large positive integer. Summing n^{-M} gives every required bound at infinity. The Fourier transform preserves Schwartz seminorms continuously, preserves evenness, and interchanges the two vanishing moments. Poisson summation therefore gives $\Theta \phi(x) = x^{-1} \Theta \hat{\phi}(1/x)$; applying D^k and the preceding bounds gives every required estimate at zero. Each resulting estimate involves finitely many source seminorms. Termwise differentiation is justified uniformly on every $x \geq \delta > 0$ by the same summable majorants.

The theta map is injective. For fixed $x > 0$, absolute convergence permits

$$\frac{1}{2} \sum_{m \geq 1} \mu(m) \Theta \phi(mx) = \sum_{r \geq 1} \left(\sum_{m|r} \mu(m) \right) \phi(rx) = \phi(x).$$

Indeed the rearranged absolute sum is bounded by $\sum_{r \geq 1} d(r) |\phi(rx)| < \infty$, since $d(r) \leq r$ and ϕ is Schwartz. Thus no theta relation is lost when its source is recovered.

For $a > 0$ define $U_a F(x) = F(x/a)$, and use the same formula on V . Direct substitutions give

$$p_{b,k}(U_a F) = a^b p_{b,k}(F), \quad q_{N,k}(U_a \phi) \leq a^{-k} \max(1, a)^N q_{N,k}(\phi), \quad (1)$$

$$\mathcal{M}U_a F(s) = a^s \mathcal{M}F(s), \quad \widehat{U_a \phi} = a U_{1/a} \hat{\phi}, \quad \Theta U_a = U_a \Theta. \quad (2)$$

The two moments of $U_a \phi$ are $\phi(0)$ and $a \int \phi$, so V is invariant. Differentiating the defining formula gives

$$\frac{d}{dt} U_{e^t} \phi = D U_{e^t} \phi = U_{e^t} D \phi.$$

This is a derivative in every Schwartz seminorm. To verify it, apply the ordinary Taylor formula with integral remainder to $U_{e^{t+h}} \phi$; its second derivative is $U_{e^{t+h}} D^2 \phi$, uniformly bounded on each compact time interval by (1). The same argument, using the exact first identity in (1), proves smoothness in $\mathcal{B}$. Repeating with $D^j \phi$ proves derivatives of all orders. In particular this is a smooth group action on the retained Frechet spaces.

The four-point source poset is

$$+ < \eta < \sigma, \quad - < \eta < \sigma.$$

For the sheaf with stalks $V, V, \mathcal{B}, 0$ and restriction maps $\Theta, \Theta\mathcal{F}$, the source projective resolution computes

$$C = A_\tau(\mathcal{F}) = \left[V \oplus V \xrightarrow{d} \mathcal{B} \right], \quad d(v, w) = \Theta(v - \widehat{w}), \quad |V \oplus V| = 0, \quad |\mathcal{B}| = 1. \quad (3)$$

Its actual chain action is

$$\mathcal{U}_a^0(v, w) = (U_a v, aU_{1/a} w), \quad \mathcal{U}_a^1 F = U_a F. \quad (4)$$

Equation (2) proves $d\mathcal{U}_a^0 = \mathcal{U}_a^1 d$. On even Schwartz functions $\mathcal{F}^2 = 1$, so the exact degree-zero cycle inclusion is

$$i_0(w) = (\widehat{w}, w), \quad \mathcal{U}_a i_0(w) = i_0(aU_{1/a} w). \quad (5)$$

It follows from theta injectivity that $H^0(C) = i_0(V)$ and $H^1(C) = Q = \mathcal{B}/\Theta V$. These are the original global cohomology groups over the absolute tau base.

2. The finite arithmetic packet and its unchanged Taylor unit

Retain the original Gaussian source

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}, \quad f_0 = \Theta \phi_*.$$

It has value zero at zero, and direct Gaussian integration gives zero integral. More precisely, with $M_+ \phi(s) = \int_0^\infty \phi(x) x^s dx/x$,

$$M_+ \phi_*(s) = \pi^{-s/2} (2\Gamma(s/2 + 2) - 3\Gamma(s/2 + 1)) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma(s/2). \quad (6)$$

Consequently the entire function $g = \mathcal{M} f_0$ agrees, first by absolutely convergent integration in a right half-plane and then by continuation, with the original function

$$g(s) = 2\xi(s) = s(s-1) \pi^{-s/2} \Gamma(s/2) \zeta(s).$$

Entireness of the Mellin transform of $f_0 \in \mathcal{B}$ follows directly from the bounds in $\mathcal{B}$. For $0 < \operatorname{Re} s < 1$, set

$$H_\phi(s) = \frac{2\pi^{s/2}}{s(s-1)\Gamma(s/2)} M_+ \phi(s).$$

Then

$$\mathcal{M} \Theta \phi = g H_\phi, \quad H_{\phi_*} = 1, \quad H_{P(D)\phi_*}(s) = P(s). \quad (7)$$

For completeness, $M_+ \phi$ is holomorphic on $\operatorname{Re} s > -2$, since smooth evenness and $\phi(0) = 0$ give $\phi(x) = O(x^2)$. In $\operatorname{Re} s > 1$, interchanging its theta series and Mellin integral gives $\mathcal{M} \Theta \phi = 2\zeta(s) M_+ \phi(s)$. Continue to the indicated strip to obtain the first equality in (7). Integrating $D\phi$ by parts proves $H_{D\phi} = s H_\phi$; the endpoint term $[x^s \phi(x)]_0^\infty$ vanishes on that strip. Equation (6) then proves the other two equalities.

Fix the original nonempty finite packet Z of actual nontrivial zeros, each with its complete order m_ρ , and put

$$h(X) = \prod_{\rho \in Z} (X - \rho)^{m_\rho}, \quad d_h = \deg h, \quad E = \mathbb{C}[X]/(h), \quad A = M_X,$$

$$v(s) = g(s)/h(s), \quad \varepsilon = j_h(h/g) \in E^\times.$$

The quotient h/g here is taken as a germ at each selected zero; these germs are holomorphic and nonzero because the complete multiplicity has been retained. The local unit is **not** replaced by 1. For $u \in E$, let $R_Z(u)$ be the unique polynomial of degree less than d_h representing εu , and put

$$\ell(u) = [X^{d_h-1}]R_Z(u). \quad (8)$$

We record the inverse used to define the source section, rather than taking an unproved inverse as an extra hypothesis. On $F \in \mathcal{B}$ satisfying $\mathcal{M}F(\rho) = 0$, define

$$S_\rho F(x) = x^{-\rho} \int_x^\infty F(y) y^{\rho-1} dy.$$

Differentiation gives $(D - \rho)S_\rho F = F$. The moment condition makes the upper integral the negative lower integral. If $\sigma = \operatorname{Re} \rho$, the upper estimate, for $N > \sigma$, is

$$|S_\rho F(x)| \leq \frac{p_{N,0}(F)}{N - \sigma} x^{-N}.$$

The lower estimate, for $N + \sigma > 0$, is

$$|S_\rho F(x)| \leq \frac{p_{-N,0}(F)}{N + \sigma} x^N.$$

Using the upper estimate on $x \geq 1$ and the lower estimate on $x \leq 1$ gives every weighted bound. The differential identity expresses $D^k S_\rho F$ as $\rho^k S_\rho F$ plus a finite linear combination of $D^j F$, proving all derivative bounds and continuity. The homogeneous solution $cx^{-\rho}$ cannot belong to $\mathcal{B}$ unless $c = 0$, so the inverse is unique. Integration by parts gives

$$\mathcal{M}S_\rho F(s) = \frac{\mathcal{M}F(s)}{s - \rho},$$

with the removable value filled in. Iterating this construction, removing one vanishing factor at a time, defines the continuous inverse S_h of $h(D)$ on $\ker(j_h \mathcal{M})$. The operation preserves the remaining zeros and lowers the order at the chosen zero by exactly one at each step. Conversely any $h(D)F$ has those vanishing jets. This proves

$$h(D)\mathcal{B} = \ker(j_h \mathcal{M}), \quad h(D) \text{ is injective.} \quad (9)$$

The section in the question is therefore the actual well-defined map

$$R = R_{\text{ref}} : E \longrightarrow \mathcal{B}, \quad Ru = S_h \Theta(R_Z(u)(D)\phi_*). \quad (10)$$

Indeed its input has Mellin transform $gR_Z(u)$, which vanishes to all the required orders. Equations (7) and (9) give

$$\mathcal{M}Ru = vR_Z(u), \quad J_Z R = 1_E, \quad J_Z = j_h \mathcal{M}, \quad J_Z \Theta = 0. \quad (11)$$

The Mellin-jet map is continuous: split its integral at 1, dominate the logarithmic factors at zero by a sufficiently negative weight and those at infinity by a sufficiently positive weight. This yields a finite sum of seminorm bounds for each derivative at each ρ . In (11), $j_h(v)\varepsilon = 1$ proves the middle identity. Hence R and $\sigma = qR : E \rightarrow Q$ are injective, since J_Z descends to Q .

The infinitesimal boundary is precisely the retained one:

$$DR - RA = \Theta \phi_* \ell = f_0 \ell. \quad (12)$$

To prove it without losing its coefficient, the polynomial $XR_Z(u) - R_Z(Au)$ vanishes modulo h and has degree at most d_h . Since h is monic, its quotient by h is exactly $\ell(u)$. Apply $h(D)$ to (12), use (10), $D\Theta = \Theta D$, and that polynomial identity. Injectivity of $h(D)$ from (9) proves (12). For a single simple zero, $R_Z(u) = u/g'(\rho)$ and $\ell(u) = u/(2\xi'(\rho))$, with the original factor of two.

3. The finite dilation cocycle as an actual convergent source integral

For every real t , define $T_t = e^{tA}$ and

$$c_t(u) = \int_0^t U_{e^{t-r}} \phi_* \ell(T_r u) dr \in V. \quad (13)$$

If $t < 0$, $\int_0^t = -\int_t^0$; no orientation is changed. Write $c_a = c_{\log a}$ and $a^A = T_{\log a}$ when using the multiplicative parameter.

Convergence and continuity. Fix any norm on the finite-dimensional coefficient space, solely for estimates, and its induced operator norms. On the oriented interval between 0 and t , both $|r|$ and $|t-r|$ are at most $|t|$. Equation (1) and $\|T_r\| \leq e^{|r|\|A\|}$ yield

$$q_{N,k}(c_t u) \leq |t| e^{(N+k+\|A\|)|t|} q_{N,k}(\phi_*) \|\ell\| \|u\|. \quad (14)$$

The integrand is continuous in every seminorm. Its Riemann sums are Cauchy in every seminorm by uniform continuity on the compact interval. Completeness of V proves their common limit exists in V , and the inequality follows first for sums and then for their limit. Thus the integral is an actual source vector with the two original zero moments and even Schwartz regularity. It is not merely an integral in a weak dual or in L^2 .

Smoothness in t follows either by the differentiable compact-interval rule justified by the same estimates for all time derivatives, or by writing (13) as $t \int_0^1 U_{e^{t(1-r)}} \phi_* \ell(T_{tr} u) dr$. This latter formula works without a sign convention change for negative t . Every derivative has a finite sum of terms involving $D^j \phi_*$, powers of A , and bounded polynomial factors in r . Inequality (14) with those finitely many seminorms proves local uniform convergence of the differentiated Riemann sums. Therefore $t \mapsto c_t$ is smooth as a map to continuous linear maps $E \rightarrow V$, with their bounded-set topology. Finite dimensionality of E makes the preceding uniform estimates on its unit ball sufficient for this statement.

The exact dilation identity. For all $t \in \mathbb{R}$,

$$\boxed{U_{e^t} R - R T_t = \Theta c_t.} \quad (15)$$

For fixed u , differentiate in the Frechet space $\mathcal{B}$:

$$\frac{d}{dr}(U_{e^{t-r}}RT_r u) = -U_{e^{t-r}}(DR - RA)T_r u = -\Theta U_{e^{t-r}}\phi_*\ell(T_r u).$$

The fundamental theorem of calculus follows in every defining seminorm from the same Riemann-sum argument; integrating with the indicated orientation proves (15), including $t < 0$. Continuity of Θ permits moving it through the integral.

Cocycle and derivative. Directly splitting the defect in (15) gives

$$\boxed{c_{t+s} = U_{e^t}c_s + c_t T_s, \quad c_0 = 0, \quad c_{-t} = -U_{e^{-t}}c_t T_{-t}.} \quad (16)$$

Indeed applying Θ to the asserted first equality yields the identity obtained by inserting $U_{e^t}RT_s$ in $U_{e^{t+s}}R - RT_{t+s}$. Injectivity of Θ proves equality in the actual source V . The other two statements follow by setting $t = s = 0$ and $s = -t$. Differentiating (13), or the cocycle in either variable, gives the two equal expressions

$$\boxed{\partial_t c_t = Dc_t + \phi_*\ell T_t = U_{e^t}\phi_*\ell + c_t A, \quad \partial_t c_t|_{t=0} = \phi_*\ell.} \quad (17)$$

All derivatives are in V . In multiplicative notation (16) is

$$c_{ab} = U_a \circ c_b + c_a \circ b^A,$$

where the final product means the map $c_a \circ b^A$; to remove any typographical ambiguity it can also be written $c_{ab}(u) = U_a(c_b(u)) + c_a(b^A u)$.

The space $\mathcal{B}_Z = \Theta V + RE$ is a direct topological sum with coordinates $(\phi, u) \mapsto \Theta\phi + Ru$. Its inverse on that subspace is $(\Theta^{-1}(F - RJ_Z F), J_Z F)$; continuity of the inverse theta map is also proved in Section 6 below. Equations (15) and (16) integrate the source block operator to the exact group representation

$$U_a|_{\mathcal{B}_Z} = \begin{pmatrix} U_a|_V & c_a \\ 0 & a^A \end{pmatrix}. \quad (18)$$

This is the explicit group-level form of the already known extension (A12). In particular (15) proves the true quotient intertwining $\overline{U}_a \sigma = \sigma a^A$.

4. All nilpotents, the Mellin multiplier, and an incomplete-gamma evaluation

The Chinese remainder decomposition is

$$E = \bigoplus_{\rho \in Z} \mathbb{C}[z_\rho]/(z_\rho^{m_\rho}), \quad A|_\rho = \rho I + N_\rho,$$

where $N_\rho z_\rho^j = z_\rho^{j+1}$ and $N_\rho^{m_\rho} = 0$. Denote the summand projector by P_ρ , extending N_ρ by zero on other summands. These are the actual full local algebras. They can be constructed explicitly: if $H_\rho(X) = h(X)/(X - \rho)^{m_\rho}$, then P_ρ is multiplication by the remainder modulo h of

$$H_\rho(X) \sum_{j=0}^{m_\rho-1} \frac{(H_\rho^{-1})^{(j)}(\rho)}{j!} (X - \rho)^j.$$

At every other centre this has all required zero jets; at ρ it has jet 1. This verifies all its projector identities by the Chinese remainder map.

The action and the entire boundary integral therefore contain the full expressions

$$T_t = \sum_{\rho \in Z} e^{t\rho} \sum_{j=0}^{m_\rho-1} \frac{t^j}{j!} N_\rho^j P_\rho, \quad (19)$$

$$c_t u = \sum_{\rho \in Z} \sum_{j=0}^{m_\rho-1} \ell(N_\rho^j P_\rho u) \int_0^t U_{e^{t-r}\phi_*} e^{r\rho} \frac{r^j}{j!} dr. \quad (20)$$

The unit entering every coefficient is still $\varepsilon = j_h(h/g)$ through (8). If the packet has one zero of order m , and $\varepsilon = \sum_{i < m} \varepsilon_i z^i$, $u = \sum_{i < m} u_i z^i$, then $\ell(u) = \sum_{i=0}^{m-1} \varepsilon_i u_{m-1-i}$. This follows because the top coefficient of a polynomial of degree less than m is the same in the X and $z = X - \rho$ bases. Thus even the functional ℓ retains the lower coefficients of the Taylor unit, rather than only its nonzero constant value.

There is a closed expression for the integral kernels in (20). For $x > 0$ let

$$K_{\rho,t}(x) = \int_0^t e^{r\rho} \phi_*(x e^{r-t}) dr.$$

Define the oriented incomplete-gamma interval by

$$\Gamma_{[L,H]}(w) = \int_L^H y^{w-1} e^{-y} dy \quad (L, H > 0),$$

where $y^{w-1} = \exp((w-1)\log y)$ uses the real logarithm. Then substitution first of $y = x e^{r-t}$ and then of $z = \pi y^2$ gives

$$K_{\rho,t}(x) = e^{t\rho} x^{-\rho} \pi^{-\rho/2} \left[2\Gamma_{[\pi x^2 e^{-2t}, \pi x^2]}(\rho/2 + 2) - 3\Gamma_{[\pi x^2 e^{-2t}, \pi x^2]}(\rho/2 + 1) \right]. \quad (21)$$

Both substitutions preserve orientation when $t < 0$. The coefficients 2 and -3 come respectively from $4\pi^2 x^4$ and $-6\pi x^2$, including $dy/y = dz/(2z)$. No Gaussian constant is removed.

Differentiation under the compact oriented r -integral now gives the finite formula

$$c_t u = \sum_{\rho \in Z} \sum_{j=0}^{m_\rho-1} \frac{\ell(N_\rho^j P_\rho u)}{j!} \partial_\rho^j K_{\rho,t}. \quad (22)$$

The derivative here acts on the complex parameter in the entire kernel $K_{\rho,t}$ before evaluation at the indicated centre, not on the arithmetic zero or on the coefficient $\ell(N_\rho^j P_\rho u)$. The formula on $x > 0$ is extended evenly to $\mathbb{R}$, with its smooth value at 0 supplied by the integral defining K . That integral and Section 3 prove smoothness and both zero moments, so no regularity is inferred from cancelling the separate singular-looking factors in (21).

The full Mellin multiplier of the source boundary is also explicit:

$$H_{c_t u}(s) = \int_0^t e^{(t-r)s} \ell(T_r u) dr = \frac{e^{ts} R_Z(u)(s) - R_Z(T_t u)(s)}{h(s)}. \quad (23)$$

For the first equality in the original strip, use (7), $H_{U_a \phi_*} = a^s$, and the continuous source integral. For the second, differentiate $e^{(t-r)s} R_Z(T_r u)(s)$ in r . The polynomial identity proving (12) says its derivative is $-e^{(t-r)s} h(s) \ell(T_r u)$. Integration proves (23) wherever $h(s) \neq 0$. The numerator has

every required vanishing jet: in the local algebra its jet is $e^{tA}\varepsilon u - \varepsilon T_t u = 0$. Hence the quotient in (23) has removable singularities at every chosen zero and is entire. This entire expression extends the original source multiplier; it is not a replacement of h or g .

For every integer $n \geq 0$ its actual Taylor coefficient at any complex centre s_0 is

$$\frac{H_{c_t u}^{(n)}(s_0)}{n!} = \int_0^t e^{(t-r)s_0} \frac{(t-r)^n}{n!} \ell(T_r u) dr. \quad (24)$$

Inserting (19) gives every mixed coefficient in $(t-r)^n r^j / (n!j!)$ and keeps all nilpotent terms. This is also a direct convergence proof for every jet, since the parameter interval is compact.

5. Actual two-leg homotopies and the support-face comparison

Regard $E[-1]$ as the complex with E in degree 1 and zero elsewhere. The finite section defines a degree-zero cochain map $r : E[-1] \rightarrow C$, with $r^1 = R$. The maps $\mathcal{U}_a r$ and ra^A have homotopies

$$\begin{aligned} H_a^+ : E[-1] &\longrightarrow C[-1], & (H_a^+)^1 u &= (c_a u, 0), \\ H_a^- : E[-1] &\longrightarrow C[-1], & (H_a^-)^1 u &= (0, -\widehat{c_a u}). \end{aligned} \quad (25)$$

Here the notation records maps of cohomological degree -1 ; the source only has degree 1. In ordinary component notation the exact identity is

$$\boxed{\mathcal{U}_a r - ra^A = dH_a^\pm + H_a^\pm d_{E[-1]} = dH_a^\pm.} \quad (26)$$

Indeed $d(c_a u, 0) = \Theta c_a u$, whereas $d(0, -\widehat{c_a u}) = \Theta \mathcal{F}^2 c_a u = \Theta c_a u$. This verifies the minus sign using the original differential. Equation (2) also gives the homotopy cocycle itself:

$$H_{ab}^\pm = \mathcal{U}_a^0 H_b^\pm + H_a^\pm b^A. \quad (27)$$

No extra Fourier scalar is inserted: $\widehat{U_a c} = aU_{1/a} \widehat{c}$ is exactly the second-leg action in (4).

The original support masks are $\lambda \in \{\{+\}, \{-\}, \{+, -\}\}$. The fibre C_λ^0 consists of the legs present in λ , $C_\lambda^1 = \mathcal{B}$, with the same differential. The $+$ homotopy in (25) is defined on $\{+\}$ and on the joint face. The $-$ homotopy is defined on $\{-\}$ and on the joint face. On the joint face their difference is the precisely retained cycle

$$H_a^+ - H_a^- = (c_a, \widehat{c_a}) = i_0(\widehat{c_a}). \quad (28)$$

It lies in $H^0(C)$ by (5) and theta injectivity. There is no term in degree -1 of C , so a nonzero such cycle is not a boundary in C .

The two singleton homotopies are forced. On $\{+\}$, injectivity of theta forces the first one; on $\{-\}$ it forces the second. Consequently a single homotopy that is strictly natural for both singleton-to-joint inclusions would require $(c_a, 0) = (0, -\widehat{c_a})$, which is equivalent to $c_a = 0$. This is the exact scope of the support-naturality obstruction; it does not remove the homotopies in (25), their difference (28), or the resulting quotient action.

For completeness, c_a as a linear map is nonzero for every $a \neq 1$. If $c_a = 0$, (15) would make the nonzero finite-dimensional subspace RE invariant under U_a , with invertible action a^A . It therefore has a nonzero eigenvector $F \in \mathcal{B}$ with $U_a F = \lambda F$, $\lambda \neq 0$. Replacing a by a^{-1} if needed, take $a > 1$. Choose $x_0 > 0$ with $F(x_0) \neq 0$. Iteration gives $F(a^n x_0) = \lambda^{-n} F(x_0)$ for $n \geq 0$. The bound

$p_{N,0}(F) < \infty$ implies $a^{nN}|\lambda|^{-n}|F(x_0)| \leq x_0^{-N}p_{N,0}(F)$. Choose N with $a^N > |\lambda|$ and let $n \rightarrow \infty$ to obtain a contradiction. Thus no such finite invariant subspace exists. This extends the previously proved failure of a strict equivariant section by giving its exact nonzero finite-dilation defect; the failure itself was already present in (A12).

At a fixed admitted mask the split lift of every displayed map is

$$\tilde{f}(\lambda, u) = (\lambda, f(u)), \quad \tilde{f}(\tau) = \tau.$$

An amplitude $f(u) = 0$ has value $(\lambda, 0)$, the supported zero, including at $a = 1$. All homotopy identities are interpreted using addition and the supported scalar $(-1)^\bullet$ in that fibre. No absent leg is inserted by the face-specific formulas.

The known synchronization comparison can be retained explicitly when a common joint target is desired. Send every active degree-one mask to $(\{+, -\}, F)$ and send the degree-zero pair to its joint-supported pair with the same two amplitudes, putting a supported zero in a formerly missing coordinate. Denote this operation by $\mathfrak{r}$. It is the source map $(a, b) \mapsto (a + eb, b + ea)$, because multiplication by e gives a supported zero at a present coordinate. It preserves the differential, fixes joint masks, and is idempotent. The pair $(\text{supp } x, \mathfrak{r}x)$ recovers the original element by selecting the originally present coordinates, proving injectivity of that comparison. After this explicit synchronization, (25)–(28) hold in the common joint fibre. The original mask must stay attached to the comparison; forgetting it is not part of the homotopy calculation.

One can also see the extra degree-zero group in an exact equivariant roof. Let $C_Z = [V \oplus V \rightarrow \mathcal{B}_Z]$ with the same differential, and define

$$p_Z^0(v, w) = w, \quad p_Z^1 = J_Z.$$

This is a cochain map to $V[0] \oplus E[-1]$. It is equivariant for the action $aU_{1/a}$ on V and a^A on E , and induces isomorphisms on both cohomology groups: the degree-zero assertion follows from (5), while the degree-one assertion follows from the direct sum $\mathcal{B}_Z = \Theta V \oplus RE$ and (11). Thus the exact two-leg version of the source roof is

$$V[0] \oplus E[-1] \xleftarrow{p_Z} C_Z \rightarrow C.$$

Its ordinary cochain section is $i_Z^0(w) = (\widehat{w}, w)$ and $i_Z^1 = R$. If $F = \Theta\phi + Ru$, the map $h_Z^1(F) = (\phi, 0)$ obeys $1 - i_Z p_Z = dh_Z + h_Z d$: in degree zero it returns $(v - \widehat{w}, 0)$ and in degree one it returns $\Theta\phi$. Its failure of equivariance in degree one is exactly (25). This proves the replacement on the entire original two-leg complex, retaining its $V[0]$ summand.

The transported minus face has an explicitly nonzero extension component

The preceding roof also computes what happens to the original support arrows when the action is retained. Work over $\mathbb{C}[X]$, with $X = D$ on $\mathcal{B}_Z$, $X = A$ on E , and $X = 1 - D$ on the degree-zero second-leg module, denoted V_- . Let $r(X) = R_Z(1)$ and $\psi = r(D)\phi_*$. Then

$$h(D)R(1) = \Theta\psi.$$

The class of ψ in $V/h(D)V$ is nonzero: applying $j_h H$ gives $j_h r = \varepsilon \neq 0$, whereas $j_h H$ vanishes on $h(D)V$ by (7). The Fourier map carries this quotient isomorphically onto $V/h(1 - D)V$, since differentiating (2) gives $\mathcal{F}D = (1 - D)\mathcal{F}$.

For the minus-face extension the injection is $i_- = -\Theta\mathcal{F}$. Its extension representative, with the convention $h(D)R(1) = i_-(w)$, is therefore

$$w = -\widehat{\psi} \quad \text{in } V/h(1-D)V.$$

The component of the minus support arrow in the $V_-[0]$ summand of the joint roof has the opposite sign to this extension representative. Here is the exact shifted-resolution calculation. The free cochain resolution of E in degrees $-1, 0$ has differential h . We use $(C[n])^j = C^{j+n}$ and $d_{C[n]} = (-1)^n d_C$; after shifting by -1 , that resolution is in degrees $0, 1$ with differential $-h$. Its degree-one lift to the minus-face complex sends 1 to $R(1)$, extended $\mathbb{C}[X]$ -linearly. Its degree-zero lift must send 1 to $-w$, because $i_-(-w) = -h(D)R(1)$. Projection to $V_-[0]$ is identity in degree zero. Thus the induced derived map $E[-1] \rightarrow V_-[0]$ represents exactly

$$[-w] = [\widehat{\psi}] \quad \text{in} \quad \text{Ext}_{\mathbb{C}[X]}^1(E, V_-) \cong V/h(1-D)V. \quad (28a)$$

The isomorphism here follows by applying $\text{Hom}_{\mathbb{C}[X]}(-, V_-)$ to that two-term free resolution; its cokernel is the stated quotient. The class (28a) is nonzero by the preceding source-jet calculation. The same arrow has identity as its $E[-1]$ component. By contrast, the plus support arrow has zero $V_-[0]$ component and identity $E[-1]$ component, because its degree-zero image has second coordinate zero. Consequently the two support arrows in the equivariant derived category do not both become inclusion of the $E[-1]$ summand: the exact difference carries (28a), with the original Taylor unit. The strict original cochain support diagram remains available before passage through the roof.

6. The exact bridge to the already constructed global section

We recall the global section in a form that makes its use here mathematically explicit. Choose the same fixed even real cutoffs $\alpha, \beta \in C_c^\infty(\mathbb{R})$, with $0 \leq \alpha, \beta \leq 1$, equal to 1 near zero. For $x \neq 0$ define

$$\mathcal{U}F(x) = \frac{1}{2} \sum_{n \geq 1} \mu(n) F(n|x|).$$

If $P_j(D) = (-1)^j D(D+1) \cdots (D+j-1) = x^j \partial_x^j$, and $C_{N,j}(F)$ is the sum of $p_{N,k}(F)$ weighted by the absolute coefficients of P_j , then for $x > 0$, $N > 1$,

$$|\partial_x^j \mathcal{U}F(x)| \leq \frac{1}{2} \zeta(N) C_{N,j}(F) x^{-N-j}. \quad (29)$$

Each summand is bounded by $\frac{1}{2} C_{N,j}(F) n^{-N} x^{-N-j}$; summation proves the estimate and local uniform convergence of every derivative. Thus the cutoff exterior functions $(1-\alpha)\mathcal{U}F$ and $(1-\beta)\mathcal{U}JF$, extended by zero near zero, are continuous Schwartz-valued maps. The theta inversion in Section 1 and Poisson give $\mathcal{U}\Theta\phi = \phi$ and $\mathcal{U}J\Theta\phi = \widehat{\phi}$ away from zero.

On $L^2(\mathbb{R}, dx)$ define

$$T = M_\alpha \mathcal{F}^{-1} M_\beta \mathcal{F}, \quad YF = (1-\alpha)\mathcal{U}F + \alpha \mathcal{F}^{-1}((1-\beta)\mathcal{U}JF).$$

The kernel of T is $\alpha(x)\check{\beta}(x-y)$; its squared integral is $\|\alpha\|_2^2 \|\beta\|_2^2$, so T is compact. Its norm c is strictly less than 1. Otherwise compactness would give a unit vector attaining norm 1. Equality in the two contraction inequalities forces its Fourier transform to be supported where $\beta = 1$, and then the vector itself to be supported where $\alpha = 1$. A compactly supported Fourier transform in L^2 is in L^1 , and its inverse Fourier transform extends to an entire function by its finite-interval integral. That entire function would vanish on a real open interval because of the compact spatial support, hence vanish identically, a contradiction.

Thus $(1 - T)^{-1} = \sum_{n \geq 0} T^n$ in operator norm, with bound $(1 - c)^{-1}$. The operator $T : L^2 \rightarrow \mathcal{S}$ is continuous: Cauchy–Schwarz on the compact Fourier support bounds every derivative of $\mathcal{F}^{-1}(\beta \hat{f})$ by a constant times $\|f\|_2$, and multiplication by α gives compact spatial support and every Schwartz seminorm. It follows that $\Phi = (1 - T)^{-1}Y$ is Schwartz-valued and continuous; the explicit identity $\Phi = Y + T\Phi$ gives

$$q_{N,k}(\Phi F) \leq q_{N,k}(YF) + \frac{C_{N,k}}{1 - c} \|YF\|_2.$$

No uniform lower bound for $1 - c$ as cutoffs vary is used. On a theta input, $Y\Theta\phi = (1 - T)\phi$, so $\Phi\Theta\phi = \phi$. The range of Φ is the even Schwartz space $\mathcal{S}(\mathbb{R})^{\text{even}}$. To check this exact domain, $\mathcal{U}F$ is even because its defining series uses $|x|$, and the same is true of $\mathcal{U}JF$. The cutoffs α, β are even, and Fourier transformation and inverse Fourier transformation preserve evenness, so YF is even. Reflection $\phi(x) \mapsto \phi(-x)$ commutes with T ; hence it commutes with every partial sum of the norm-convergent inverse $(1 - T)^{-1}$ and with its limit. Thus ΦF is even as an L^2 vector, and its Schwartz representative is even pointwise by continuity. With the original Gaussian moment corrections, define the map on this precise domain by

$$P_V\phi = \phi - \phi(0)e^{-\pi x^2} - \left(\int_{\mathbb{R}} \phi - \phi(0) \right) 2\pi x^2 e^{-\pi x^2}, \quad \Lambda = P_V\Phi.$$

The moment columns of the two displayed Gaussians are $(1, 1)$ and $(0, 1)$ by direct Gaussian integration. The two subtractions therefore set the value at zero and the integral to zero, while preserving evenness. On V both correction coefficients vanish. Consequently $P_V : \mathcal{S}(\mathbb{R})^{\text{even}} \rightarrow V$ is a continuous projection and $P_V|_V = 1$. This is its stated domain; no projection of the entire Schwartz space onto the even subspace is asserted by that formula. It follows that

$$\Lambda\Theta = 1_V, \quad K = 1 - \Theta\Lambda, \quad s : Q \rightarrow \mathcal{B}, \quad s[F] = KF. \quad (30)$$

This proves all needed existence and continuity claims for the particular global construction used in A1694. Since $\ker K = \Theta V$, that image is closed. The expression for s is well-defined, $qs = 1$, $\Lambda s = 0$, and it is continuous by the definition of the quotient topology. In particular Θ^{-1} on its image is Λ , verifying the topological inverse asserted after (17).

The already existing global cocycle is $k_a = \Lambda U_a s$, with

$$U_a s - s\bar{U}_a = \Theta k_a.$$

This follows by applying $1 = K + \Theta\Lambda$ to $U_a s$. The finite/global comparison is now fully determined, with no new choice of arithmetic unit:

$$b_Z = \Lambda R : E \rightarrow V, \quad R^s = s\sigma = R - \Theta b_Z. \quad (31)$$

Define $c_a^s = k_a \sigma$. Substitution in (15) yields

$$\boxed{c_a^s = c_a - U_a b_Z + b_Z a^A.} \quad (32)$$

To verify the equality in V , use $\bar{U}_a \sigma = \sigma a^A$ from (18) to compute

$$\begin{aligned} U_a R^s - R^s a^A &= U_a R - R a^A - \Theta U_a b_Z + \Theta b_Z a^A \\ &= \Theta(c_a - U_a b_Z + b_Z a^A) = \Theta k_a \sigma. \end{aligned}$$

Theta injectivity proves (32). Its derivative is the exact infinitesimal bridge

$$\boxed{k_D \sigma = \phi_* \ell - D b_Z + b_Z A, \quad k_D = \Lambda D s.} \quad (33)$$

All terms are defined continuous maps; differentiation is allowed by Section 1. This proves that the explicitly evaluated Duhamel integral and the transcript's global continuous cocycle represent the same actual arithmetic extension, through the displayed coboundary.

The full extension class is unchanged. Applying $h(D)$ to (31) gives

$$h(D)R^s(u) = \Theta(R_Z(u)(D)\phi_* - h(D)b_Z u).$$

Modulo $h(D)V$, its source class equals that of $R_Z(u)(D)\phi_*$. Under $j_h H$ its value is exactly εu , by (7). There is no quotient operation that discards the nonconstant coefficients of ε .

At cochain level the two finite representative maps r^s and r have the difference

$$r^s - r = dL^\pm, \quad L^+(u) = (-b_Z u, 0), \quad L^-(u) = (0, \widehat{b_Z u}). \quad (34)$$

The minus-face sign follows by applying $d(0, \hat{b}) = -\Theta b$. Equations (25), (32), and (34) give the exact homotopy comparison

$$H_a^{s,\pm} = H_a^\pm + \mathcal{U}_a^0 L^\pm - L^\pm a^A.$$

These identities retain the joint degree-zero difference as in (28), rather than asserting strict agreement of the two face homotopies.

7. Tensor powers with every cohomological sign and jet coefficient

Let $k \geq 1$. On the original ordered tensor complex $C^{\otimes k}$ use

$$d(x_1 \otimes \cdots \otimes x_k) = \sum_{j=1}^k (-1)^{\sum_{i<j} |x_i|} x_1 \otimes \cdots \otimes dx_j \otimes \cdots \otimes x_k. \quad (35)$$

This is exactly the tau-base product resolution differential. It squares to zero because the two orders of applying differentials in different factors have opposite signs; a repeated differential in one factor is zero.

Write $f_a = U_a R : E \rightarrow C^1$, $g_a = R a^A : E \rightarrow C^1$, and $h_a = H_a^\pm : E \rightarrow C^0$ for any admitted choice of face in each factor. The explicit degree- -1 homotopy on the top packet is

$$\mathcal{H}_{a,k}(u_1 \otimes \cdots \otimes u_k) = \sum_{j=1}^k (-1)^{j-1} f_a u_1 \otimes \cdots \otimes f_a u_{j-1} \otimes h_a u_j \otimes g_a u_{j+1} \otimes \cdots \otimes g_a u_k. \quad (36)$$

In this evaluated formula the factor $(-1)^{j-1}$ is displayed exactly once. If the same formula is instead written as a tensor of graded linear maps, it is the Koszul evaluation sign of the degree- -1 map in slot j crossing the preceding $j-1$ inputs, each of degree 1; it must not be inserted a second time.

Only the slot containing $h_a u_j$ contributes to the differential, because the other slots are already in top degree. Its differential sign from (35) is $(-1)^{j-1}$, which cancels the prefactor. Since $dh_a = f_a - g_a$, the result is the telescoping identity

$$\boxed{d\mathcal{H}_{a,k} = (U_a R)^{\otimes k} - (R a^A)^{\otimes k}.} \quad (37)$$

For $k = 2$ this reads $\mathcal{H}_{a,2} = h_a \otimes g_a - f_a \otimes h_a$ on evaluated inputs, and its differential is $(f_a - g_a) \otimes g_a + f_a \otimes (f_a - g_a)$. This displays the original minus sign without relying on an implicit convention. Define

$$R_k = R^{\otimes k}, \quad A_k = \sum_{j=1}^k 1^{\otimes(j-1)} \otimes A \otimes 1^{\otimes(k-j)}, \quad T_{t,k} = T_t^{\otimes k} = e^{tA_k}.$$

For the ordered local block $(\rho_1, \dots, \rho_k)$ the action is exactly

$$T_{t,k} = e^{t \sum_i \rho_i} \sum_{0 \leq j_i < m_{\rho_i}} \frac{t^{j_1 + \dots + j_k}}{j_1! \dots j_k!} N_{\rho_1}^{j_1} \otimes \dots \otimes N_{\rho_k}^{j_k}. \quad (38)$$

The formula follows by multiplying (19); the nilpotents in different factors commute, and no term has been discarded. The representative is still the tensor of the maps using ε in (10), hence the coefficient unit is $\varepsilon^{\otimes k}$ in the ordered local algebra.

For the multivariable Mellin transform $\mathcal{M}_k$ with measure $\prod_i dx_i/x_i$, the top boundary $B_{a,k} = d\mathcal{H}_{a,k}$ has, on a pure coefficient tensor, the exact expression

$$\begin{aligned} \mathcal{M}_k B_{a,k}(\mathbf{s}) &= \prod_{i=1}^k [a^{s_i} v(s_i) R_Z(u_i)(s_i)] \\ &\quad - \prod_{i=1}^k [v(s_i) R_Z(a^A u_i)(s_i)]. \end{aligned} \quad (39)$$

All integrals factor by absolute convergence of the Schwartz Mellin integrals. Applying the full packet jet in each factor gives zero, as also follows from (37). The homotopy (36) exhibits the actual tensor relation giving that zero. It is not inferred from a dimension count or an assumed vanishing tensor kernel.

Every map in (36) is continuous. More explicitly, for seminorms $p_1, \dots, p_k$ on the appropriate target factors, the projective tensor seminorm of one summand on a pure input is bounded by the product of the corresponding operator bounds from (1), (10), and (14), times $\prod_i \|u_i\|$. Sum over the k terms and use the defining infimum for the projective seminorm on arbitrary finite sums. Thus all maps extend to completed projective tensor products, and (37) persists by density and continuity. The coefficient domain $E^{\otimes k}$ is finite dimensional, but this continuity statement identifies its targets in the same completed topology used by the original global construction.

The support rule is applied to each admitted factor before the tensor balancing quotient: an external absent factor is absorbing, while a zero amplitude in an active factor remains its supported tensor zero. The chosen face in that factor determines which of (25) occurs. Changing that face replaces its homotopy by (28); substitution into (36) gives an explicit tensor of the retained H^0 cycle with the other degree-one factors. No identification of external tensor support with a Cartesian product of labels is needed or asserted.

This tensor homotopy difference can be detected without discarding a possible lower-degree boundary. Define cochain maps $\pi_0 : C \rightarrow V[0]$ by $\pi_0^0(v, w) = w$, $\pi_0^1 = 0$, and $\pi_1 : C \rightarrow Q[-1]$ by $\pi_1^0 = 0$, $\pi_1^1 = q$. In the difference between the all-plus and all-minus versions of (36), apply π_0 in slot j and π_1 in the others. Every summand but the j -th vanishes. The result is

$$(-1)^{j-1} (\sigma a^A)^{\otimes(j-1)} \otimes \hat{c}_a \otimes (\sigma a^A)^{\otimes(k-j)}.$$

It is a nonzero map whenever $a \neq 1$: σa^A is injective by (11), $\hat{c}_a$ is nonzero by Section 5, and a tensor of nonzero vectors over $\mathbb{C}$ is nonzero, as witnessed by a tensor of linear functionals nonzero on

those vectors. The target complex has zero differential, so the detected class cannot be a boundary. This proves the exact survival of the homotopy difference in degree $k - 1$, where one factor carries the original $H^0(C)$ and the others carry Q .

8. The complete dx boundary pairing, its derivative, and tensor mass

The inclusion $\mathcal{B} \hookrightarrow L^2(\mathbb{R}_{>0}, dx)$ is continuous. For example on $(0, 1)$ use $|F(x)| \leq p_{-1,0}(F)x$, and on $(1, \infty)$ use $|F(x)| \leq p_{1,0}(F)x^{-1}$. Both squared majorants are integrable. Use the inner product conjugate-linear in its first argument. Direct substitution, retaining dx , gives

$$\langle U_a F, U_a G \rangle = a \langle F, G \rangle. \quad (40)$$

Let $G = R^* R$ in the retained coefficient basis and let $B_a = \Theta c_a : E \rightarrow L^2(dx)$. Since $J_Z R = 1$, G is positive definite. Equation (15), $U_a R = R a^A + B_a$, gives the exact finite boundary identity

$$\boxed{(a^A)^* G a^A - a G = -(U_a R)^* B_a - B_a^* (U_a R) + B_a^* B_a.} \quad (41)$$

On vectors u, w , its right side is

$$-\langle U_a R u, \Theta c_a w \rangle - \langle \Theta c_a u, U_a R w \rangle + \langle \Theta c_a u, \Theta c_a w \rangle.$$

To check the signs, solve $R a^A = U_a R - B_a$ and expand its squared pairing. Equation (40) supplies $a G$. Equivalently the right side is $-(R a^A)^* B_a - B_a^* R a^A - B_a^* B_a$; these two forms agree by (15). Thus the positive sign in the version using $U_a R$ is forced by the expansion, not a sign choice.

Every term is finite and depends smoothly on a , by the Frechet bounds already proved and the continuous Hilbert inclusion. For a full generalized packet vector u , the left side is evaluated using (19), retaining every power of $\log a$. For a true eigenvector $Au = \rho u$ it is $(a^{2\text{Re } \rho} - a) u^* G u$.

The infinitesimal identity obtained by differentiating (41) at $t = \log a = 0$ is

$$W = A^* G + G A - G = -R^* f_0 \ell - \ell^* f_0^* R. \quad (42)$$

This agrees exactly with (12) and the original integration by parts. Indeed for $F, H \in \mathcal{B}$,

$$\langle DF, H \rangle + \langle F, DH \rangle = - \int_0^\infty x (\overline{F} H)' dx = \langle F, H \rangle,$$

since the endpoint term $[x \overline{F}(x) H(x)]_0^\infty$ vanishes by the displayed weighted bounds. This independently verifies the mass term $-G$ and the two negative boundary cross terms in (42).

The finite defect is exactly the integrated infinitesimal one:

$$\boxed{T_t^* G T_t - e^t G = \int_0^t e^{t-r} T_r^* W T_r dr.} \quad (43)$$

Set $F(t) = T_t^* G T_t - e^t G$. Direct differentiation gives $F'(t) = F(t) + T_t^* W T_t$ and $F(0) = 0$. Multiplying by e^{-t} and integrating proves (43) with the original orientation for $t < 0$. This is a finite-dimensional exact matrix integration, valid with all the nilpotents in (19).

For the global representative R^s from (31), replace R by R^s and c_a by the fully computed c_a^s in (32). The finite expansion (41) then holds with

$$G^s = (R - \Theta b_Z)^* (R - \Theta b_Z), \quad \Theta c_a^s = \Theta(c_a - U_a b_Z + b_Z a^A).$$

Its infinitesimal boundary is the changed map

$$B_D^s = \Theta k_D \sigma = \Theta(\phi_* \ell - Db_Z + b_Z A). \quad (43a)$$

Indeed differentiating the explicit $c_{e^t}^s$ at $t = 0$ gives $k_D \sigma$ by (33), and direct differentiation of $R^s = R - \Theta b_Z$ gives $DR^s - R^s A = B_D^s$ by (12). Apply the same dx integration-by-parts identity to $R^s u, R^s w$ for arbitrary coefficient vectors. Substituting $DR^s = R^s A + B_D^s$ and moving the two boundary terms to the other side proves exactly

$$W^s = A^* G^s + G^s A - G^s = -(R^s)^* B_D^s - (B_D^s)^* R^s. \quad (43b)$$

Thus the factor $f_0 \ell$ in (42) is replaced by B_D^s , not retained as the old infinitesimal source boundary. Finally differentiating $T_t^* G^s T_t - e^t G^s$ as in the proof of (43) gives $F'_s(t) = F_s(t) + T_t^* W^s T_t$, $F_s(0) = 0$, and therefore

$$T_t^* G^s T_t - e^t G^s = \int_0^t e^{t-r} T_r^* W^s T_r dr. \quad (43c)$$

This is exactly the restriction of the A1694 global pairing identity to the actual packet σE . It proves the finite/global relationship between both the cocycle and its boundary pairings, without assuming an unproved vanishing of either cross term.

For k factors use the Hilbert pairing with measure $dx_1 \cdots dx_k$. Fubini gives its value on pure tensors as the product of the original pairings, hence

$$\langle U_a^{\otimes k} F, U_a^{\otimes k} H \rangle = a^k \langle F, H \rangle.$$

Let $G_k = G^{\otimes k}$, $B_{a,k}$ be the actual boundary (37), and $\mathcal{R}_a = (U_a R)^{\otimes k}$. Then the exact tensor version of (41) is

$$\boxed{(a^A)^{* \otimes k} G_k (a^A)^{\otimes k} - a^k G_k = -\mathcal{R}_a^* B_{a,k} - B_{a,k}^* \mathcal{R}_a + B_{a,k}^* B_{a,k}.} \quad (44)$$

Indeed $R_k(a^A)^{\otimes k} = \mathcal{R}_a - B_{a,k}$ by (37), and expansion proves the formula. Its derivative has the exact ordered sum

$$W_k = A_k^* G_k + G_k A_k - k G_k = \sum_{j=1}^k G^{\otimes(j-1)} \otimes W \otimes G^{\otimes(k-j)}, \quad (45)$$

because each of the k copies of the term $-G$ in (42) contributes one copy of $-G_k$. The finite integral is

$$T_{t,k}^* G_k T_{t,k} - e^{kt} G_k = \int_0^t e^{k(t-r)} T_{r,k}^* W_k T_{r,k} dr. \quad (46)$$

The proof differentiates the left side exactly as in (43), with mass exponent k . Equations (38) and (46) retain all tensor nilpotents. They are actual formulas for the original representative boundary contribution on the original tau-base tensor complex.

9. Source pins and status of the calculation

The complete input files read for this calculation were:

- `arithmetic_input.tex` (original September 12 cumulative source), SHA256 `a50b0fb587b644c4ea94dba45b2b423d038f2bfbe7fef188543939ba850e58d2`.
- *`Tau_Base_Cohomology`*, NOTE.md, SHA256 `d03e71ad18f187bd89880cb8ca6fc9c5d6f260b3de8e8e5702c27db97e7b63c3`.
- Original transcript turn A1694.md, SHA256 `a87bff4960c0a1742734e9e711780e0533ea76ba6f46ceab1fea115a2fd66f18`; its original locator is recorded in Section 0.
- *`Tau_Global_Comparison`*, NOTE.tex, SHA256 `50284fcebade0dc859ac398a00839a29e15be60c008e02314ac1f35c31950ed01`; its equations are cited at the beginning.

The completed new formulas are (13)–(24), their exact finite/global comparison (31)–(34), the evaluated tensor homotopy and jets (36)–(39), and the integrated finite and tensor boundary expressions (43)–(46). Sections 1, 2, 5, 6, and the unintegrated boundary pairing explicitly reprove the source interfaces needed to type and verify those formulas. The global retraction, global cocycle, and support comparison in A1694 remain credited to that earlier calculation.

The resulting boundary terms have been computed, including their orientations, topologies, full arithmetic coefficients, and product masses. None has been set to zero by calling it a theta boundary. No estimate forcing the sign or vanishing of the original global weight defect is asserted by these identities. That precise limit on the conclusion does not delete the displayed maps, the full quotient Q , or the original tau-base construction.

Chapter 4

The full-source balanced Mellin kernel

The original balanced Mellin kernel: full polynomial comparison and meromorphic remainder

The authoritative shared transcript constructs the raw Mellin realization in A1167, retains its kernel complex in A1361, and replaces it by the $\mathbb{C}[t]$ -balanced realization in A1427. This section computes the polynomial action on the remaining kernel on those original objects. It uses no replacement arithmetic function and no auxiliary zero packet.

Original objects and full-plane Euler inverses

Retain, exactly as in A1167 and A1427,

$$\begin{aligned} V &= \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \int_{\mathbb{R}} \phi(x) dx = 0\}, \\ \mathcal{B} &= \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^k F(x)| < \infty \quad (b \in \mathbb{Z}, k \geq 0)\}, \quad D = -x\partial_x, \\ \Theta\phi(x) &= 2 \sum_{n \geq 1} \phi(nx), \quad \mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}, \quad M_+\phi(s) = \int_0^\infty \phi(x) x^s \frac{dx}{x}, \quad (\text{BK.1}) \\ g(s) &= 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \\ H_\phi(s) &= \frac{2\pi^{s/2}M_+\phi(s)}{s(s-1)\Gamma(s/2)}. \end{aligned}$$

The original Fourier convention is $\hat{\phi}(\nu) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \nu} dx$. The original theta construction proves

$$\mathcal{M}\Theta\phi = gH_\phi, \quad \Theta D = D\Theta, \quad \widehat{D\phi} = (1-D)\hat{\phi}, \quad H_{D\phi}(s) = sH_\phi(s), \quad H_{\hat{\phi}}(s) = H_\phi(1-s). \quad (\text{BK.2})$$

The last equality follows directly from Poisson's identity $\Theta\hat{\phi}(x) = x^{-1}\Theta\phi(1/x)$, Mellin substitution, and $g(1-s) = g(s)$; cancellation holds on the nonempty open set $g \neq 0$ and then throughout the plane by holomorphy. Here H_ϕ is entire: Taylor subtraction of the even smooth function at zero gives only simple possible poles of $M_+\phi$ at the negative even integers; the corresponding zeros of $1/\Gamma(s/2)$ cancel them. At $s = 0$, $M_+\phi$ is holomorphic because $\phi(0) = 0$, and $s\Gamma(s/2)$ tends to 2. At $s = 1$, $M_+\phi(1) = 0$ is exactly the retained integral condition. In particular

$$H_\phi(1) = 2(M_+\phi)'(1). \quad (\text{BK.3})$$

The topology and all these maps are the original Schwartz ones.

For every $\rho \in \mathbb{C}$ one has the exact range formulas

$$(D - \rho)\mathcal{B} = \ker(\mathcal{M}(\cdot)(\rho)), \quad (D - \rho)V = \ker(H_{(\cdot)}(\rho)). \quad (\text{BK.4})$$

Both operators are injective. We now prove the full-plane source formula, including the endpoint $\rho = 1$ that cannot be handled by dividing by $1 - \rho$.

On the target zero-Mellin domain the inverse is the original integral

$$S_\rho F(x) = x^{-\rho} \int_x^\infty F(y) y^{\rho-1} dy. \quad (\text{BK.5})$$

Differentiating gives $(D - \rho)S_\rho F = F$. Since the total Mellin moment is zero, the same expression is $-x^{-\rho} \int_0^x F(y) y^{\rho-1} dy$ near zero. For $\sigma = \Re \rho$, an upper bound $|F(y)| \leq C_N y^{-N}$ yields $|S_\rho F(x)| \leq C_N x^{-N}/(N - \sigma)$ when $N > \sigma$. A lower bound $|F(y)| \leq C'_N y^N$ yields $|S_\rho F(x)| \leq C'_N x^N/(N + \sigma)$ when $N + \sigma > 0$. The differential identity controls each further Euler derivative. Thus every original seminorm is finite and the inverse is continuous on the indicated kernel. The only homogeneous solution is $cx^{-\rho}$, which belongs to $\mathcal{B}$ only when $c = 0$. Integration by parts proves $\mathcal{M}((D - \rho)F) = (s - \rho)\mathcal{M}F$, so the range is exactly the first kernel in (BK.4).

For the source, first let $\Re \rho > 0$ and $H_\phi(\rho) = 0$. For $\rho \neq 1$, the multiplier of $M_+ \phi(\rho)$ in (BK.1) is finite and nonzero, so this Mellin moment vanishes. For $\rho = 1$ the same moment already vanishes by the definition of V . The integral (BK.5) is consequently also

$$S_\rho \phi(x) = - \int_0^1 u^{\rho-1} \phi(xu) du. \quad (\text{BK.6})$$

For every ordinary derivative of order k , differentiating this integral inserts $u^k \phi^{(k)}(xu)$. Uniform domination on bounded x intervals follows from $\Re \rho + k > 0$. Therefore the result extends smoothly and evenly across zero with value zero there. The upper-tail expression gives Schwartz decay, with every derivative, at infinity. If $\rho \neq 1$, integration of $(D - \rho)S_\rho \phi = \phi$ over $\mathbb{R}$ gives

$$(1 - \rho) \int_{\mathbb{R}} S_\rho \phi = \int_{\mathbb{R}} \phi = 0.$$

If $\rho = 1$, integration by parts in a right half-plane gives

$$M_+(S_1 \phi)(s) = \frac{M_+ \phi(s)}{s - 1}.$$

Taking its removable value at 1 and using (BK.3) gives $\int_{\mathbb{R}} S_1 \phi = 2(M_+ \phi)'(1) = H_\phi(1) = 0$. Thus the original source integral condition is retained at every ρ with positive real part.

If $\Re \rho \leq 0$, put $\lambda = 1 - \rho$, so $\Re \lambda \geq 1$. Fourier preserves V , and (BK.2) gives $H_{\widehat{\phi}}(\lambda) = H_\phi(\rho) = 0$. The exact inverse is

$$S_\rho^V \phi = -\widehat{S_\lambda \phi}. \quad (\text{BK.7})$$

On even functions Fourier squares to the identity. The intertwiner in (BK.2) therefore gives $(D - \rho)(-\widehat{S_\lambda \phi}) = \phi$, with every source condition preserved. For positive real part the inverse is (BK.6). The homogeneous solution cannot be Schwartz at infinity unless zero, so the inverse is unique. Identity (BK.2) proves the reverse containment in the second formula of (BK.4). This completes the full-plane calculation without excluding either endpoint or negative even Taylor coordinate.

Every polynomial kernel and cokernel is computed on the actual theta sequence

Set $A = \mathbb{C}[t]$, with t acting as the original D , and $Q = \mathcal{B}/\Theta V$. For a nonconstant monic polynomial

$$h(t) = \prod_\rho (t - \rho)^{n_\rho}, \quad E_h = A/(h),$$

let j_h retain all Taylor jets of orders less than n_ρ . Iterating (BK.4) gives

$$h(D)V = \ker(j_h H), \quad h(D)\mathcal{B} = \ker(j_h \mathcal{M}). \quad (\text{BK.8})$$

Indeed after dividing by $D - \rho$, the Mellin multiplier or H is divided by $s - \rho$ by (BK.2); the remaining zero orders are exactly the orders needed at the next division. Injectivity proves that the iterated inverse is independent of the chosen ordering.

The source jet map is onto because the original vector

$$\phi_* = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2} \quad \text{satisfies} \quad H_{P(D)\phi_*} = P. \quad (\text{BK.9})$$

The target jet map is onto because its functionals on compactly supported smooth logarithmic functions are the independent functions $y^j e^{\rho y} / j!$. For precision, a dependence vanishing on an interval extends to an entire exponential polynomial. Applying $\prod_{\sigma \neq \rho} (\partial_y - \sigma)^{n_\sigma}$ removes the other summands; its restriction to $e^{\rho y}$ times polynomials of degree less than n_ρ is invertible, since it is a product of nonzero constants plus nilpotent derivatives. Then $(\partial_y - \rho)$ repeatedly separates the polynomial coefficients. All coefficients vanish. If the finite vector-valued jet map were not onto, a nonzero linear functional on its missing quotient would give such a dependence, a contradiction.

Consequently the quotient square and exact theta sequence are

$$\begin{array}{ccc} V/h(D)V & \xrightarrow{\Theta} & \mathcal{B}/h(D)\mathcal{B} \\ j_h H \downarrow \wr & & \wr \downarrow j_h \mathcal{M} \\ E_h & \xrightarrow{M_{j_h g}} & E_h, \end{array} \quad (\text{BK.10})$$

$$0 \longrightarrow Q[h(D)] \xrightarrow{\delta_h} E_h \xrightarrow{M_{j_h g}} E_h \longrightarrow Q/h(D)Q \longrightarrow 0. \quad (\text{BK.11})$$

These identities now hold for every polynomial h , without a strip restriction. The map δ_h sends $[F]$ to $j_h H_\phi$ where $h(D)F = \Theta\phi$. The source ϕ is unique because Θ is injective. Changing F by a theta relation changes ϕ by an $h(D)$ multiple, so the map is well-defined. If its value is zero, (BK.8) writes $\phi = h(D)\psi$ and injectivity of $h(D)$ on $\mathcal{B}$ gives $F = \Theta\psi$. A vector in the kernel of $M_{j_h g}$ is represented by a polynomial P ; the zero-jet condition permits applying the actual inverse of $h(D)$ to $\Theta(P(D)\phi_*)$, producing its preimage. Quotienting the right target in (BK.10) proves the last cokernel. This proves the entire sequence with its original maps.

The balanced kernel has invertible polynomial action

Fix any analytic stalk $O = \mathcal{O}_{\mathbb{C}, s_0}$, with t acting on O by multiplication by s . Write

$$M = O \otimes_A Q, \quad N = O/(g), \quad b : M \rightarrow N, \quad b(a \otimes [F]) = [aMF], \quad K = \ker b. \quad (\text{BK.12})$$

The map is balanced by (BK.2). The ring O is torsion-free over the PID A : a nonzero polynomial times a holomorphic germ is zero only when that germ is zero. It is therefore flat. One direct justification of this familiar PID fact is to write a torsion-free module as the directed union of its finitely generated submodules; each such submodule is free by the PID structure theorem. Tensoring with a free module preserves injections, and filtered colimits preserve exactness, proving flatness.

The map b is onto. In fact the original test function

$$F_0(x) = \exp[-(\log x)^2], \quad H_0(s) = \mathcal{M}F_0(s) = \sqrt{\pi} \exp(s^2/4) \quad (\text{BK.13})$$

belongs to $\mathcal{B}$ and H_0 is an invertible germ at every point. Euler derivatives of F_0 are polynomials in $\log x$ times F_0 , and $x^b F_0$ is bounded for every integer b . Its Mellin integral becomes $\int_{\mathbb{R}} e^{-y^2 + sy} dy$; real completion of the square gives the formula for real s , and entire continuation gives it for every complex s . Thus $b(aH_0^{-1} \otimes [F_0]) = [a]$, proving surjectivity.

For every nonzero polynomial h , the two induced maps are isomorphisms:

$$b : M[h] \xrightarrow{\sim} N[h], \quad \bar{b} : M/hM \xrightarrow{\sim} N/hN. \quad (\text{BK.14})$$

For a nonzero constant both statements are immediate. For a nonconstant polynomial flatness gives $M[h] = O \otimes_A Q[h(D)]$ and $M/hM = O \otimes_A (Q/h(D)Q)$. Also $O \otimes_A E_h = O/(h)$ by the explicit map $a \otimes [P(t)] \mapsto [aP(s)]$, with inverse $[a] \mapsto a \otimes 1$. The cokernel assertion in (BK.11) consequently gives

$$M/hM \cong O/(h, g) = N/hN,$$

and the isomorphism is precisely the Mellin map $\bar{b}$.

For the kernel map retain the actual local units. Put $z = s - s_0$ and write

$$h(s) = a(z)z^n, \quad g(s) = u(z)z^m, \quad a(0)u(0) \neq 0, \quad n, m \geq 0. \quad (\text{BK.15})$$

Under (BK.11), a local vector v in $\ker(g : O/(h) \rightarrow O/(h))$ corresponds to a torsion class with $h\mathcal{M}F = gH_\phi$, and its image under b is exactly

$$v \mapsto \left[\frac{g}{h} v \right] \in N[h]. \quad (\text{BK.16})$$

This is well-defined: changing v by h times a germ changes the displayed product by g times a germ. The kernel condition makes the product holomorphic. Explicitly its domain is $z^{\max(n-m, 0)}O/z^nO$ and its target is $z^{\max(m-n, 0)}O/z^mO$. Formula (BK.16) is multiplication by the actual unit u/a , together with the power z^{m-n} . It sends precisely the indicated source powers to the indicated target powers and has inverse multiplication by a/u and the power z^{n-m} . It is an isomorphism, including $m = 0$ or $n = 0$ when both spaces are zero. This proves the first map in (BK.14) without dropping a local factor of g or a repeated-zero order.

Apply the kernel-cokernel exact sequence for multiplication by h to $0 \rightarrow K \rightarrow M \xrightarrow{b} N \rightarrow 0$. It is

$$0 \rightarrow K[h] \rightarrow M[h] \rightarrow N[h] \rightarrow K/hK \rightarrow M/hM \rightarrow N/hN \rightarrow 0. \quad (\text{BK.17})$$

Both maps in (BK.14) are isomorphisms, so $K[h] = 0$ and $K/hK = 0$. We have proved

$$h : K \xrightarrow{\sim} K \quad \text{for every nonzero } h \in \mathbb{C}[t]. \quad (\text{BK.18})$$

This is a computed property of the original balanced kernel.

It has a stronger local form. Every nonzero holomorphic germ is a unit times z^r . Units already act bijectively, and (BK.18) applied to $(t - s_0)^r$ makes z^r bijective. Hence K has a unique module structure over the meromorphic germ field $\text{Frac } O$, obtained by taking these actual inverses:

$$\frac{a}{b} \cdot k = a(b|_K)^{-1}k. \quad (\text{BK.19})$$

If $a/b = c/d$, the equality $ad = bc$ and invertibility of bd show that the two prescribed actions agree. The field laws follow by multiplying through their nonzero denominators.

Canonical finite spectral part and an explicit nonzero kernel class

At a point where $m = \text{ord}_{s_0} g > 0$, (BK.14) for $h = (t - s_0)^m$ gives a unique inverse $j : N \xrightarrow{\sim} M[z^m]$ of b on that kernel. Therefore

$$M = M[z^m] \oplus K, \quad x = j(bx) + (x - j(bx)), \quad M[z^m] \cong O/(g). \quad (\text{BK.20})$$

The first summand is all the O -torsion of M . Indeed every torsion element is killed by a power of z . For larger powers than m , (BK.14) still gives the same unique preimage of each element of N , because K has no z -torsion. When $m = 0$, $N = 0$ and the formula is $M = K$. All the full nilpotent jets of an actual zero occur in the first summand of (BK.20).

The kernel K is nonzero. The only classical zero-set input needed for the following assertion is the already-retained Hardy theorem: there are distinct $T_j \rightarrow +\infty$ with $g(1/2 + iT_j) = 0$. The cumulative reader uses this exact input in `theta_gamma_reference.tex`, (TG.26), and `sum_connection_stieltjes.tex`, (FC.21), with Hardy's original 1914 source. No assumption of RH is used.

Let $q_0 = [F_0] \in Q$. It is A -nontorsion. If $P(D)F_0 = \Theta\phi$ with a polynomial P , then (BK.2) and (BK.13) give

$$P(s)\sqrt{\pi}e^{s^2/4} = g(s)H_\phi(s).$$

At every $1/2 + iT_j$, the nonzero exponential forces $P(1/2 + iT_j) = 0$. A nonzero polynomial cannot have infinitely many distinct roots, so $P = 0$. Thus $A \rightarrow Q$, $P \mapsto P(D)q_0$, is injective. Flatness now proves that $O \rightarrow M$, $a \mapsto a \otimes q_0$, is injective. In particular

$$0 \neq g \otimes q_0 \in K. \quad (\text{BK.21})$$

The corresponding cycle is explicit on the original balanced complex. Put $b_* = \Theta\phi_*$, so $\mathcal{M}b_* = g$. In $O \otimes_A \mathcal{B}$, set

$$Z_0 = g \otimes F_0 - H_0 \otimes b_*. \quad \beta Z_0 = gH_0 - H_0g = 0. \quad (\text{BK.22})$$

Its degree-one class is exactly the nonzero class in (BK.21), since its second term is the original theta boundary. For the original invariant source label $W_* = \mathbb{C}[D]\phi_*$, the map $O \otimes_A W_* \rightarrow O$ induced by H is an isomorphism by (BK.9). Thus the kernel complex at that label has degree-zero term zero, and (BK.22) is an explicitly nonzero degree-one kernel cycle. Passing to the full label V preserves its nonzero class, as proved by the A -nontorsion argument. This resolves the status left open in A1427: balancing removes the earlier displayed spurious relation, but its full degree-one kernel is not zero.

This statement does not create a spectral obstruction to RH. For every finite-dimensional A -module E , let p_E be the characteristic polynomial of its t action. Cayley–Hamilton gives $p_E E = 0$. If $u : E \rightarrow K$ is A -linear, then $p_E u(E) = 0$ and injectivity in (BK.18) gives $u = 0$. If $v : K \rightarrow E$ is A -linear, every $k \in K$ is $p_E k'$ for some k' , and $v(k) = p_E v(k') = 0$. Consequently

$$\text{Hom}_A(E, K) = 0 = \text{Hom}_A(K, E) \quad (\dim_{\mathbb{C}} E < \infty). \quad (\text{BK.23})$$

The entire kernel has no finite-dimensional invariant subspace and no nonzero finite-dimensional equivariant quotient. This does not assert the same for arbitrary subquotients of its proper submodules. Its calculated contribution to every full finite spectral jet observation is zero, whereas (BK.20) proves that the actual finite spectral torsion survives bijectively.

Finally every construction occurs in the original supported quotient. The structure assertions (BK.18)–(BK.23) concern the full source label V used in (BK.12). They are not asserted for an arbitrary proper D -invariant source label. For such a label $W \subset V$, the exact comparison that must be retained is

$$0 \rightarrow V/W \xrightarrow{[\phi] \mapsto [\Theta\phi]} Q_W \rightarrow Q \rightarrow 0, \quad Q_W = \mathcal{B}/\Theta W. \quad (\text{BK.24})$$

Injectivity of Θ proves the first map injective, and the quotient definitions prove its image is the last map's kernel. All maps commute with D when W is invariant. Tensoring this sequence with the flat O retains the entire kernel $O \otimes_A (V/W)$, which can have additional finite spectral torsion before transport to the full label. No such proper-label torsion is discarded by the scope of the structure theorem. The inclusion of labels $W_* \hookrightarrow V$ carries $\mathcal{B}/\Theta W_* \rightarrow \mathcal{B}/\Theta V$ with the original kernel $\Theta V/\Theta W_*$. Balancing imposes the explicit linear relations $sa \otimes v - a \otimes Dv$ within each such label. A linear map L at a fixed label ℓ lifts as $(\ell, x) \mapsto (\ell, Lx)$ and sends external τ to external τ . Addition joins labels and applies their given linear transports; the displayed formulas commute with those transports. Therefore (BK.22) is sent to the supported zero $(\ell, 0)$ by Mellin evaluation, while its nonzero kernel class remains at that label. Equations (BK.18)–(BK.24), including every actual nilpotent jet and the proper-label transition kernel, describe precisely which full-source kernel assertions survive the complete original quotient diagram.

Chapter 5

Finite derived observations and source-label transitions

The exact derived comparison at every finite spectral thickening

This calculation uses the balanced comparison of the original theta complex, as constructed in the shared conversation at A1427, and the polynomial-invertibility theorem proved in the accompanying balanced-kernel chapter. It uses no replacement polynomial in place of an actual zeta packet. Write $\mathcal{O} = \mathcal{O}_{\mathbb{C}, s_0}$ for the local ring of holomorphic germs at the specified spectral coordinate s_0 , $A = \mathbb{C}[t]$, $M = \mathcal{O} \otimes_A Q$, and

$$Q = \mathcal{B}/\Theta V, \quad g = 2\xi, \quad \beta : M \longrightarrow \mathcal{O}/(g), \quad \beta(a \otimes [F]) = [aMF].$$

The A -action is exactly $tF = DF = -x\partial_x F$ on the original function space and $ta = sa$ on $\mathcal{O}$. Let $K = \ker \beta$. The preceding calculation on these original spaces proves that

$$0 \longrightarrow K \longrightarrow M \xrightarrow{\beta} \mathcal{O}/(g) \longrightarrow 0 \quad \text{is exact,} \quad (s - s_0) : K \xrightarrow{\sim} K. \quad (\text{DP.1})$$

In particular this is not an assumption of injectivity of β ; the preceding chapter exhibits an explicit nonzero element of K .

Proposition 5.1 (Derived finite spectral observations). *For every nonzero holomorphic germ $h \in \mathcal{O}$, the original comparison induces a quasi-isomorphism*

$$M \otimes_{\mathcal{O}}^{\mathbf{L}} \mathcal{O}/(h) \xrightarrow{\beta \otimes 1} (\mathcal{O}/(g)) \otimes_{\mathcal{O}}^{\mathbf{L}} \mathcal{O}/(h). \quad (\text{DP.2})$$

Both the degree-minus-one kernel and the degree-zero quotient are retained. The map commutes with multiplication by every holomorphic germ, in particular the original action a^s for every real $a > 0$.

Proof. Write the specified germ as $h = (s - s_0)^n u$, where $n \geq 0$ is its actual vanishing order and $u(s_0) \neq 0$. This identity retains the entire unit u , not merely its value at s_0 . Multiplication by u is invertible on every $\mathcal{O}$ -module. By (DP.1), multiplication by $(s - s_0)^n$ is invertible on K . Thus the literal multiplication map $h : K \rightarrow K$ is invertible.

The integral domain $\mathcal{O}$ has the free resolution

$$0 \longrightarrow \mathcal{O} \xrightarrow{\times h} \mathcal{O} \longrightarrow \mathcal{O}/(h) \longrightarrow 0.$$

Place its two free terms in degrees $-1, 0$. Tensoring it with the three modules in (DP.1) gives a short exact sequence of complexes

$$0 \longrightarrow [K \xrightarrow{h} K] \longrightarrow [M \xrightarrow{h} M] \longrightarrow [\mathcal{O}/(g) \xrightarrow{h} \mathcal{O}/(g)] \longrightarrow 0. \quad (\text{DP.3})$$

The first complex has zero cohomology in both degrees because its differential is invertible. The cohomology exact sequence of (DP.3) therefore proves (DP.2), including the degree -1 terms $\ker(h : M \rightarrow M)$ and $\ker(h : \mathcal{O}/(g) \rightarrow \mathcal{O}/(g))$. Its degree-zero terms are the actual quotients by h . All maps in (DP.3) are multiplication or $\mathcal{O}$ -linear maps; they commute with multiplication by every germ. This proves the claimed action compatibility without removing a nilpotent term. $\square$

At an actual zero ρ of complete order m_ρ , retain $g(s) = (s - \rho)^{m_\rho} v_\rho(s)$ with its nonvanishing full unit. For the exact thickening $h = (s - \rho)^n$, the right complex in (DP.3) has

$$H^{-1} = \{z \in \mathcal{O}_\rho/(g) : (s - \rho)^n z = 0\}, \quad H^0 = \mathcal{O}_\rho/(g, (s - \rho)^n). \quad (\text{DP.4})$$

When $n \geq m_\rho$, its differential is zero, so both terms are the entire length- m_ρ arithmetic local algebra. When $0 < n < m_\rho$, its kernel is exactly $(s - \rho)^{m_\rho - n} \mathcal{O}_\rho/(g)$ and its quotient is $\mathcal{O}_\rho/((s - \rho)^n)$, each of length n . These statements follow by division by the retained unit v_ρ , which is an isomorphism of the specified ideals and does not change the action. On every retained full block the action is still

$$a^\rho \sum_{j=0}^{m_\rho-1} \frac{(\log a)^j}{j!} N_\rho^j, \quad N_\rho[z^j] = [z^{j+1}], \quad z = s - \rho. \quad (\text{DP.5})$$

Here is the exact bridge to the original dilation, rather than only to scalar multiplication on a balanced module. Define $R_a F(x) = F(x/a)$ and $R_a \phi(x) = \phi(x/a)$ for $a > 0$. These maps preserve the original spaces, commute with D , and $\Theta R_a = R_a \Theta$ term by term. Substitution $x = ay$ gives $\mathcal{M}(R_a F)(s) = a^s \mathcal{M}F(s)$; hence the induced $\mathcal{O}$ -linear map $1 \otimes R_a$ on M satisfies $\beta(1 \otimes R_a) = a^s \beta$. It preserves $M[h]$ and hM , since it commutes with h . The proved comparison on kernels and quotients therefore transports its action to literal multiplication by a^s , giving (DP.5). This does not assert equality of $1 \otimes R_a$ and scalar a^s on the entire balanced module: their difference has image in K , which is retained.

For the counterfactual under examination, ρ is an actual off-critical nontrivial zero of g ; its existence is the temporary premise, not a conclusion. Equations (DP.2)–(DP.5) prove that its complete finite block persists through this balanced comparison, even after taking every finite derived spectral thickening. It cannot be an element of the nonzero kernel K , since multiplication by its annihilating polynomial is invertible on K .

On each of the three original nonempty two-leg masks whose admitted theta image is the full ΘV , these complexes have the split lifts $(\lambda, x) \mapsto (\lambda, f(x))$, and external τ maps to external τ . A zero in (DP.3) is the supported zero at that same mask. This assertion uses the full admitted source V . The arithmetic spectral coordinate s_0 and the absolute base point τ are connected by this labelled comparison; they are not identified by it. Smaller admitted sources have the following additional, explicitly computed finite spectral data.

Proposition 5.2 (All finite defects of a smaller source label). *Let $W \subseteq V$ be an original D -invariant admitted source, and retain $Q_W = \mathcal{B}/\Theta W$. Put*

$$M_W = \mathcal{O} \otimes_A Q_W, \quad L_W = \mathcal{O} \otimes_A (V/W), \quad \beta_W(a \otimes [F]_W) = [a \mathcal{M}F], \quad K_W = \ker \beta_W.$$

Then the source inclusion induces the exact sequences

$$0 \longrightarrow L_W \xrightarrow{[\phi] \mapsto [\Theta \phi]} M_W \xrightarrow{\pi_W} M \longrightarrow 0, \quad 0 \longrightarrow L_W \longrightarrow K_W \longrightarrow K \longrightarrow 0. \quad (\text{DP.6})$$

For every nonzero germ h , their induced maps give

$$L_W[h] \xrightarrow{\sim} K_W[h], \quad L_W/hL_W \xrightarrow{\sim} K_W/hK_W. \quad (\text{DP.7})$$

More fully, inclusion is a quasi-isomorphism

$$[L_W \xrightarrow{h} L_W] \xrightarrow{\sim} [K_W \xrightarrow{h} K_W]. \quad (\text{DP.8})$$

Proof. Injectivity of the original $\Theta : V \rightarrow \mathcal{B}$ gives the exact A -module sequence $0 \rightarrow V/W \xrightarrow{\Theta} Q_W \rightarrow Q \rightarrow 0$. The flatness proved in the balanced-kernel chapter yields the first sequence in (DP.6). The equality $\beta_W = \beta\pi_W$ follows directly from the retained Mellin formula. Every element of L_W maps to zero under β_W , since $\mathcal{M}\Theta\phi = gH_\phi$. If $k \in K$, lift it to $x \in M_W$ using the surjection π_W ; then $\beta_W x = \beta k = 0$. This proves the second sequence in (DP.6), including surjectivity of its last map. Tensor its terms with the same two-term free resolution of $\mathcal{O}/(h)$ used in (DP.3). The quotient complex is $[K \xrightarrow{h} K]$, whose differential is invertible by (DP.1) and its unit-factor argument. Its cohomology vanishes. The resulting exact cohomology sequence proves both (DP.7) and (DP.8). Every arrow comes from the actual source inclusion, and commutes with the given actions. $\square$

Thus a finite spectral class in a smaller label's Mellin kernel is precisely a finite spectral class in its retained relation quotient V/W . It vanishes upon passing to the full arithmetic source V , by (DP.6). This is an actual transition map, not a deletion of the smaller label or its extension. The next chapter constructs such a nonzero class explicitly from the original dilates of ϕ_* , so that the restriction to the full source in (DP.1)–(DP.5) is also tested by a concrete example. At a smaller label only D -invariance has been imposed. Accordingly (DP.6)–(DP.8) commute with the D and holomorphic-scalar actions; an original dilation R_a acts between the labels W and $R_a W$, and is an endomorphism at W only when $R_a W = W$. This follows from the exact map $[F]_W \mapsto [R_a F]_{R_a W}$ and $\Theta R_a = R_a \Theta$.

Chapter 6

An explicit proper-label spectral kernel

An actual proper source label with a finite spectral kernel

The full-source calculation (BK.18) applies to $Q = \mathcal{B}/\Theta V$. It does not assert the same result for $Q_W = \mathcal{B}/\Theta W$ at an arbitrary proper, D -invariant source label $W \subset V$. The following construction proves the distinction and its exact transition map on the original source functions. It does not replace $g = 2\xi$ by a model function and does not assume an off-critical zero.

Retain $A = \mathbb{C}[t]$, $t = D = -x\partial_x$, and the original source vector and its literal dilation

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, \quad \psi(x) = \phi_*(2x). \quad (\text{PL.1})$$

Both functions belong to V : dilation preserves evenness, Schwartz decay, and value zero at the origin, while $\int_{\mathbb{R}} \psi(x) dx = \frac{1}{2} \int_{\mathbb{R}} \phi_*(y) dy = 0$. The change of variable $y = 2x$ in the original positive Mellin integral gives

$$M_+ \psi(s) = 2^{-s} M_+ \phi_*(s), \quad H_\psi(s) = 2^{-s}, \quad \mathcal{M}\Theta\psi(s) = g(s)2^{-s}. \quad (\text{PL.2})$$

The first equality holds in its initial half-plane of absolute convergence; the entire continuation of H proved in (BK.1)–(BK.3) extends the second equality to the full plane. Here $2^{-s} = \exp(-s \log 2)$ uses the positive real logarithm of the actual dilation constant 2.

Define the original invariant source label

$$W = A\phi_* + A\psi \subset V. \quad (\text{PL.3})$$

This is a free rank-two A -module on the two displayed vectors. Indeed an equality $P(D)\phi_* + Q(D)\psi = 0$, for polynomials $P, Q \in \mathbb{C}[t]$, implies by (BK.2) and (PL.2)

$$P(s) + Q(s)2^{-s} = 0 \quad (s \in \mathbb{C}).$$

As the real variable s tends to positive infinity, $Q(s)2^{-s}$ tends to zero. Thus $P(s)$ tends to zero. A nonzero complex polynomial cannot tend to zero along that ray: division by its highest real power gives its nonzero leading coefficient. Consequently $P = 0$, and the nonvanishing exponential forces $Q = 0$. This also proves the exponential-polynomial independence used below.

Fix any $\rho \in \mathbb{C}$ and retain its exact nonzero scalar $c_\rho = 2^{-\rho}$. Set

$$\phi_\rho = \psi - c_\rho \phi_* \in W, \quad H_{\phi_\rho}(s) = 2^{-s} - c_\rho. \quad (\text{PL.4})$$

The last entire function vanishes at $s = \rho$. The already-proved full-plane inverse (BK.4), including its endpoint and Fourier-reflected cases, therefore gives the unique actual vector

$$\chi_\rho = S_\rho^V \phi_\rho \in V, \quad (D - \rho)\chi_\rho = \phi_\rho, \quad H_{\chi_\rho}(s) = \frac{2^{-s} - 2^{-\rho}}{s - \rho}. \quad (\text{PL.5})$$

Its displayed quotient is entire with its retained value $H_{\chi_\rho}(\rho) = -(\log 2)2^{-\rho} \neq 0$. The vector is specified without an implicit choice: for $\operatorname{Re} \rho > 0$ its positive-half-line formula is

$$\chi_\rho(x) = x^{-\rho} \int_x^\infty \phi_\rho(y) y^{\rho-1} dy = - \int_0^1 u^{\rho-1} \phi_\rho(xu) du, \quad (\text{PL.6})$$

extended evenly; for $\operatorname{Re} \rho \leq 0$ it is $-\widehat{S_{1-\rho}\phi_\rho}$, with the same positive-half-line integral for $S_{1-\rho}$. The proof in (BK.4)–(BK.7) establishes all original source conditions for these formulas.

The vector χ_ρ is not in W . Suppose otherwise and write $\chi_\rho = P(D)\phi_* + Q(D)\psi$. Multiplying the equality of their H transforms by $s - \rho$ gives the entire identity

$$((s - \rho)P(s) + 2^{-\rho}) + ((s - \rho)Q(s) - 1)2^{-s} = 0. \quad (\text{PL.7})$$

The proved exponential-polynomial independence makes both polynomial coefficients identically zero. Evaluating the first at $s = \rho$ would give $2^{-\rho} = 0$, a contradiction. This proves $\chi_\rho \notin W$, so in particular $W \neq V$.

Now take the actual theta representative and its original supported quotient class:

$$F_\rho = \Theta\chi_\rho \in \mathcal{B}, \quad q_\rho = [F_\rho]_W \in \mathcal{Q}_W = \mathcal{B}/\Theta W. \quad (\text{PL.8})$$

It is nonzero because $F_\rho = \Theta w$, $w \in W$, would imply $\chi_\rho = w$ by injectivity of the original theta map. Nevertheless

$$(t - \rho)q_\rho = 0, \quad \mathcal{M}F_\rho(s) = g(s) \frac{2^{-s} - 2^{-\rho}}{s - \rho}. \quad (\text{PL.9})$$

The first equality follows from $(D - \rho)F_\rho = \Theta\phi_\rho \in \Theta W$; the second is the original identity $\mathcal{M}\Theta = gH$ applied to χ_ρ . In fact $Aq_\rho \cong A/(t - \rho)$. For any polynomial P , division by $t - \rho$ gives $P(t)q_\rho = P(\rho)q_\rho$. Since $q_\rho \neq 0$, this vanishes exactly when $P(\rho) = 0$.

Let $O = \mathcal{O}_{\mathbb{C}, \rho}$, $z = s - \rho$, and retain the actual proper-label comparison

$$M_W = O \otimes_A \mathcal{Q}_W, \quad b_W : M_W \longrightarrow O/(g), \quad b_W(a \otimes [F]_W) = [a\mathcal{M}F], \quad K_W = \ker b_W. \quad (\text{PL.10})$$

It is well-defined because $\mathcal{M}\Theta w = gH_w$ for every $w \in W$; it is balanced because $\mathcal{M}DF = s\mathcal{M}F$. It is onto by the original F_0 and its invertible transform H_0 in (BK.13): $b_W(aH_0^{-1} \otimes [F_0]_W) = [a]$. The inclusion $Aq_\rho \hookrightarrow \mathcal{Q}_W$ remains injective after tensoring with the flat A -module O . Since

$$O \otimes_A Aq_\rho \cong O/(z) \neq 0,$$

the element $k_\rho = 1 \otimes q_\rho$ is nonzero in M_W . Its original image (PL.9) is divisible by g , giving

$$0 \neq k_\rho \in K_W, \quad zk_\rho = 0. \quad (\text{PL.11})$$

Thus K_W has a nonzero one-dimensional spectral submodule on an actual proper source label containing the original arithmetic generator ϕ_* . Invariance of W and inclusion of that generator do not imply the polynomial-invertibility theorem for K_W .

We record the exact morphism that transports this class to the proved full-source comparison. Inclusion of source labels gives

$$0 \longrightarrow V/W \xrightarrow{[v] \mapsto [\Theta v]_W} \mathcal{Q}_W \xrightarrow{[F]_W \mapsto [F]_V} \mathcal{Q} \longrightarrow 0. \quad (\text{PL.12})$$

The first arrow is injective by theta injectivity. The final map is onto, and its kernel consists exactly of those representatives in ΘV , modulo ΘW . These statements prove exactness and identify its kernel with the original V/W , without changing the source quotient. The arrows commute with D because theta does. Flatness therefore gives the actual balanced exact sequence

$$0 \longrightarrow O \otimes_A (V/W) \longrightarrow M_W \xrightarrow{\pi} M \longrightarrow 0, \quad M = O \otimes_A \mathcal{Q}, \quad b_W = b\pi. \quad (\text{PL.13})$$

Restricting π to the indicated kernels gives

$$0 \longrightarrow O \otimes_A (V/W) \longrightarrow K_W \xrightarrow{\pi} K \longrightarrow 0. \quad (\text{PL.14})$$

Indeed the first term is already in K_W since its Mellin values are multiples of g . Every element of K has a lift in M_W by (PL.13), and every such lift is in K_W because $b_W = b\pi$. This proves surjectivity of the last arrow and identifies its kernel. The explicit k_ρ is the image of $1 \otimes [\chi_\rho]$ in the left term of (PL.14), and its image in K is exactly zero. Its finite spectral contribution is therefore removed by the specified enlargement from W to V . It does not contradict the polynomial invertibility of the full-source K .

This difference is also detected by the finite derived observations. Tensor the free resolution $[O \xrightarrow{z} O]$ with the proper-label exact sequence $0 \rightarrow K_W \rightarrow M_W \xrightarrow{b_W} O/(g) \rightarrow 0$. It produces the short exact sequence of complexes

$$0 \rightarrow [K_W \xrightarrow{z} K_W] \rightarrow [M_W \xrightarrow{z} M_W] \rightarrow [O/(g) \xrightarrow{z} O/(g)] \rightarrow 0. \quad (\text{PL.15})$$

The first complex has nonzero degree-minus-one cohomology, since $k_\rho \in \ker(z : K_W \rightarrow K_W) \setminus \{0\}$. Equivalently the degree-minus-one map on the last two complexes has the nonzero kernel containing k_ρ . Thus $b_W \otimes_O^{\mathbf{L}} O/(z)$ is not a quasi-isomorphism. The finite derived comparison (DP.2) is therefore a theorem for its stated full-source Q and its specified copies in the supported construction, not for arbitrary proper source labels.

Every element above retains its original label. The source-label transport carries $[\Theta\chi_\rho]_W$ to the supported zero in the V fibre; it does not send that element to the external absence point τ . Equations (PL.12)–(PL.15) prove the exact relation between the proper-label finite spectral kernel and the full-source meromorphic kernel, including the class that the transition kills. Nothing in the construction asserts $g(\rho) = 0$; it works at every specified ρ . If ρ is an actual off-critical zero, its arithmetic block in $O/(g)$ remains the one computed in (DP.4), and the proper-label class (PL.11) is the separately computed kernel of the displayed source-label transition.

Chapter 7

The arithmetic packet through coefficient purity

The actual arithmetic packet through mixed-support synchronization

This calculation starts with the mixed-support construction in original shared-thread record A0961, node a7bbd422-311c-4380-ba6c-a4bdd61fce2f, chain 2032, and its full authored proof in original record A0953, node 0d2934ea-14ac-48c1-a1f3-cd918c1f5fe5, chain 1987. It does not replace the actual arithmetic function by a symmetric test polynomial. The polynomial below is the full divisor polynomial of an actual packet in the counterfactual under study.

Let Z be a nonempty finite set of actual nontrivial zeros of the original completed function $g(s) = 2\xi(s)$, with their full orders m_ρ . Retain

$$h_Z(x) = \prod_{\rho \in Z} (x - \rho)^{m_\rho}, \quad d = \sum_{\rho \in Z} m_\rho, \quad E_Z = \mathbb{C}[x]/(h_Z(x)).$$

The variable x is the original spectral variable. The additional variable t below is the cyclic coefficient coordinate; it is not substituted for x . For $a > 0$ let

$$A_a = M_{j_{h_Z}(a^s)} : E_Z \longrightarrow E_Z.$$

Here $a^s = \exp(s \log a)$ uses the real logarithm and j_{h_Z} retains all Taylor coefficients at the actual zero centres. On the summand

$$E_\rho = \mathbb{C}[z_\rho]/(z_\rho^{m_\rho}), \quad x = \rho + z_\rho,$$

this is exactly

$$A_a|_{E_\rho} = a^\rho \sum_{j=0}^{m_\rho-1} \frac{(\log a)^j}{j!} N_\rho^j, \quad N_\rho = M_{z_\rho}. \quad (7.1)$$

Thus no multiplicity or nilpotent term is omitted.

The original mixed-support object, with its unit

Fix $n \geq 1$. Form the source object $D_{E_Z, n} = G(E_Z)^n$ with coordinatewise addition and the source product

$$(b \star c)_k = \bigoplus_{i+j=k} b_i c_j \oplus \bigoplus_{i+j=k+n} (-1)^{\bullet} b_i c_j, \quad 0 \leq k < n.$$

An empty sum is τ , and the original unit is $(1^\bullet, \tau, \dots, \tau)$. Write $e = 0_{E_Z}^\bullet$. The exact amplitude target is

$$B_n = E_Z[t]/(t^n + 1), \quad \pi_n(b) = \sum_{i=0}^{n-1} p(b_i)t^i.$$

The support mask of b has its i th entry equal to one precisely when $b_i \neq \tau$. The pair consisting of the mask and $\pi_n(b)$ is injective: the monic polynomial gives the free E_Z -basis $1, t, \dots, t^{n-1}$, and the mask then distinguishes e from τ in each zero coefficient.

Let

$$E_n = (1^\bullet, e, \dots, e).$$

Multiplication by E_n sends the wholly absent input to itself and every active input to the fully supported vector with the same ordinary coefficients. To check the support assertion, fix an active input coefficient at index j . At every output index k the summand using the E_n coefficient at $k - j$ modulo n is supported; all wrap factors are supported. Thus every output coordinate is supported, including a coordinate whose resulting amplitude is zero. The amplitude assertion follows from $\pi_n(E_n) = 1$. It follows at once that $E_n^2 = E_n$ and

$$D_{E_Z, n}[E_n^{-1}] \simeq E_n D_{E_Z, n} \simeq G(B_n). \quad (7.2)$$

The image has unit E_n . For completeness, a unital semiring map making E_n invertible sends it to 1 by idempotence and therefore sends b and $E_n b$ to the same value. This is the localization universal property. The inclusion of the image into the original unsynchronized object is not falsely declared to preserve the original unit.

A proved split injection and its exact retraction

Define

$$\iota_n : E_Z \longrightarrow B_n, \quad f \longmapsto f, \quad r_n : B_n \longrightarrow E_Z, \quad b \longmapsto \frac{1}{n} \text{Tr}_{B_n/E_Z}(M_b).$$

The first is a unital ring homomorphism. The second is an E_Z -linear map of modules; it is not asserted to be multiplicative. In the basis $1, t, \dots, t^{n-1}$, multiplication by t^k has no diagonal entries for $1 \leq k < n$, since it shifts every basis index by the nonzero class $k \pmod{n}$. Its wrap entries carry the original minus signs. Multiplication by $b_0 \in E_Z$ is the scalar matrix with n diagonal entries b_0 . Consequently

$$\text{Tr}_{B_n/E_Z} \left(M_{\sum_{i=0}^{n-1} b_i t^i} \right) = n b_0, \quad r_n \left(\sum_{i=0}^{n-1} b_i t^i \right) = b_0, \quad r_n \iota_n = 1_{E_Z}. \quad (7.3)$$

In particular the trace factor n and its inverse in the retraction are both retained. No arithmetic trace form has been rescaled.

For a linear map $T : M \rightarrow N$ over E_Z , its split lift is $T^\tau(m^\bullet) = T(m)^\bullet$ and $T^\tau(\tau) = \tau$. It is $G(E_Z)$ -linear: addition of supported inputs is the original module addition, scalar multiplication is the original module action, and the cases containing an absent scalar or input are absorbed by τ . A zero result is $(0_N)^\bullet$, not τ . Thus (7.3) lifts to an actual retract

$$E_Z^\tau \xrightarrow{\iota_n^\tau} G(B_n) \xrightarrow{r_n^\tau} E_Z^\tau, \quad r_n^\tau \iota_n^\tau = 1. \quad (7.4)$$

Through (7.2), the first map is the fully supported mixed vector $(f^\bullet, e, \dots, e)$ for supported f , and the wholly absent vector for τ . Therefore every nonzero amplitude packet class remains nonzero through this specific mixed-support realization and localization.

Write $\sigma_Z : E_Z \hookrightarrow Q$ and $J_Z : Q \rightarrow E_Z$ for the original arithmetic packet injection and observation, whose proved identity is $J_Z \sigma_Z = 1_{E_Z}$. Their composite with the maps just computed is

$$\sigma_Z r_n : B_n \longrightarrow Q, \quad (\sigma_Z r_n) \iota_n = \sigma_Z.$$

This is a composite of the established maps, not a new assertion that the spectral coefficients are unrelated to arithmetic cohomology.

The scaling intertwiner used here comes from the original source. Retain $Q = \mathcal{B}/\Theta V$, $D = -x\partial_x$, and $\Theta\phi(x) = \sum_{j \in \mathbb{Z} \setminus \{0\}} \phi(jx)$. On both original test spaces define

$$(\mathcal{R}_a F)(x) = F(x/a), \quad a > 0.$$

Positive dilation preserves even Schwartz functions and their two zero moments: the value at zero is unchanged and the integral is multiplied by a . Direct substitution in the theta sum proves $\Theta\mathcal{R}_a = \mathcal{R}_a\Theta$. Thus $\mathcal{R}_a$ induces the original action $\overline{\mathcal{R}}_a$ on Q . Substitution $x = ay$ in the Mellin integral gives

$$\mathcal{M}(\mathcal{R}_a F)(s) = a^s \mathcal{M}F(s), \quad J_Z \overline{\mathcal{R}}_a = A_a J_Z.$$

The source's complete finite-packet theorem identifies $\sigma_Z(E_Z) = Q[h_Z(D)]$ and makes J_Z its inverse there. Since $\mathcal{R}_a D = D\mathcal{R}_a$, that subspace is invariant. Applying the bijection J_Z to its two vectors therefore proves

$$\overline{\mathcal{R}}_a \sigma_Z = \sigma_Z A_a.$$

This carries the full factor a^ρ and every Taylor coefficient in (7.1) from the actual theta quotient.

The exact coefficient Frobenius comparison

Let q be an odd prime power with $\gcd(q, n) = 1$. On the integral coefficient algebra

$$C_{n, \mathbb{Z}} = \mathbb{Z}[t]/(t^n + 1)$$

define the unital algebra map

$$\Psi_q(t) = t^q, \quad \Psi_q(c) = c \quad (c \in \mathbb{Z}).$$

It is well-defined since $(t^q)^n = (-1)^q = -1$. Because $\gcd(q, 2n) = 1$, choose $r > 0$ with $q^r \equiv 1 \pmod{2n}$. The identity $t^{2n} = 1$ then gives $\Psi_q^r = 1$, proving it is an automorphism. Its matrix in the original coefficient basis is the signed permutation

$$t^i \mapsto (-1)^{\lfloor qi/n \rfloor} t^{qi \bmod n}. \quad (7.5)$$

Reduction of this integral map to $\mathbb{F}_q$ is exactly the source's q -power Frobenius on $\mathbb{F}_q[t]/(t^n + 1)$, because every scalar of $\mathbb{F}_q$ has q th power equal to itself. Its extension to $\mathbb{C}$ is a coefficient-linear lift of that integral matrix. It is not the map $c \mapsto c^q$ on complex scalars. These two specializations are linked by the displayed integral map; no ring homomorphism $\mathbb{F}_q \rightarrow \mathbb{C}$ is asserted.

There is also a map on the full unsynchronized mixed-support object: send coefficient b_i to $((-1)^\bullet)^{\lfloor qi/n \rfloor} b_i$ in slot $qi \bmod n$, fixing E_Z scalars and preserving absent inputs. It is additive. On two supported monomials, reducing the exponent before or after the map gives the same wrap factor because $((-1)^\bullet)^q = (-1)^\bullet$; the absent cases are absorbed, and distributivity proves the product identity for arbitrary vectors. Hence this is a semiring morphism, fixes E_n , preserves every mask up to the stated permutation, and induces Ψ_q through (7.2).

Extend Ψ_q E_Z -linearly to B_n . It fixes the subspace $\iota_n(E_Z)$ pointwise. It preserves r_n : formula (7.5) moves no index $i \in \{1, \dots, n-1\}$ to zero modulo n , and fixes index zero. Hence

$$\Psi_q \iota_n = \iota_n, \quad r_n \Psi_q = r_n. \quad (7.6)$$

Every eigenvalue of the complex coefficient operator Ψ_q is a root of unity, because $\Psi_q^r = 1$. Over $\mathbb{C}$ the polynomial $X^r - 1$ has no repeated roots, so the operator is diagonalizable. This is the exact finite coefficient purity calculation transported through its integral matrix.

The original arithmetic action, full jets, and modulus

Extend A_a to B_n by coefficient multiplication:

$$A_a^B \left(\sum b_i t^i \right) = \sum A_a(b_i) t^i.$$

Since A_a is multiplication by $j_{h_Z}(a^s) \in E_Z$, this commutes with the E_Z -linear automorphism Ψ_q . The maps in (7.3) intertwine the arithmetic action exactly:

$$A_a^B \iota_n = \iota_n A_a, \quad r_n A_a^B = A_a r_n.$$

Consequently they also intertwine A_a with $A_a^B \Psi_q$ because of (7.6). On the coefficient eigenspace of Ψ_q with eigenvalue ω , the original ρ -packet action is

$$\omega a^\rho \sum_{j=0}^{m_\rho-1} \frac{(\log a)^j}{j!} N_\rho^j. \quad (7.7)$$

The entire nilpotent polynomial is retained. For $a \neq 1$, its nilpotent part has the same nilpotence index as N_ρ :

$$\exp((\log a) N_\rho) - 1 = N_\rho \left((\log a) I + \sum_{j=2}^{m_\rho-1} \frac{(\log a)^j}{j!} N_\rho^{j-1} \right),$$

and the factor in parentheses is invertible because its constant coefficient is nonzero. The expression also states the case $m_\rho = 1$ directly.

Let e_ρ be the full Chinese-remainder idempotent in E_Z . The vector

$$v_\rho = e_\rho(x - \rho)^{m_\rho-1}$$

is nonzero: in its ρ -local quotient its coefficient of $z_\rho^{m_\rho-1}$ is one. It is killed by N_ρ and satisfies $A_a v_\rho = a^\rho v_\rho$. The vector $\iota_n(v_\rho)$ is nonzero by the retraction, lies in the fixed coefficient line of Ψ_q , and is therefore an eigenvector of $A_a^B \Psi_q$ with the unchanged eigenvalue a^ρ .

Suppose now the actual counterfactual supplies $\rho = \frac{1}{2} + \delta + i\gamma$, $0 < \delta < \frac{1}{2}$. For every $a > 1$,

$$\frac{|a^\rho|^2}{a} = a^{2\delta} > 1, \quad \log \frac{|a^\rho|^2}{a} = 2\delta \log a. \quad (7.8)$$

For the other coefficient eigenspaces in (7.7), $|\omega| = 1$ gives exactly the same modulus. At $a = q$, these eigenvalues are $q^\rho \omega$. The coefficient pure spectrum therefore preserves the actual arithmetic weight defect in this specified construction.

Take every tensor power in this paragraph over $\mathbb{C}$. The vector $\iota_n(v_\rho)^{\otimes_{\mathbb{C}} k}$ is nonzero because $(r_n)^{\otimes_{\mathbb{C}} k}$ sends it to $v_\rho^{\otimes_{\mathbb{C}} k}$. The latter is nonzero: a complex linear functional taking v_ρ to 1 has k -fold complex tensor power taking that tensor to 1. Its eigenvalue under $(A_a^B \Psi_q)^{\otimes_{\mathbb{C}} k}$ is $a^{k\rho}$, and

$$\log \frac{|a^{k\rho}|^2}{a^k} = 2k\delta \log a.$$

This is an exact input to the user's selected Deligne amplification argument. The computation does not import a cohomological upper estimate from another object. It identifies which actual nonzero class and which unchanged eigenvalue such an estimate must control.

The trace obstruction also survives the pure coefficient factor

Assume Z is stable under $\rho \mapsto 1 - \bar{\rho}$ and under conjugation, with the original multiplicities. Retain the actual derivative-residue trace contraction

$$\mathcal{T}_Z(f, u) = \sum_{\rho \in Z} m_\rho \overline{f(1 - \bar{\rho})} u(\rho).$$

This formula is the contraction of the original residue pairing using g' ; the factors in $g = 2\xi$ are not modified here. In a genuine off-line quartet let m be its common multiplicity and set $f = e_\rho - e_{1-\bar{\rho}}$. The two nonzero evaluations contribute $-m$ each, so $\mathcal{T}_Z(f, f) = -2m$.

On $C_n = \mathbb{C}[t]/(t^n + 1)$ retain the positive form

$$H_n(c, d) = \sum_{j=0}^{n-1} \overline{c(r_j)} d(r_j), \quad r_j = \exp\left(\frac{(2j+1)\pi i}{n}\right).$$

The distinct roots make evaluation an isomorphism $C_n \simeq \mathbb{C}^n$, proving positive definiteness. The operator Ψ_q permutes the evaluations, so it preserves H_n . The original constant vector has $H_n(1, 1) = n$. Under $B_n \simeq E_Z \otimes_{\mathbb{C}} C_n$ the tensor form therefore satisfies

$$(\mathcal{T}_Z \otimes_{\mathbb{C}} H_n)(\iota_n f, \iota_n f) = -2mn < 0.$$

This proves the actual negative trace vector survives this exact positive coefficient attachment, with the factor n retained. It does not assert that the global arithmetic polarization is automatically this tensor form. Any different global comparison has its own map and must be calculated on this surviving input.

Chapter 8

The full-multiplicity arithmetic boundary floor

The multiplicity boundary floor

This section proves an exact lower bound on the relative control defect of the theta-generated representatives. The bound counts every retained zero with its complete multiplicity. The source constructions are A1644, lines 2576–2718, and A1771, lines 4477–4835, in the authoritative `audit_segment_U0041_U0054.md`. The rank-two identity itself is A1771, lines 4688–4725. The invariant trace and the invariant-subspace argument below are new deductions from those source constructions.

All matrix coordinates on the packet algebra are the original polynomial coordinates. The generator is $D = -x\partial_x$, the arithmetic multiplier is $g = 2\xi$, and the Hilbert-space measure is dx on the positive half-line. The local extension unit, the unscaled orthogonal vectors, their norm squares, and all nilpotent jets are retained.

The original source, packet algebra, and arithmetic section

Write

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \quad g(s) = 2\xi(s),$$

with the entire continuations understood at the apparent singularities. Let

$$\begin{aligned} V &= \left\{ \phi \in \mathcal{S}(\mathbb{R}) : \phi \text{ is even, } \phi(0) = 0, \int_{\mathbb{R}} \phi(x) dx = 0 \right\}, \\ \mathcal{B} &= \left\{ F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^j F(x)| < \infty \text{ for all } b \in \mathbb{Z}, j \geq 0 \right\}, \\ D &= -x\partial_x, \quad (\Theta\phi)(x) = \sum_{a \neq 0} \phi(ax) = 2 \sum_{a \geq 1} \phi(ax). \end{aligned} \tag{BF.01}$$

Here $\mathcal{S}(\mathbb{R})$ is the Schwartz space, and the summation indices a in Θ are integers. The symbol $\mathcal{B}$ denotes exactly the strong-Schwartz target specified in A1694, lines 3049–3073. Put

$$Q = \mathcal{B}/\Theta V, \quad q : \mathcal{B} \longrightarrow Q, \quad \mathcal{H}_+ = L^2(\mathbb{R}_{>0}, dx).$$

The scalar product convention is

$$\langle F, H \rangle_+ = \int_0^\infty \overline{F(x)} H(x) dx,$$

antilinear in the first argument. Adjoint signs on maps into $\mathcal{H}_+$ use this scalar product and the standard scalar product in the stated polynomial coordinates on their finite-dimensional domains.

Retain the actual source function and its derivative sequence:

$$\begin{aligned} \phi_*(x) &= (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, \\ f_0 &= \Theta\phi_*, \quad \phi_j = D^j\phi_*, \quad f_j = \Theta\phi_j = D^j f_0 \quad (j \geq 0), \\ (\mathcal{M}F)(s) &= \int_0^\infty F(x)x^s \frac{dx}{x}, \quad \mathcal{M}f_0 = g. \end{aligned} \tag{BF.02}$$

The source identity $\mathcal{M}(D^j f_0)(s) = s^j g(s)$, with these exact constants, is A1644, lines 2566–2574, and A1771, lines 4479–4529. The commutation $D\Theta\phi = \Theta D\phi$ uses the same generator on the source and target. Differentiation of the defining series is justified on each compact subinterval of $(0, \infty)$ by the Schwartz estimates.

Let Z be a nonempty finite set of actual zeros of g . For $\rho \in Z$, let $m_\rho \geq 1$ be its complete order of vanishing. Define

$$h_Z(t) = \prod_{\rho \in Z} (t - \rho)^{m_\rho}, \quad d = \deg h_Z = \sum_{\rho \in Z} m_\rho, \quad E_Z = \mathbb{C}[t]/(h_Z), \quad A = M_t : E_Z \longrightarrow E_Z. \tag{BF.03}$$

The ordered coordinates on E_Z are the classes of $1, t, \dots, t^{d-1}$. Multiplication by t is not replaced by a shifted generator. The empty packet is treated separately at the end.

For a function holomorphic near every member of Z , j_{h_Z} means its complete collection of jets of orders $0, \dots, m_\rho - 1$, identified with E_Z by the Chinese remainder isomorphism. In particular, the finite Mellin-jet map is

$$J_Z : \mathcal{B} \longrightarrow E_Z, \quad J_Z F = j_{h_Z}(\mathcal{M}F).$$

In the local coordinate $z = s - \rho$, the ρ -component is

$$\sum_{j=0}^{m_\rho-1} \frac{(\mathcal{M}F)^{(j)}(\rho)}{j!} z^j \pmod{z^{m_\rho}}.$$

The map retains the factorials and the full jet length.

The actual extension unit is

$$\varepsilon_Z = \left(j_{h_Z}(g/h_Z) \right)^{-1} = j_{h_Z}(h_Z/g) \in E_Z^\times. \tag{BF.04}$$

To see why it is a unit, write

$$g(\rho + z) = z^{m_\rho} w_\rho(z), \quad w_\rho(0) \neq 0.$$

The polynomial $h_Z(\rho + z)$ has precisely the same zero order. Thus g/h_Z and h_Z/g are mutually inverse holomorphic units near every selected zero. Equation (BF.04) keeps their complete jets. This finite-algebra unit is distinct from the supported-zero scalar e of the original split-support construction.

For $u \in E_Z$, let $R_Z(u)$ be the unique polynomial of degree less than d representing $\varepsilon_Z u$, and define the extension row

$$\ell_Z : E_Z \longrightarrow \mathbb{C}, \quad \ell_Z(u) = [t^{d-1}]R_Z(u). \tag{BF.05}$$

The established arithmetic section in A1644, lines 2594–2718, and A1771, lines 4499–4510, is the map

$$s_Z : E_Z \longrightarrow \mathcal{B}$$

with the identities

$$J_Z s_Z = I_{E_Z}, \quad J_Z \Theta = 0, \quad q s_Z = \sigma_Z : E_Z \hookrightarrow Q, \quad D s_Z - s_Z A = f_0 \ell_Z. \tag{BF.06}$$

These are the inherited source identities used below, with the actual source section and its analytic construction. For a one-point packet at a simple zero, their coefficient is exactly

$$\ell_Z(1) = \frac{1}{g'(\rho)} = \frac{1}{2\xi'(\rho)}.$$

No replacement of this coefficient by a unit-free scalar is made.

Theta relation spaces and the exact rank-two boundary

Define

$$V_n = \text{span}_{\mathbb{C}}\{\phi_0, \dots, \phi_n\}, \quad L_n = \Theta V_n = \text{span}_{\mathbb{C}}\{f_0, \dots, f_n\}, \quad n \geq 0. \quad (\text{BF.07})$$

The vectors f_j are linearly independent. Indeed, a finite dependence has Mellin transform $g(s)P(s) = 0$, where $P(s) = \sum_j c_j s^j$. Since g is not identically zero, it is nonzero on a nonempty open set. Therefore P vanishes on that open set and is the zero polynomial. Every coefficient c_j is zero.

Let $F_n : \mathbb{C}^{n+1} \rightarrow \mathcal{H}_+$ have columns $f_0, \dots, f_n$, and set

$$H_n = F_n^* F_n, \quad P_n = F_n H_n^{-1} F_n^*, \quad R_n = (I - P_n) s_Z, \quad G_n = R_n^* R_n. \quad (\text{BF.08})$$

Independence gives $H_n \succ 0$. Direct multiplication gives $P_n^2 = P_n = P_n^*$, and its range is L_n ; it is the actual orthogonal projection onto the specified finite theta relation space. The expression

$$b_n(u) = \sum_{j=0}^n (H_n^{-1} F_n^* s_Z u)_j \phi_j$$

defines a map $b_n : E_Z \rightarrow V_n$ satisfying $P_n s_Z = \Theta b_n$. Consequently, by (BF.06),

$$qR_n = \sigma_Z, \quad J_Z R_n = I_{E_Z}, \quad G_n = s_Z^* F_n H_n^{-1} F_n^* s_Z \succ 0. \quad (\text{BF.09})$$

For the strict inequality, $v^* G_n v = \|R_n v\|_+^2 = 0$ implies $R_n v = 0$ and hence $v = J_Z R_n v = 0$. Thus all inverses of G_n used below are inverses of strictly positive matrices at the stated finite level.

The original theta boundary and the associated control matrix are

$$\begin{aligned} B_n &= D R_n - R_n A = \Theta K_n, \\ K_n &= \phi_* \ell_Z - D_V b_n + b_n A, \\ W_n &= A^* G_n + G_n A - G_n. \end{aligned} \quad (\text{BF.10})$$

Here D_V is D on V . The boundary formula follows by substituting $R_n = s_Z - \Theta b_n$ into (BF.06). It uses only original theta boundaries.

For $F, H \in \mathcal{B}$, integration by parts gives

$$\langle DF, H \rangle_+ + \langle F, DH \rangle_+ = - \left[x \overline{F(x)} H(x) \right]_0^\infty + \langle F, H \rangle_+ = \langle F, H \rangle_+. \quad (\text{BF.11})$$

The defining estimates of $\mathcal{B}$ make the boundary expression zero at both endpoints. Applying (BF.11) to columns of R_n , and substituting $DR_n = R_n A + B_n$, proves

$$W_n = -(R_n^* B_n + B_n^* R_n). \quad (\text{BF.12})$$

The coefficient of G_n in (BF.10) is exactly one because the measure in (BF.11) is dx . Even extension to the full line has squared norm twice this norm; it is not silently substituted here.

Construct the unscaled orthogonal vectors

$$u_0 = f_0, \quad u_n = (I - P_{n-1})f_n \quad (n \geq 1), \quad \mathfrak{h}_n = \langle u_n, u_n \rangle_+ > 0, \quad r_n = \mathfrak{h}_n^{-1} u_n^* s_Z : E_Z \rightarrow \mathbb{C}. \quad (\text{BF.13})$$

The operator $u_n^* : \mathcal{H}_+ \rightarrow \mathbb{C}$ sends F to $\langle u_n, F \rangle_+$. Each u_n has leading coefficient one on f_n ; none is divided by its norm. Since the f_j are real functions, their Gram matrices are real, and each u_n is real.

Degree and orthogonality determine the recurrence

$$\begin{aligned} Du_0 &= u_1 + \frac{1}{2}u_0, \\ Du_n &= u_{n+1} + \frac{1}{2}u_n - \frac{\mathfrak{h}_n}{\mathfrak{h}_{n-1}}u_{n-1} \quad (n \geq 1). \end{aligned} \quad (\text{BF.14})$$

Here is the complete coefficient calculation. As $u_n = f_n + \sum_{j < n} c_j f_j$, Du_n belongs to L_{n+1} and has coefficient one on u_{n+1} . For $j \leq n-2$, (BF.11) gives

$$\langle u_j, Du_n \rangle_+ = \langle u_j, u_n \rangle_+ - \langle Du_j, u_n \rangle_+ = 0,$$

because $Du_j \in L_{j+1} \subseteq L_{n-1}$. Reality and (BF.11) give

$$2\langle u_n, Du_n \rangle_+ = \mathfrak{h}_n,$$

so the full diagonal coefficient is $1/2$, not merely its real part. For $n \geq 1$,

$$\langle u_{n-1}, Du_n \rangle_+ = -\langle Du_{n-1}, u_n \rangle_+ = -\mathfrak{h}_n.$$

Dividing only this coefficient by the corresponding original norm square $\mathfrak{h}_{n-1}$ yields the last term of (BF.14).

The section coefficients satisfy

$$P_n s_Z = \sum_{j=0}^n u_j r_j, \quad \frac{\mathfrak{h}_{n+1}}{\mathfrak{h}_n} r_{n+1} = r_n \left(\frac{1}{2}I - A \right) + r_{n-1} - \mathbf{1}_{\{n=0\}} \ell_Z, \quad r_{-1} = 0. \quad (\text{BF.15})$$

The indicator $\mathbf{1}_{\{n=0\}}$ equals one for $n = 0$ and zero otherwise. To check the row equation, pair $Ds_Z = s_Z A + u_0 \ell_Z$ against u_n . The right side is

$$\mathfrak{h}_n r_n A + \mathbf{1}_{\{n=0\}} \mathfrak{h}_0 \ell_Z.$$

By (BF.11) and (BF.14), the left side is

$$\frac{1}{2} \mathfrak{h}_n r_n - \mathfrak{h}_{n+1} r_{n+1} + \mathfrak{h}_n r_{n-1},$$

where the last term is zero for $n = 0$. Rearrangement gives (BF.15). The initial extension functional is therefore fixed by the original arithmetic section.

We now obtain the source's rank-two formula with its exact sign. Since

$$B_n = f_0 \ell_Z - DP_n s_Z + P_n s_Z A,$$

the first and third terms have image in L_n , orthogonal to the image of R_n . Equation (BF.12) becomes

$$W_n = R_n^* DP_n s_Z + (DP_n s_Z)^* R_n.$$

Equations (BF.14) and (BF.15) give

$$(I - P_n)DP_n s_Z = u_{n+1} r_n, \quad u_{n+1}^* R_n = u_{n+1}^* s_Z = \mathfrak{h}_{n+1} r_{n+1}.$$

It follows that

$$\boxed{W_n = \mathfrak{h}_{n+1} (r_{n+1}^* r_n + r_n^* r_{n+1})}. \quad (\text{BF.16})$$

Also $R_{n+1} = R_n - u_{n+1} r_{n+1}$, with R_{n+1} orthogonal to u_{n+1} . Taking Gram matrices proves the exact decrement

$$G_n - G_{n+1} = \mathfrak{h}_{n+1} r_{n+1}^* r_{n+1} \succeq 0. \quad (\text{BF.17})$$

These formulas are A1771, lines 4688–4734. The rank bound in (BF.16) concerns the boundary control form on E_Z ; the dimension d , and the full operator A , have not been reduced to two dimensions.

For clarity, the retained source-support transition is also exact. Write $Q_{V_n} = \mathcal{B}/\Theta V_n = \mathcal{B}/L_n$. Since $D\Theta V_n \subseteq \Theta V_{n+1}$, differentiation gives a well-defined map

$$\overline{D}_n : Q_{V_n} \longrightarrow Q_{V_{n+1}}, \quad [F] \longmapsto [DF].$$

Before that transition, the boundary class is

$$[B_n v]_{Q_{V_n}} = -[u_{n+1}]_{Q_{V_n}} r_n(v).$$

The layer map

$$\Theta V_{n+1}/\Theta V_n \longrightarrow L_{n+1} \cap L_n^\perp, \quad [z] \longmapsto (I - P_n)z$$

is well-defined and injective because its kernel in L_{n+1} is L_n . It is surjective because every vector in the displayed orthogonal complement is fixed by $I - P_n$. The subsequent support transport makes this particular class the supported zero in $Q_{V_{n+1}}$. Its earlier pairing in (BF.16) is retained. These are the actual maps of A1771, lines 4736–4773.

Relative control spectrum and its invariant trace

For $G_n \succ 0$, define its least two-sided control excess by

$$\epsilon_n = \inf\{\epsilon \geq 0 : -\epsilon G_n \preceq W_n \preceq \epsilon G_n\}. \quad (\text{BF.18})$$

Here $X \preceq Y$ means that $Y - X$ is a positive semidefinite Hermitian matrix. The operator $G_n^{-1}W_n$ is self-adjoint in the positive inner product $\langle u, v \rangle_{G_n} = u^* G_n v$:

$$\langle u, G_n^{-1}W_n v \rangle_{G_n} = u^* W_n v = \langle G_n^{-1}W_n u, v \rangle_{G_n}.$$

Thus ϵ_n is the largest absolute value of its eigenvalues. This uses the actual metric and operator and makes no assumption that A is normal for this metric.

In the unchanged coordinates, set

$$\mathbf{a}_n = r_n G_n^{-1} r_n^*, \quad \mathbf{b}_n = r_{n+1} G_n^{-1} r_{n+1}^*, \quad \mathbf{c}_n = r_n G_n^{-1} r_{n+1}^*. \quad (\text{BF.19})$$

The exact maps

$$\begin{aligned} \mathcal{J}_n : E_Z &\longrightarrow \mathbb{C}^2, & \mathcal{J}_n v &= (r_n v, r_{n+1} v), \\ \mathcal{U}_n : \mathbb{C}^2 &\longrightarrow E_Z, & & \\ \mathcal{U}_n(a, b) &= \mathfrak{h}_{n+1} (G_n^{-1} r_{n+1}^* a + G_n^{-1} r_n^* b) \end{aligned} \quad (\text{BF.20})$$

satisfy

$$\mathcal{U}_n \mathcal{J}_n = G_n^{-1} W_n, \quad \mathcal{J}_n \mathcal{U}_n = \mathfrak{h}_{n+1} \begin{pmatrix} \mathbf{c}_n & \mathbf{a}_n \\ \mathbf{b}_n & \overline{\mathbf{c}_n} \end{pmatrix}.$$

For every nonzero eigenvalue, applying $\mathcal{J}_n$ and $\mathcal{U}_n$ gives mutually inverse eigenspace maps up to that nonzero scalar. Therefore the nonzero eigenvalues of the two composites agree. Cauchy–Schwarz

for the positive form on row covectors induced by G_n^{-1} gives $\mathbf{a}_n \mathbf{b}_n \geq |\mathbf{c}_n|^2$. The two roots of the displayed 2-by-2 characteristic polynomial are consequently

$$\lambda_{n,\pm} = \mathfrak{h}_{n+1} \left(\operatorname{Re} \mathbf{c}_n \pm \sqrt{\mathbf{a}_n \mathbf{b}_n - (\operatorname{Im} \mathbf{c}_n)^2} \right), \quad \lambda_{n,+} \geq 0 \geq \lambda_{n,-}. \quad (\text{BF.21})$$

Zero roots are included in this notation even when the corresponding eigenspace does not occur in E_Z . There are no other nonzero eigenvalues by the factorization (BF.20). Thus

$$\epsilon_n = \mathfrak{h}_{n+1} \left(|\operatorname{Re} \mathbf{c}_n| + \sqrt{\mathbf{a}_n \mathbf{b}_n - (\operatorname{Im} \mathbf{c}_n)^2} \right). \quad (\text{BF.22})$$

This is the exact expression in A1771, lines 4775–4821. In particular, G_n^{-1} remains present in every relative coefficient.

The trace of this control operator is independent of the chosen positive representative metric. Cyclicity of finite-dimensional trace and (BF.10) give

$$\begin{aligned} \operatorname{tr}(G_n^{-1} W_n) &= \operatorname{tr}(G_n^{-1} A^* G_n) + \operatorname{tr} A - d \\ &= 2 \operatorname{Re} \operatorname{tr} A - d \\ &= \sum_{\rho \in Z} m_\rho (2 \operatorname{Re} \rho - 1). \end{aligned} \quad (\text{BF.23})$$

For the final equality, the Chinese remainder decomposition preserves the local coordinates $z = t - \rho$. On the length- m_ρ summand, multiplication by t is $\rho I + M_z$. In the ordered basis $1, z, \dots, z^{m_\rho-1}$, M_z has zero diagonal and trace zero. Thus its full trace contribution is $m_\rho \rho$. No generalized eigenvector has been deleted.

Taking the trace directly in (BF.16), and using (BF.19), yields the exact row constraint

$$\boxed{2 \mathfrak{h}_{n+1} \operatorname{Re} \mathbf{c}_n = \sum_{\rho \in Z} m_\rho (2 \operatorname{Re} \rho - 1).} \quad (\text{BF.24})$$

Define the reflection $\iota \rho = 1 - \bar{\rho}$. A reflection-stable packet means that $\iota Z = Z$ and that $m_{\iota \rho} = m_\rho$ at every member. For such a packet the contributions of every two-point orbit in (BF.23) cancel. At a fixed point of ι , $\operatorname{Re} \rho = 1/2$, so its contribution is zero. Since $\mathfrak{h}_{n+1} > 0$,

$$\begin{aligned} \operatorname{tr}(G_n^{-1} W_n) &= 0, \quad \operatorname{Re} \mathbf{c}_n = 0, \\ \lambda_{n,+} &= \epsilon_n = -\lambda_{n,-}, \\ \epsilon_n &= \mathfrak{h}_{n+1} \sqrt{\mathbf{a}_n \mathbf{b}_n - (\operatorname{Im} \mathbf{c}_n)^2}. \end{aligned} \quad (\text{BF.25})$$

If $W_n \neq 0$, the two nonzero eigenvalues are ϵ_n and $-\epsilon_n$, each once. If $\epsilon_n = 0$, the self-adjoint operator $G_n^{-1} W_n$ has only the zero eigenvalue and hence is zero. Trace cancellation alone does not set the matrix W_n to zero.

The invariant-subspace projection argument

The next estimates hold for an arbitrary actual packet Z ; reflection stability will be used only for their final combination. Define

$$Z_+ = \{\rho \in Z : \operatorname{Re} \rho > 1/2\}, \quad Z_- = \{\rho \in Z : \operatorname{Re} \rho < 1/2\}.$$

Let

$$E_+ = \bigoplus_{\rho \in Z_+} \mathbb{C}[z]/(z^{m_\rho}), \quad E_- = \bigoplus_{\rho \in Z_-} \mathbb{C}[z]/(z^{m_\rho}). \quad (\text{BF.26})$$

The inclusions $I_\pm : E_\pm \rightarrow E_Z$ are the actual Chinese remainder inclusions, not coordinate projections chosen to make A normal. Explicitly, put $h_\rho = (t - \rho)^{m_\rho}$ and $H_\rho = h_Z/h_\rho$. Since H_ρ and h_ρ are

coprime, choose $b_\rho(t)$ with $H_\rho b_\rho \equiv 1 \pmod{h_\rho}$, and let e_ρ be the class of $H_\rho b_\rho$ in E_Z . It has residue one at the ρ -summand and zero at all other summands. The maps in (BF.26) are

$$I_\pm((u_\rho)_\rho) = \sum_{\rho \in Z_\pm} e_\rho u_\rho(t - \rho).$$

Changing the representative b_ρ changes nothing in E_Z . All original polynomial coordinates and every full local length remain in these maps.

Let $A_\pm$ be multiplication by $\rho + z$ on each local summand of $E_\pm$. The intertwining identities are

$$AI_\pm = I_\pm A_\pm. \quad (\text{BF.27})$$

In particular the images are invariant generalized eigenspaces, with their nilpotent parts unchanged.

Fix $n \geq 0$. For a nonzero $E_\pm$, define

$$G_\pm = I_\pm^* G_n I_\pm, \quad P_\pm = I_\pm G_\pm^{-1} I_\pm^* G_n. \quad (\text{BF.28})$$

The notation $G_\pm, P_\pm$ in this subsection refers to this fixed value of n . Since $I_\pm$ is injective and $G_n \succ 0, G_\pm \succ 0$. Multiplication gives $P_\pm^2 = P_\pm, P_\pm I_\pm = I_\pm$, and

$$P_\pm^* G_n = G_n P_\pm.$$

Thus $P_\pm$ is the G_n -orthogonal projection onto $\text{im } I_\pm$. When $E_\pm = \{0\}$, define $P_\pm = 0$ directly; no inverse of a nonpositive-dimensional Gram matrix is needed.

For nonzero $E_\pm$, the restricted defect is exact. By (BF.27),

$$I_\pm^* W_n I_\pm = A_\pm^* G_\pm + G_\pm A_\pm - G_\pm. \quad (\text{BF.29})$$

For nonzero $E_\pm$, cyclicity of the trace of the rectangular composites, followed by (BF.29), gives

$$\begin{aligned} \text{tr}(P_\pm G_n^{-1} W_n) &= \text{tr}(G_\pm^{-1} I_\pm^* W_n I_\pm) \\ &= 2 \text{Re tr } A_\pm - \dim E_\pm \\ &= 2 \sum_{\rho \in Z_\pm} m_\rho (\text{Re } \rho - 1/2). \end{aligned} \quad (\text{BF.30})$$

For $E_\pm = \{0\}$, the first and last expressions in (BF.30) are both zero and remain equal; the inverse-containing expression is not invoked. The last trace evaluation again uses $\rho I + M_z$ on each full local summand.

The rank-two factorization gives a stronger estimate for these compressed traces than a generic rank bound would give. Indeed, the Hermitian form $v^* W_n v$ is the pullback under $v \mapsto (r_n v, r_{n+1} v)$ of

$$\mathfrak{h}_{n+1} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (\text{BF.31})$$

This form has one positive and one negative direction. To verify the inherited inertia bound directly, a positive definite subspace for W_n must map injectively under $v \mapsto (r_n v, r_{n+1} v)$; otherwise its kernel would contain a nonzero vector on which the form is zero. Its image is a positive definite subspace of (BF.31), so has dimension at most one. The same reasoning applies to a negative definite subspace. Therefore $G_n^{-1} W_n$, self-adjoint in the G_n metric, has at most one strictly positive and at most one strictly negative eigenvalue.

Use $\lambda_{n,+}$ and $\lambda_{n,-}$ from (BF.21). Let Π_+ and Π_- be their G_n -orthogonal spectral projections when the corresponding eigenvalue is strictly positive or strictly negative. When a sign is absent, set both its eigenvalue and its projection equal to zero. The finite-dimensional spectral theorem gives

$$G_n^{-1} W_n = \lambda_{n,+} \Pi_+ + \lambda_{n,-} \Pi_-, \quad \text{rank } \Pi_\pm \leq 1. \quad (\text{BF.32})$$

All remaining spectral directions have eigenvalue zero.

Let P be any G_n -orthogonal projection. For either of the nonzero projections $\Pi_\pm$, $\Pi_\pm P \Pi_\pm$ acts on a one-dimensional subspace as a positive contraction. Positivity follows because

$$\langle x, \Pi_\pm P \Pi_\pm x \rangle_{G_n} = \langle P \Pi_\pm x, P \Pi_\pm x \rangle_{G_n} \geq 0,$$

and the upper bound follows from $\|P \Pi_\pm x\|_{G_n} \leq \|\Pi_\pm x\|_{G_n}$. Its eigenvalue belongs to $[0, 1]$. Cyclicity and $\Pi_\pm^2 = \Pi_\pm$ then prove

$$0 \leq \text{tr}(P \Pi_\pm) = \text{tr}(\Pi_\pm P \Pi_\pm) \leq 1. \quad (\text{BF.33})$$

The equality and both inequalities also hold when $\Pi_\pm = 0$. Combining (BF.32) and (BF.33) gives

$$\lambda_{n,-} \leq \text{tr}(P G_n^{-1} W_n) \leq \lambda_{n,+}. \quad (\text{BF.34})$$

For example the upper bound is $\lambda_{n,+} \text{tr}(P \Pi_+) + \lambda_{n,-} \text{tr}(P \Pi_-) \leq \lambda_{n,+}$; the lower bound follows with the two signs interchanged.

Apply (BF.34) to the actual projections P_+ and P_- in (BF.30). For every packet and every $n \geq 0$, this proves

$$\boxed{\begin{aligned} \lambda_{n,+} &\geq 2 \sum_{\rho \in Z_+} m_\rho (\text{Re } \rho - 1/2), \\ -\lambda_{n,-} &\geq 2 \sum_{\rho \in Z_-} m_\rho (1/2 - \text{Re } \rho). \end{aligned}} \quad (\text{BF.35})$$

These estimates do not require reflection stability or semisimplicity.

For a reflection-stable packet, reflection bijects Z_+ and Z_- , preserves multiplicity, and negates $\text{Re } \rho - 1/2$. Thus the two positive distance sums in (BF.35) are equal. Equation (BF.25) identifies the two control eigenvalues with ϵ_n and $-\epsilon_n$. Their combination is the claimed multiplicity boundary floor:

$$\boxed{\begin{aligned} \epsilon_n &= \mathfrak{h}_{n+1} \sqrt{\mathfrak{a}_n \mathfrak{b}_n - (\text{Im } \mathfrak{c}_n)^2} \\ &\geq \sum_{\rho \in Z} m_\rho \left| \text{Re } \rho - \frac{1}{2} \right| \quad (n \geq 0). \end{aligned}} \quad (\text{BF.36})$$

The restriction traces in (BF.30) show explicitly why complete multiplicities occur. The proof includes every generalized eigenspace and uses its induced metric. No normality of A , deletion of nilpotent jets, or positivity of the arithmetic Weil form has entered.

Exact consequences for a putative off-line zero

An individual eigenvector gives a useful comparison. If $0 \neq v \in E_Z$ satisfies $Av = \rho v$, then directly from (BF.10),

$$v^* W_n v = (2 \text{Re } \rho - 1) v^* G_n v. \quad (\text{BF.37})$$

Such a vector exists in every local summand: the class $z^{m_\rho-1}$, embedded by e_ρ , is nonzero and is annihilated by $A - \rho I$. By (BF.18), (BF.37) gives $\epsilon_n \geq |2 \text{Re } \rho - 1|$. Equation (BF.36) includes the additional constraints from all other roots and their multiplicities.

For a reflected off-line pair $\{\rho, 1 - \bar{\rho}\}$, with complete multiplicity m at both points, put $\delta = \text{Re } \rho - 1/2 \neq 0$. The two points are distinct, and their contributions to the sum in (BF.36) give

$$\epsilon_n \geq 2m|\delta| = m|2 \text{Re } \rho - 1| \quad (n \geq 0). \quad (\text{BF.38})$$

The real structure and functional equation of the retained xi-function are $g(\bar{s}) = \overline{g(s)}$ and $g(1-s) = g(s)$. Composition of a local Taylor expansion with conjugation or the affine map $s \mapsto 1-s$ preserves its first nonzero degree, so these symmetries preserve the complete vanishing multiplicities.

Consequently, when the four points $\rho, \bar{\rho}, 1 - \rho, 1 - \bar{\rho}$ are distinct, they form a reflection-stable packet with the same complete multiplicity m . Their exact bound is

$$\epsilon_n \geq 4m|\delta| = 2m|2\operatorname{Re} \rho - 1| \quad (n \geq 0). \quad (\text{BF.39})$$

For a collapsed symmetry orbit, only the actual distinct points with their actual multiplicities are counted. Formula (BF.38) already treats the two-point reflection orbit.

In particular, a reflection-stable actual packet containing an off-line root forces a strictly positive right side of (BF.36) at every finite level. Then W_n is nonzero, indefinite, and of rank exactly two, with generalized eigenvalues $\epsilon_n, -\epsilon_n$. The strict positivity follows from the packet sum; both signs and exact rank follow from (BF.25) and (BF.16). For an arbitrary packet without reflection stability, the valid statements are instead the separate inequalities (BF.35); its trace need not vanish.

This lower bound is on the relative expression containing G_n^{-1} . The absolute monotonicity (BF.17) does not supply an upper bound on that inverse-weighted expression. The homotopy freedom in A1771, lines 4432–4475, does not alter this conclusion. Retain the Fourier transform

$$(\mathcal{F}\phi)(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx.$$

It restricts to an isomorphism $V \rightarrow V$: it preserves evenness and the Schwartz space, and interchanges evaluation at zero with the integral, so preserves the two zero moments in (BF.01). Its inverse on V is the inverse Fourier transform with the same convention. For the two-leg differential and its cycle map,

$$\begin{aligned} d_{\text{leg}} : V \oplus V &\longrightarrow \mathcal{B}, & d_{\text{leg}}(v, w) &= \Theta(v - \mathcal{F}w), \\ \gamma : V &\longrightarrow V \oplus V, & \gamma(v) &= (v, \mathcal{F}^{-1}v), \end{aligned}$$

one has $d_{\text{leg}}\gamma = 0$ by direct substitution. For a linear map $L : E_Z \rightarrow V$, adding γL to a primitive therefore leaves B_n unchanged, and hence leaves W_n unchanged for fixed R_n . The support transition described above likewise records where the boundary becomes a supported zero while retaining the pairing that produced (BF.16).

There is also an exact tensor obstruction independent of the rank-two argument. For an integer $k \geq 1$, keep

$$A_k = \sum_{j=1}^k I^{\otimes(j-1)} \otimes A \otimes I^{\otimes(k-j)} \quad \text{on } E_Z^{\otimes k}. \quad (\text{BF.40})$$

For any positive definite metric H on this same tensor space, set $W_{k,H} = A_k^* H + H A_k - kH$. The nonzero tensor $v^{\otimes k}$ of the eigenvector above satisfies $A_k v^{\otimes k} = k\rho v^{\otimes k}$. Consequently

$$\frac{(v^{\otimes k})^* W_{k,H} v^{\otimes k}}{(v^{\otimes k})^* H v^{\otimes k}} = k(2\operatorname{Re} \rho - 1). \quad (\text{BF.41})$$

The denominator is strictly positive. Thus the least two-sided excess for $H, W_{k,H}$ is at least $k|2\operatorname{Re} \rho - 1|$. In particular this applies to the actual orbit metric of A1721, lines 4102–4139,

$$H_{k,T} = \int_0^T e^{-kt} \exp(tA_k)^* G_n^{\otimes k} \exp(tA_k) dt, \quad T > 0.$$

It is positive definite because the integrand has positive quadratic form on every nonzero vector at every t ; $\exp(tA_k)$ is invertible. The same time parameter is used in every factor, and the full weight e^{-kt} remains. Changing T cannot change the Rayleigh value (BF.41).

For the empty packet, $h_Z = 1$, $E_Z = \{0\}$, and every packet sum is zero. Define its control excess to be zero. Then the boundary-floor assertion is the equality $0 = 0$. The extension row $[t^{d-1}]R_Z$ and the positive-dimensional Gram inverses above are not invoked in this empty case.

Equations (BF.35) and (BF.36) are lower bounds proved on the actual arithmetic objects. They neither assert an off-line zero nor provide the arithmetic upper estimate needed to exclude one. The source explicitly retains that missing estimate in A1721, line 4259, and A1771, line 4890. Finite regression checks, the stated integer-sum tail bound, and absolute Gram decrements do not supply it.

Chapter 9

The original symmetric frontier and determinant growth

Arithmetic symmetric frontier cost and determinant growth

This section proves finite restrictions on the spectral positions of the original arithmetic packet. It retains the actual theta source, the measure $|2\xi/h|^2 dy/(2\pi)$, all zero multiplicities, and the unscaled orbit coordinates. No purity assertion is used. Stars denote conjugate transpose in the displayed finite coefficient bases, and Hilbert adjoint for maps into the source Hilbert space. For Hermitian matrices, $U \preceq V$ means that $V - U$ is positive semidefinite. The one-factor quotient coefficient basis is always $[1]_h, [t]_h, \dots, [t^{d-1}]_h$, with its specified tensor and orbit inclusions below.

Authoritative source pins. The retained construction is in the original messages A1829 and A1859 of `audit_segment_U0041_U0054.md`. The line numbers here are literal lines in that source, whose SHA-256 is recorded in the accompanying audit. A1829, lines 4956–5071, specifies the source, the arithmetic function, the full-jet map, and its exact polynomial kernel; lines 5073–5169 give the canonical minimum and its metric; lines 5383–5561 give the joint tensor source and actual next relation layer. A1859, lines 5693–5874, gives the unscaled polynomial coordinates, the relation isomorphism, and the exact metric update; lines 5876–6040 give the recurrence and finite lowering bound; lines 6042–6197 give the signed symmetric summand and its unscaled orbit factors. The aggregate inequalities proved below are new consequences of those retained objects.

The retained arithmetic source and its canonical finite metric

Keep the original operator, theta map, and arithmetic function

$$\begin{aligned} D &= -x\partial_x, & \Theta\phi(x) &= \sum_{m \neq 0} \phi(mx), \\ \phi_*(x) &= (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, & f_0 &= \Theta\phi_*, \\ g(s) &= \mathcal{M}f_0(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s). \end{aligned} \tag{FD.1}$$

Here $\mathcal{M}F(s) = \int_0^\infty F(x)x^{s-1} dx$, on its original source domain. Choose a nonempty finite packet of actual zeros and retain the full order m_ρ at every selected centre:

$$\begin{aligned} h(t) &= \prod_{\rho \in Z} (t - \rho)^{m_\rho}, & d &= \deg h, \\ E_h &= \mathbb{C}[t]/(h), & v_h &= [g/h]_h \in E_h^\times. \end{aligned} \tag{FD.2}$$

The bracket $[\cdot]_h$ means the complete local Taylor jets modulo h ; it includes every nilpotent jet at a multiple zero. Since each selected order is the full order of that zero, g/h is nonzero at every selected centre, proving that v_h is a unit in the finite quotient. Let A_h be multiplication by $[t]_h$ on E_h .

The retained actual source inverse supplies F_h with

$$h(D)F_h = f_0, \quad \mathcal{M}F_h = g/h, \quad \mathcal{T}_h(P) = P(D)F_h. \quad (\text{FD.3})$$

Its exact identities, retained from the original source construction, are

$$\mathcal{M}\mathcal{T}_h(P) = (g/h)P, \quad \mathcal{T}_h(hP) = \Theta(P(D)\phi_*), \quad J_h\mathcal{T}_h(P) = v_h[P]_h. \quad (\text{FD.4})$$

Here $J_h F = [\mathcal{M}F]_h$ is the full Mellin-jet map on the original analytic source. The second identity gives an actual theta primitive, and the last identity includes the arithmetic coefficient v_h . The map $\mathcal{T}_h$ is linear and injective: a zero image has identically zero Mellin transform $(g/h)P$, which forces $P = 0$.

The Hilbert observation remains $\mathcal{H} = L^2((0, \infty), dx)$. The exact change of variables and Fourier map

$$F \mapsto (y \mapsto e^{y/2}F(e^y)) \mapsto \left(t \mapsto \int_{\mathbb{R}} e^{y/2}F(e^y)e^{ity} dy \right)$$

have target measure $dt/(2\pi)$ and composite $F \mapsto \mathcal{M}F(1/2 + it)$. Consequently the arithmetic polynomial measure is exactly

$$d\nu_h(y) = \left| \frac{g(\frac{1}{2} + iy)}{h(\frac{1}{2} + iy)} \right|^2 \frac{dy}{2\pi}. \quad (\text{FD.5})$$

At selected zeros on the integration line, the quotient is evaluated as the entire function g/h . The finiteness of every polynomial moment can also be checked directly from the retained factors. If $\eta(s) = \sum_{n \geq 1} (-1)^{n-1} n^{-s}$, its alternating coefficient partial sums have absolute value at most one. Partial summation gives $|\eta(s)| \leq |s|/\Re s$ for $\Re s > 0$. The identity $\eta(s) = (1 - 2^{1-s})\zeta(s)$, first obtained by splitting the absolutely convergent series and then continued analytically, implies

$$|\zeta(\frac{1}{2} + iy)| \leq \frac{2|\frac{1}{2} + iy|}{\sqrt{2} - 1},$$

because $|1 - 2^{1/2 - iy}| \geq \sqrt{2} - 1$. Fix $0 < \theta < \pi/2$. Rotating the Euler gamma integral through angle $\text{sgn}(y)\theta$ gives

$$|\Gamma(\frac{1}{4} + iy/2)| \leq \Gamma(\frac{1}{4})(\cos \theta)^{-1/4} e^{-\theta|y|/2}.$$

For $y = 0$, the same bound follows from the unrotated integral. The rotation follows by integrating $z^{1/4 + iy/2 - 1} e^{-z}$ around a truncated sector using its continuous logarithm there. The small circular arc tends to zero as its radius to the power $1/4$; the large arc tends to zero because $\cos \theta > 0$. Taking absolute values on the rotated ray yields the displayed bound. Retaining the factors in (FD.1) therefore gives

$$|g(\frac{1}{2} + iy)|^2 \leq \frac{4\pi^{-1/2}\Gamma(\frac{1}{4})^2}{(\sqrt{2} - 1)^2 \sqrt{\cos \theta}} (\frac{1}{4} + y^2)^3 e^{-\theta|y|}.$$

Monicity of the fixed degree- d polynomial h gives $|h(\frac{1}{2} + iy)| \geq c_h(1 + |y|)^d$ for some $c_h > 0$ outside a sufficiently large interval: divide its value by $(\frac{1}{2} + iy)^d$, whose quotient tends to one. On the remaining compact interval, the entire function g/h is bounded. These estimates prove finiteness of every moment of ν_h . The measure is positive away from a discrete set. Therefore the norm of a nonzero polynomial is strictly positive: zero norm would make the polynomial vanish almost everywhere on a line and hence identically.

Fix $k \geq 1$, and let

$$\begin{aligned} E^{(k)} &= E_h^{\otimes k}, & A_k &= \sum_{j=1}^k A_{h,j}, & D^{(k)} &= \sum_{j=1}^k D_j, \\ \mathcal{P}_M^{(k)} &= \mathbb{C}[t_1, \dots, t_k]_{\deg \leq M}, & \mathcal{T}^{(k)}(P) &= P(D_1, \dots, D_k) F_h^{\otimes k}, \\ \mathcal{J}(P) &= v_h^{\otimes k}[P]_{(h(t_1), \dots, h(t_k))}. \end{aligned} \quad (\text{FD.6})$$

All tensor generators in this section are additive sums, as displayed in (FD.6). The multiplicative symmetric power of a linear map is a different construction and is not used here.

For $M \geq k(d-1)$, the map $\mathcal{J}$ on $\mathcal{P}_M^{(k)}$ is onto: each reduced monomial has each coordinate degree below d , hence total degree at most $k(d-1)$, and multiplication by $v_h^{\otimes k}$ is invertible. Its kernel and its actual theta image are

$$\mathcal{J}_M = \mathcal{P}_M^{(k)} \cap (h(t_1), \dots, h(t_k)), \quad \mathcal{B}_M = \mathcal{T}^{(k)} \mathcal{J}_M. \quad (\text{FD.7})$$

Successive monic division in the variables expresses every polynomial in the kernel as $\sum_j h(t_j) P_j$. Under $\mathcal{T}^{(k)}$, each summand is an actual theta boundary in factor j , by (FD.4). Its cochain primitive has coefficient $(-1)^{j-1}$; the tensor differential has the same coefficient, so their product gives the displayed positive summand $+h(t_j) P_j$.

Let $R_M : E^{(k)} \rightarrow \mathcal{T}^{(k)} \mathcal{P}_M^{(k)}$ be the unique least-norm lift of the full jet, using the product measure induced by (FD.5), and write $G_M = R_M^* R_M$, $\mathcal{C}_M = G_M^{-1}$. The existence, uniqueness, and inverse are also obtained directly below. The minimum is orthogonal to the specified $\mathcal{B}_M$, and its full jet is identity. Its arithmetic class is that of the original product source lift with the specified jet, since all admissible changes are the actual boundaries in (FD.7).

Start with the monic, unscaled orthogonal polynomials and their actual norms

$$p_0 = 1, \quad p_n = t^n - \text{proj}_{\mathbb{C}[t]_{\leq n-1}} t^n, \quad \omega_n = \int |p_n(\tfrac{1}{2} + iy)|^2 d\nu_h(y) > 0. \quad (\text{FD.8})$$

For a multi-index $\alpha \in \mathbb{N}_0^k$, set

$$p_\alpha = \prod_{j=1}^k p_{\alpha_j}(t_j), \quad \omega_\alpha = \prod_{j=1}^k \omega_{\alpha_j}, \quad z_\alpha = \mathcal{J}(p_\alpha). \quad (\text{FD.9})$$

Let T_M have columns $\mathcal{T}^{(k)} p_\alpha$ for $|\alpha| \leq M$, let $J_M = [z_\alpha]_{|\alpha| \leq M}$, and let $\Omega_{\leq M} = \text{diag}_{|\alpha| \leq M} \omega_\alpha$. Product orthogonality gives $T_M^* T_M = \Omega_{\leq M}$. Surjectivity of J_M implies

$$\begin{aligned} \mathcal{C}_M &= J_M \Omega_{\leq M}^{-1} J_M^* = \sum_{|\alpha| \leq M} \frac{z_\alpha z_\alpha^*}{\omega_\alpha} > 0, \\ G_M &= \mathcal{C}_M^{-1}, \quad R_M = T_M \Omega_{\leq M}^{-1} J_M^* G_M. \end{aligned} \quad (\text{FD.10})$$

Indeed $J_M \Omega_{\leq M}^{-1} J_M^* G_M = I$. Any other coefficient vector with jet u is $\Omega_{\leq M}^{-1} J_M^* G_M u + w$, where $J_M w = 0$. Its cross term with the first vector is $w^* J_M^* G_M u = 0$, so its squared norm is exactly $u^* G_M u + w^* \Omega_{\leq M} w$. This proves the unique minimum, its orthogonality to $\mathcal{B}_M$, and $R_M^* R_M = G_M$.

The actual next relation and its recurrence bound

For $|\beta| = M+1$, define the frontier matrices

$$\Omega = \text{diag}_{|\beta|=M+1} \omega_\beta, \quad F = [z_\beta]_{|\beta|=M+1}, \quad T_+ = [\mathcal{T}^{(k)} p_\beta]_{|\beta|=M+1}. \quad (\text{FD.11})$$

There are $r = \binom{M+k}{k-1}$ columns. They are orthogonal to the entire degree- M source image. If $P_{\mathcal{B}_M}$ denotes orthogonal projection onto the actual boundary space, put

$$\mathcal{E}_M = (1 - P_{\mathcal{B}_M})\mathcal{B}_{M+1}, \quad b_M = T_+ - R_M F : \mathbb{C}^r \longrightarrow \mathcal{E}_M. \quad (\text{FD.12})$$

The full jet of b_M is $F - F = 0$, and its two terms are orthogonal to $\mathcal{B}_M$, so its range is contained in $\mathcal{E}_M$. Its highest orthogonal polynomial component is T_+ , proving injectivity. Conversely, extract the highest component of a member of $\mathcal{E}_M$ and subtract its corresponding b_M -image. The remainder is simultaneously in $\mathcal{B}_M$ and orthogonal to $\mathcal{B}_M$, and is therefore zero. This proves that b_M is an isomorphism and constructs its inverse.

The exact Gram and cross-pairing are

$$H_M^{\text{rel}} := b_M^* b_M = \Omega + F^* G_M F, \quad b_M^* R_M = -F^* G_M. \quad (\text{FD.13})$$

The superscript distinguishes this relation Gram from the Hermitian control matrix introduced later. The projection of R_M onto the new layer and the updated representative are

$$Y_M = -b_M (H_M^{\text{rel}})^{-1} F^* G_M, \quad R_{M+1} = R_M - Y_M. \quad (\text{FD.14})$$

These corrections are actual boundaries. In particular, the full jet and the original arithmetic class stay fixed. The quotient-layer map

$$\mathcal{B}_{M+1}/\mathcal{B}_M \longrightarrow \mathcal{E}_M, \quad [z] \mapsto (1 - P_{\mathcal{B}_M})z$$

is an isomorphism, whose inverse is the inclusion followed by quotient. Its further map into the quotient by $\mathcal{B}_{M+1}$ is zero. Write λ_M for the specified original joint support label recording the relation space $\mathcal{B}_M$, and write 0 for the zero amplitude in that support fibre. On the original supported reconstruction this is $(\lambda_M, u) \mapsto (\lambda_{M+1}, 0)$. For $u \neq 0$ the preceding class is nonzero. The final class is the supported zero of that fibre; it is distinct from τ , the original external-absence element. The notation $e = 0^\bullet$ denotes the nonabsent represented scalar zero in the original scalar reconstruction. Thus the relation Gram in (FD.13) measures the actual relations becoming represented zero at the next support.

Orthogonal projection and the finite sum in (FD.10) give

$$G_{M+1} = G_M - G_M F (\Omega + F^* G_M F)^{-1} F^* G_M, \quad (\text{FD.15})$$

$$\mathcal{C}_{M+1} - \mathcal{C}_M = F \Omega^{-1} F^*.$$

The second equality also follows simply by adding the degree- $M + 1$ terms to (FD.10); it does not require a limiting argument.

Set $p_{-1} = 0$. Multiplication by t on the integration line has adjoint multiplication by $1 - t$. Orthogonality and monicity therefore give

$$tp_n = p_{n+1} + b_n p_n - a_n p_{n-1}, \quad a_n = \omega_n / \omega_{n-1} \quad (n \geq 1), \quad a_0 = 0, \quad (\text{FD.16})$$

$$b_n = \frac{\int (\frac{1}{2} + iy) |p_n(\frac{1}{2} + iy)|^2 d\nu_h(y)}{\omega_n}, \quad \Re b_n = \frac{1}{2}.$$

For $n \geq 1$, pairing against a polynomial of degree below $n - 1$ in the expansion of tp_n vanishes by the adjoint identity. Pairing against p_{n-1} gives $-\omega_n / \omega_{n-1}$, and pairing against p_n gives the displayed b_n . For $n = 0$, monicity gives $tp_0 = p_1 + b_0 p_0$, and the term $a_0 p_{-1} = 0$ is exactly the declared zero term. This proves every coefficient in the recurrence. The minus sign and the possibly nonzero imaginary part of b_n are retained.

Define

$$w_\beta = \sum_{j: \beta_j > 0} a_{\beta_j} z_{\beta - e_j}, \quad E_+ = [w_\beta]_{|\beta|=M+1}. \quad (\text{FD.17})$$

Here e_j is the standard multi-index vector; it is not the supported scalar $e = 0^\bullet$. The full arithmetic jet is a module map, so $\mathcal{J}((\sum_j t_j)P) = A_k \mathcal{J}(P)$. Inserting (FD.16) into (FD.10) proves

$$\mathcal{Z}_M := A_k \mathcal{C}_M + \mathcal{C}_M A_k^* - k \mathcal{C}_M = F \Omega^{-1} E_+^* + E_+ \Omega^{-1} F^*. \quad (\text{FD.18})$$

To verify the cancellation explicitly, for every internal edge $\alpha \rightarrow \alpha + e_j$, the raising term has coefficient $1/\omega_\alpha$, while the lowering term has coefficient $-a_{\alpha_j+1}/\omega_{\alpha+e_j} = -1/\omega_\alpha$. Their adjoints cancel as well. The diagonal contribution is $2 \sum_j \Re b_{\alpha_j} = k$. The only uncanceled edges have $|\alpha| = M$ and $|\alpha + e_j| = M + 1$, and their coefficients are exactly those in the right side of (FD.18).

For completeness, the arithmetic derivative defect has the same actual relation coordinates. Set

$$B_M^{\text{der}} = D^{(k)} R_M - R_M A_k, \quad C_M^{\text{rel}} = (1 - P_{\mathcal{B}_M}) B_M^{\text{der}}.$$

Its full jet is zero, and its highest polynomial component is $T_+ \Omega^{-1} E_+^* G_M$. The latter assertion follows from the coefficient of p_β in $(\sum_j t_j) R_M$: it is $\sum_{j: \beta_j > 0} z_{\beta - e_j}^* G_M / \omega_{\beta - e_j} = w_\beta^* G_M / \omega_\beta$. The layer isomorphism (FD.12) therefore proves

$$\begin{aligned} C_M^{\text{rel}} &= b_M \Omega^{-1} E_+^* G_M, \\ W_M &:= A_k^* G_M + G_M A_k - k G_M \\ &= G_M \mathcal{Z}_M G_M \\ &= -(Y_M^* C_M^{\text{rel}} + (C_M^{\text{rel}})^* Y_M). \end{aligned} \quad (\text{FD.19})$$

The last equality follows by substituting (FD.13) and (FD.14); in particular its minus sign is retained.

The recurrence constant is

$$\Gamma_{k,M} = \max_{|\alpha|=M} \sum_{j=1}^k (a_{\alpha_j+1} + (k-1)a_{\alpha_j}) > 0. \quad (\text{FD.20})$$

Strict positivity follows from $a_{n+1} > 0$ for every $n \geq 0$. We now prove its complete lowering estimate.

For coefficient calculations only, use the exact coordinate map that divides the basis vector p_α by $\sqrt{\omega_\alpha}$. Its inverse multiplies by $\sqrt{\omega_\alpha}$; the original polynomials, their norms, and the measure are unchanged. The degree- (-1) coefficient space is by definition $\{0\}$; in particular every lowering map from degree zero has that zero codomain. This retains the admitted endpoint $d = 1, M = 0$. In these coefficient spaces let R_j be the raising map $e_\alpha \mapsto \sqrt{a_{\alpha_j+1}} e_{\alpha+e_j}$, and let R_j^* be its lowering adjoint. Put

$$Q_M^{\text{jet}} = [z_\alpha / \sqrt{\omega_\alpha}]_{|\alpha|=M}, \quad B^{\text{inc}} = (\sum_j R_j)^*|_{M+1 \rightarrow M}.$$

The retained coefficients in (FD.17) give exactly $E_+ \Omega^{-1/2} = Q_M^{\text{jet}} B^{\text{inc}}$. Raising and lowering in distinct coordinates commute. Hence, on the degree- M coefficient space,

$$B^{\text{inc}} (B^{\text{inc}})^* = \text{diag}_{|\alpha|=M} \sum_j (a_{\alpha_j+1} - a_{\alpha_j}) + L^* L, \quad L = \sum_j R_j^*|_{M \rightarrow M-1}.$$

For every coefficient vector c , Cauchy-Schwarz gives $\|Lc\|^2 \leq k \sum_j \|R_j^* c\|^2$. Thus

$$L^* L \preceq k \text{diag}_{|\alpha|=M} \sum_j a_{\alpha_j}, \quad B^{\text{inc}} (B^{\text{inc}})^* \preceq \Gamma_{k,M} I.$$

Multiplication by Q_M^{jet} and its adjoint, followed by inclusion of the positive degree- M sum in (FD.10), proves

$$E_+ \Omega^{-1} E_+^* \preceq \Gamma_{k,M} Q_M^{\text{jet}} (Q_M^{\text{jet}})^* \preceq \Gamma_{k,M} \mathcal{C}_M. \quad (\text{FD.21})$$

Exact symmetric restriction with every orbit factor

Let P_π be ordinary factor permutation. The relevant module is

$$E_k^{\text{sym}} = (E_h^{\otimes k})^{S_k}, \quad N = \dim E_k^{\text{sym}} = \binom{k+d-1}{d-1}.$$

It is the actual top-degree summand selected on the original cochain tensor product by $S_k = (1/k!) \sum_{\pi \in S_k} \text{sgn}(\pi) T_\pi$, where T_π is permutation with its Koszul sign. In top degree $T_\pi = \text{sgn}(\pi) P_\pi$, so the two signs cancel. The numerator squares to $k!$ times itself, proving the idempotent with its literal coefficient $1/k!$. In particular, every repeated tensor $v^{\otimes k}$ remains in this summand.

Use the quotient basis formed by the unscaled orbit sums of $[t^{a_1}] \otimes \cdots \otimes [t^{a_k}]$, $0 \leq a_i < d$. Let I_k be its coefficient inclusion in the full tensor basis. If n_j counts factors with exponent j , its orbit size is $k! / \prod_{j=0}^{d-1} n_j!$. The orbit supports are disjoint, so

$$D_k = I_k^* I_k = \text{diag} \left(\frac{k!}{\prod_{j=0}^{d-1} n_j!} \right), \quad L_k = D_k^{-1} I_k^*, \quad L_k I_k = I. \quad (\text{FD.22})$$

The projector onto invariant coefficient vectors is $\Pi_k = I_k L_k$. Permutations preserve the source, the relation ideal, the product measure, and $v_h^{\otimes k}$. Uniqueness of the minimum therefore gives equivariance of R_M , and consequently of G_M and $\mathcal{C}_M$. The exact restricted matrices are

$$G_M^{\text{sym}} = I_k^* G_M I_k, \quad \mathcal{C}_M^{\text{sym}} = L_k \mathcal{C}_M L_k^*, \quad A_k I_k = I_k A_k^{\text{sym}}. \quad (\text{FD.23})$$

The first two matrices are inverse. For example,

$$G_M^{\text{sym}} \mathcal{C}_M^{\text{sym}} = I_k^* G_M \Pi_k \mathcal{C}_M L_k^* = I_k^* \Pi_k L_k^* = I,$$

because both G_M and $\mathcal{C}_M$ commute with Π_k , their product is identity, $\Pi_k L_k^* = L_k^*$, and $I_k^* L_k^* = (L_k I_k)^* = I$.

In the source frontier let Q_+ be the coefficient inclusion matrix whose columns are the 0/1 orbit indicators in the ordered basis $(p_\beta)_{|\beta|=M+1}$. Thus each column represents the unscaled orbit sum of those p_β . If $m_j(\beta)$ counts the entries of β equal to j , the corresponding source orbit norm is exactly

$$\omega_{[\beta]} = \frac{k!}{\prod_{j \geq 0} m_j(\beta)!} \prod_{i=1}^k \omega_{\beta_i}. \quad (\text{FD.24})$$

This is a sum of equal squared norms of pairwise orthogonal distinct product polynomials. Define the actual restricted frontier matrices

$$\Omega_s = Q_+^* \Omega Q_+, \quad F_s = L_k F Q_+, \quad E_s = L_k E_+ Q_+. \quad (\text{FD.25})$$

Thus Ω_s is the positive diagonal matrix with entries (FD.24); none of its orbit factors are suppressed.

The source invariant projector is $\Pi_+ = Q_+ (Q_+^* Q_+)^{-1} Q_+^*$. Since Ω is constant on each orbit, it commutes with Π_+ , and direct calculation in one orbit gives

$$Q_+ \Omega_s^{-1} Q_+^* = \Omega^{-1} \Pi_+.$$

Indeed, on an orbit of size m and weight ω , the left side is the all-ones matrix divided by $m\omega$; this is precisely $\omega^{-1} \Pi_+$ there. Equivariance implies $L_k F = L_k F \Pi_+$ and $L_k E_+ = L_k E_+ \Pi_+$: averaging any source vector before applying an equivariant map has the same invariant component as applying the map first. Congruence of (FD.15), (FD.18), and (FD.21) therefore gives, with

$$\begin{aligned} \mathcal{C} &= \mathcal{C}_M^{\text{sym}}, \quad G = G_M^{\text{sym}} = \mathcal{C}^{-1}, \quad A = A_k^{\text{sym}}, \\ \Delta \mathcal{C} &= \mathcal{C}_{M+1}^{\text{sym}} - \mathcal{C}_M^{\text{sym}}, \quad \mathcal{Z} = A \mathcal{C} + \mathcal{C} A^* - k \mathcal{C}, \end{aligned} \quad (\text{FD.26})$$

the exact formulas

$$\begin{aligned}\Delta\mathcal{C} &= F_s\Omega_s^{-1}F_s^*, \\ \mathcal{Z} &= F_s\Omega_s^{-1}E_s^* + E_s\Omega_s^{-1}F_s^*, \\ E_s\Omega_s^{-1}E_s^* &\preceq \Gamma_{k,M}\mathcal{C}.\end{aligned}\tag{FD.27}$$

For the operator term, equivariance also gives $L_k A_k = A_k^{\text{sym}} L_k$, proving that the congruence of $A_k \mathcal{C}_M + \mathcal{C}_M A_k^* - k \mathcal{C}_M$ is exactly the $\mathcal{Z}$ in (FD.26).

The aggregate spectral-position restriction

Positive square roots below are the unique positive square roots of the retained positive matrices. They are used as explicit comparison maps; their inverses are retained, and neither the source measure nor the arithmetic metric is redefined. Set

$$\begin{aligned}X &= G^{1/2}F_s\Omega_s^{-1/2}, & Y &= G^{1/2}E_s\Omega_s^{-1/2}, \\ K = XX^* &= G^{1/2}\Delta\mathcal{C}G^{1/2}, & H &= G^{1/2}\mathcal{Z}G^{1/2} = XY^* + YX^*.\end{aligned}\tag{FD.28}$$

The last inequality of (FD.27) gives $YY^* \preceq \Gamma_{k,M}I$, and hence $Y^*Y \preceq \Gamma_{k,M}I$ on its own coefficient space. Here the latter statement follows from $\|Y\|_{\text{op}}^2 = \|YY^*\|_{\text{op}}$, or directly from the singular values. For the Frobenius norm $\|T\|_F^2 = \text{tr}(T^*T)$,

$$\|XY^*\|_F^2 = \text{tr}(X^*XY^*Y) \leq \Gamma_{k,M} \text{tr}(X^*X).$$

The norm of a matrix equals the norm of its adjoint, so the triangle inequality gives the proved upper estimate

$$\|H\|_F^2 \leq 4\Gamma_{k,M} \text{tr} K.\tag{FD.29}$$

List the roots of h , repeated with their original algebraic multiplicities, as $\rho_1, \dots, \rho_d$, and put

$$\delta_i = \Re \rho_i - \frac{1}{2}, \quad \eta_{\mathbf{n}} = \sum_{i=1}^d n_i \rho_i, \quad \mathbf{n} \in \mathbb{N}_0^d, \quad |\mathbf{n}| = k.\tag{FD.30}$$

The eigenvalues of A , with algebraic multiplicity, are exactly the $\eta_{\mathbf{n}}$. To prove this while retaining nilpotents, triangularize A_h by induction using an eigenvector and the induced operator on the quotient. In a basis with $A_h e_j = \rho_j e_j + \sum_{i < j} c_{ij} e_i$, applying the additive generator to a symmetric monomial of occupations $\mathbf{n}$ gives diagonal coefficient $\sum_j n_j \rho_j$. Every other term replaces an index j by $i < j$, reducing the sum of the indices. Ordering monomials by that sum, and resolving ties arbitrarily, makes the operator triangular with precisely the displayed diagonal. Each weak composition contributes once; equal eigenvalues remain repeated with their algebraic multiplicities. The triangular off-diagonal entries are retained, including those caused by nontrivial Jordan blocks.

Let $B = G^{1/2}AG^{-1/2}$. Substitution into (FD.26) gives $H = B + B^* - kI$. Unitary Schur triangularization gives $T = U^*BU$, upper triangular with diagonal entries $\eta_{\mathbf{n}}$. Such a triangularization is obtained inductively by taking a unit eigenvector as the first basis vector and applying the induction to the compression on its orthogonal complement. Unitary conjugation preserves the Frobenius norm. Its diagonal and off-diagonal entries give the exact identity

$$\|H\|_F^2 = \sum_{|\mathbf{n}|=k} (2\Re \eta_{\mathbf{n}} - k)^2 + 2 \sum_{i < j} |T_{ij}|^2 \geq 4 \sum_{|\mathbf{n}|=k} \left(\sum_i n_i \delta_i \right)^2.\tag{FD.31}$$

For $i < j$, the two corresponding entries of $T + T^* - kI$ are T_{ij} and $\overline{T_{ij}}$; this proves the factor two. Thus the retained nilpotent and nonnormal contributions cannot weaken the displayed lower bound.

We evaluate its sum exactly. For $d \geq 2$, the ordinary generating functions for weak compositions give

$$\begin{aligned} \sum_{|\mathbf{n}|=k} n_1 &= \binom{k+d-1}{d}, \\ \sum_{|\mathbf{n}|=k} n_1(n_1-1) &= 2 \binom{k+d-1}{d+1}, \\ \sum_{|\mathbf{n}|=k} n_1 n_2 &= \binom{k+d-1}{d+1}. \end{aligned} \tag{FD.32}$$

Indeed their generating functions are respectively

$$z(1-z)^{-d-1}, \quad 2z^2(1-z)^{-d-2}, \quad z^2(1-z)^{-d-2}.$$

These follow by differentiating the appropriate geometric-series factors. Extracting the coefficient of z^k gives (FD.32). Write $a = \binom{k+d-1}{d}$ and $b = \binom{k+d-1}{d+1}$. Then the square expands as

$$(a+2b) \sum_i \delta_i^2 + 2b \sum_{i<j} \delta_i \delta_j = (a+b) \sum_i \delta_i^2 + b \left(\sum_i \delta_i \right)^2.$$

Pascal's identity therefore proves

$$\begin{aligned} \mathcal{S}_{h,k} &:= \sum_{|\mathbf{n}|=k} \left(\sum_i n_i \delta_i \right)^2 \\ &= \binom{k+d}{d+1} \sum_i \delta_i^2 + \binom{k+d-1}{d+1} \left(\sum_i \delta_i \right)^2. \end{aligned} \tag{FD.33}$$

Binomial coefficients with smaller upper than nonnegative lower index are zero. For $d=1$, direct evaluation gives $k^2 \delta_1^2$; the right side of (FD.33) is $[(\binom{k+1}{2} + \binom{k}{2}) \delta_1^2 = k^2 \delta_1^2$, so the same formula holds in that case.

Combining (FD.29), (FD.31), and (FD.33) proves the aggregate arithmetic restriction

$$\boxed{\Gamma_{k,M} \operatorname{tr}(G_M^{\text{sym}}(\mathcal{C}_{M+1}^{\text{sym}} - \mathcal{C}_M^{\text{sym}})) \geq \mathcal{S}_{h,k}.} \tag{FD.34}$$

The trace in this formula is exactly $\operatorname{tr} K$, by cyclicity. All its matrices were defined from the actual measure, complete arithmetic jet unit, and unscaled orbit norms.

Coordinate invariance and determinant growth

The comparison maps have an exact coordinate-invariance statement. For invertible changes S in quotient coordinates and R in frontier coordinates, set

$$\begin{aligned} A' &= SAS^{-1}, & \mathcal{C}' &= S\mathcal{C}S^*, & G' &= S^{-*}GS^{-1}, \\ F'_s &= SF_sR, & E'_s &= SE_sR, & \Omega'_s &= R^*\Omega_sR. \end{aligned} \tag{FD.35}$$

Here $S^{-*} = (S^{-1})^*$. Cancellation of R, R^{-1} in the factorizations proves $\Delta\mathcal{C}' = S\Delta\mathcal{C}S^*$, and substitution in the generator identity gives $\mathcal{Z}' = S\mathcal{Z}S^*$. The lowering inequality is carried through the same congruence. Direct cyclic cancellation and determinant multiplicativity give

$$\begin{aligned} \operatorname{tr}(G'\Delta\mathcal{C}') &= \operatorname{tr}(G\Delta\mathcal{C}), \\ \frac{\det(\mathcal{C}' + \Delta\mathcal{C}')}{\det \mathcal{C}'} &= \frac{\det(\mathcal{C} + \Delta\mathcal{C})}{\det \mathcal{C}}. \end{aligned} \tag{FD.36}$$

Furthermore $U_S = (\mathcal{C}')^{-1/2} S \mathcal{C}'^{1/2}$ satisfies $U_S U_S^* = I$, and hence is unitary, with $H' = U_S H U_S^*$. This follows by substituting $\mathcal{C}' = S \mathcal{C} S^*$ and $\mathcal{Z}' = S \mathcal{Z} S^*$. Thus the exact trace, determinant ratio, and Frobenius quantity are preserved by the comparison-coordinate maps; no arithmetic normalization has been imposed.

The exact update and (FD.28) give

$$\frac{\det \mathcal{C}_{M+1}^{\text{sym}}}{\det \mathcal{C}_M^{\text{sym}}} = \det(I + K). \quad (\text{FD.37})$$

Indeed congruence by $\mathcal{C}^{-1/2} = G^{1/2}$ takes $\mathcal{C} + \Delta \mathcal{C}$ to $I + K$, and its determinant multiplier is $1/\det \mathcal{C}$. If $\kappa_1, \dots, \kappa_N \geq 0$ are the eigenvalues of K , expanding $\prod_j (1 + \kappa_j)$ gives $\det(I + K) \geq 1 + \sum_j \kappa_j = 1 + \text{tr } K$. Since $\Gamma_{k,M} > 0$ was proved in (FD.20), (FD.34) now proves

$$\boxed{\frac{\det \mathcal{C}_{M+1}^{\text{sym}}}{\det \mathcal{C}_M^{\text{sym}}} \geq 1 + \frac{\mathcal{S}_{h,k}}{\Gamma_{k,M}}.} \quad (\text{FD.38})$$

For any integers $M_1 > M_0 \geq k(d-1)$, multiplying the finitely many positive inequalities (FD.38) telescopes to

$$\boxed{\frac{\det \mathcal{C}_{M_1}^{\text{sym}}}{\det \mathcal{C}_{M_0}^{\text{sym}}} \geq \prod_{M=M_0}^{M_1-1} \left(1 + \frac{\mathcal{S}_{h,k}}{\Gamma_{k,M}} \right).} \quad (\text{FD.39})$$

This is a finite product over the specified actual support degrees. It requires no limit or unproved asymptotic estimate.

For a reflection-stable packet, the original reflection maps each zero ρ to $1 - \bar{\rho}$, retaining its full order. Each paired displacement δ_i is sent to $-\delta_i$, so $\sum_i \delta_i = 0$. Hence the exact coefficient becomes

$$\boxed{\mathcal{S}_{h,k} = \binom{k+d}{d+1} \sum_{i=1}^d (\Re \rho_i - \tfrac{1}{2})^2.} \quad (\text{FD.40})$$

In particular, the presence of any actual off-axis zero makes this coefficient strictly positive. It forces a positive relative frontier increment and the determinant lower bound at every admitted support degree, within the actual symmetric summand.

The exact signed eigenline witness

Let $v \neq 0$ be an actual eigenvector of A_h , with $A_h v = \rho v$. In the unscaled symmetric quotient coordinates put $u = L_k(v^{\otimes k})$, so $I_k u = v^{\otimes k}$. This vector is nonzero: choose a linear functional ℓ with $\ell v = 1$; then $\ell^{\otimes k}(v^{\otimes k}) = 1$. The additive generator satisfies $Au = k\rho u$. With the exact positive number $q = u^* G u$, put

$$x = \Omega_s^{-1/2} F_s^* G u, \quad y = \Omega_s^{-1/2} E_s^* G u. \quad (\text{FD.41})$$

The symmetric restriction of (FD.19) is $W = A^* G + G A - kG = G \mathcal{Z} G$. On the eigenvector,

$$u^* W u = k(2\Re \rho - 1)q.$$

On the other hand, (FD.27) and (FD.41) give $u^* G \mathcal{Z} G u = x^* y + y^* x = 2\Re(x^* y)$. Equating these two exact expressions proves the signed identity

$$\boxed{\Re(x^* y) = k(\Re \rho - \tfrac{1}{2})q.} \quad (\text{FD.42})$$

Thus this correlation has the retained sign of the actual real-part displacement. No absolute-value estimate was used in obtaining it.

Define the two directional costs and the actual symmetric maximum cost

$$\alpha = \frac{\|x\|^2}{q}, \quad \beta = \frac{\|y\|^2}{q}, \quad \lambda_{k,M}^{\text{sym}} = \max_{w \neq 0} \frac{w^* \Delta \mathcal{C} w}{w^* \mathcal{C} w} = \lambda_{\max}(K). \quad (\text{FD.43})$$

The last equality follows by the invertible substitution $w = \mathcal{C}^{-1/2}z$ in the Rayleigh quotient. The lowering estimate in (FD.27), evaluated at Gu , gives $\beta \leq \Gamma_{k,M}$, because $(Gu)^* \mathcal{C}(Gu) = q$. The same substitution into the maximum defining $\lambda_{k,M}^{\text{sym}}$ gives $\alpha \leq \lambda_{k,M}^{\text{sym}}$. Cauchy–Schwarz applied to (FD.42) gives

$$\alpha\beta \geq k^2(\Re \rho - \tfrac{1}{2})^2.$$

Consequently the actual arithmetic costs satisfy

$$\boxed{\Gamma_{k,M} \lambda_{k,M}^{\text{sym}} \geq k^2(\Re \rho - \tfrac{1}{2})^2.} \quad (\text{FD.44})$$

Every term in the signed identity and this bound is attached to the actual next original relation through (FD.12), (FD.19), and the exact symmetric restrictions.

Exact scope of the resulting restriction. Equations (FD.34)–(FD.40) and (FD.42)–(FD.44) prove the stated frontier and determinant restrictions for the original arithmetic family, including all multiplicities and nilpotent actions. The source explicitly records the unestimated quantity: A1829, line 5580, states that no uniform sublinear excess was established; A1859, line 6013, states that the finite bound’s constants were not bounded sublinearly in tensor degree; A1859, line 6327, states that the uniform sublinear estimate for the actual arithmetic family was not proved. No upper estimate incompatible with (FD.38) or (FD.44) is asserted here. These are proved necessary restrictions on an actual RH counterexample, not an assertion that their arithmetic values cannot satisfy them.

Chapter 10

Conormal images, arithmetic Jacobians, and complete symmetric trace fibres

The original conormal map, the arithmetic Jacobian, and complete symmetric trace fibres

Source pin and the unchanged arithmetic objects

The inherited source is the authoritative shared transcript, A1912 (chain 3986, segment lines 6410–7016) and A1959 (chain 4075, lines 7086–7810), together with the signed chain projector in A1859 (chain 3887, lines 6042–6098). These locators refer to the complete segment U0041–U0054; all 43 message bodies have been matched against the original transcript records. The calculations below extend the source’s actual morphisms. Their arithmetic coefficient object is the same one over the constructed absolute base $\mathfrak{b}_\tau = \text{Spec}\{\tau, 1\}$, with its structural square

$$\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array}$$

Here j is the original unital inclusion, $G(R) = \{\tau\} \sqcup \{a^\bullet : a \in R\}$, $e_R = 0_R^\bullet$, and $p_R(\tau) = 0$, $p_R(a^\bullet) = a$. A linear map is lifted by $a^\bullet \mapsto f(a)^\bullet$, $\tau \mapsto \tau$. Every coefficient map below has this lift on its original support fibre.

Retain the actual entire function

$$g(s) = 2\xi(s), \quad \xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s),$$

with the entire continuation in the inherited arithmetic construction. Its source is

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi \text{ even}, \phi(0) = 0, \hat{\phi}(0) = 0\}, \quad \hat{\phi}(\eta) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \eta} dx,$$

$$\mathcal{B} = \left\{ F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |(-x\partial_x)^a F(x)| < \infty \text{ for every } b \in \mathbb{Z}, a \geq 0 \right\},$$

$$C_+ = [V \xrightarrow{\Theta} \mathcal{B}], \quad Q = \mathcal{B}/\Theta V, \quad D = -x\partial_x.$$

$$\Theta\phi(x) = \sum_{n \in \mathbb{Z} \setminus \{0\}} \phi(nx), \quad \mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}.$$

The degrees of C_+ are zero and one. The original seed is

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, \quad \mathcal{M}\Theta\phi_* = g.$$

For a nonempty finite actual zero packet Z , with its complete positive multiplicities m_ρ , write

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad d = \deg h, \quad r = |Z|, \quad E_h = \mathbb{C}[s]/(h), \quad v_h = g/h.$$

The full CRT idempotents have an explicit polynomial construction. For each centre put $h_\rho = h/(s - \rho)^{m_\rho}$ and $q_\rho(z) = h_\rho(\rho + z)/h_\rho(\rho) - 1$. Since $q_\rho(0) = 0$, the polynomial class

$$a_\rho(\rho + z) = \frac{1}{h_\rho(\rho)} \sum_{j=0}^{m_\rho-1} (-q_\rho(z))^j \mod z^{m_\rho}$$

is the inverse of h_ρ modulo $(s - \rho)^{m_\rho}$: multiplication of the finite geometric sum leaves $1 - (-q_\rho)^{m_\rho}$, whose final term vanishes modulo z^{m_ρ} . Then $e_\rho = [h_\rho a_\rho]_h$ equals one on the ρ factor and zero on every other factor. Pairwise coprimality of the powers $(s - \rho)^{m_\rho}$ implies that a polynomial divisible by each is divisible by their product h . It follows that $\sum_\rho e_\rho = 1$, $e_\rho e_\sigma = 0$ for $\rho \neq \sigma$, and $e_\rho^2 = e_\rho$. These identities prove the stated CRT inverse by summing e_ρ times each local representative. The inherited arithmetic realization constructs $F_h \in \mathcal{B}$ with $\mathcal{M}F_h = v_h$, and its literal polynomial source map is

$$\mathcal{T}_h(P) = P(D)F_h, \quad \mathcal{T}_h(hP) = \Theta(P(D)\phi_*), \quad J_h \mathcal{T}_h(P) = v_h[P]_h, \quad v_h = j_h(v_h) \in E_h^\times.$$

Here j_h takes full Taylor germs modulo the factors of h , with the inverse Chinese-remainder map into E_h , and $J_h F = j_h(\mathcal{M}F)$. These identities, with their analytic domains, are the inherited source pins, not new purity assumptions. Their tensor products use the original order of variables and the cochain differential signs. The source identity $\mathcal{M}\Theta\phi = gH_\phi$, with H_ϕ holomorphic at the actual packet, proves $J_h\Theta\phi = 0$, including every prescribed multiplicity. Thus J_h induces $\bar{J}_h : Q \rightarrow E_h$. Put $\varepsilon_h = v_h^{-1}$, and use its fixed polynomial representative in the original coordinates. The map

$$\sigma_h : E_h \longrightarrow Q, \quad [P]_h \longmapsto [\mathcal{T}_h(\varepsilon_h P)]_{\Theta V}$$

is well-defined: changing P by hA , or the representative of ε_h by hB , changes its source by an explicitly displayed original theta relation. The identity $\bar{J}_h \sigma_h = 1_{E_h}$ follows by multiplication by $v_h \varepsilon_h = 1$, so σ_h is injective. Its tensor powers and the signed symmetric projector below retain these finite arithmetic classes and the same left inverses. For $k \geq 1$, put

$$\mathcal{P} = \mathbb{C}[s_1, \dots, s_k], \quad I = (h(s_1), \dots, h(s_k)), \quad E = \mathcal{P}/I = E_h^{\otimes k}, \quad E^{[2]} = \mathcal{P}/I^2,$$

$$\mathcal{T}_h^{(k)} P = P(D_1, \dots, D_k) F_h^{\otimes k}, \quad J^{(k)} = J_h^{\otimes k}, \quad U(s) = \prod_{i=1}^k v_h(s_i).$$

The tensor expressions here act on finite sums of the displayed source functions, so all stated Mellin derivatives have those original rapid-decrease domains. In coefficient multiplication U means its full class modulo I ; in a differentiated formula before reduction it means the displayed entire function. Its full class modulo I^2 is denoted $\widehat{U}$. These classes retain every Taylor coefficient allowed by their respective ideals.

The exact conormal domain and its arithmetic observation

The ring $\mathcal{P}$ is free over $\mathbb{C}[h(s_1), \dots, h(s_k)]$, with basis s^α , $0 \leq \alpha_i < d$. Successive monic division in each variable proves spanning. Independence follows by taking a largest multidegree: the leading monomials $s^{d\beta+\alpha}$, with $0 \leq \alpha_i < d$, are distinct for different pairs (β, α) . The same leading-term

argument proves that the $h(s_i)$ are algebraically independent. Modulo I^2 , precisely the terms of degree zero or one in these generators survive. This proves the exact sequence and coordinate map

$$0 \longrightarrow I/I^2 \longrightarrow E^{[2]} \xrightarrow{\pi} E \longrightarrow 0, \quad \iota_N : E^k \xrightarrow{\sim} I/I^2, \quad (a_i) \mapsto \left[\sum_i h(s_i) \tilde{a}_i \right].$$

Changing a lift $\tilde{a}_i$ by an element of I changes the sum by an element of I^2 , proving well-definedness directly.

The original relative-sum derivative and logarithmic source observable are

$$\partial_S^{\text{rel}} = \frac{1}{k} \sum_i \partial_{s_i}, \quad \mathcal{L}_k = \frac{1}{k} \sum_i \log x_i.$$

Leibniz's rule gives $\partial_S^{\text{rel}} I^2 \subseteq I$, hence defines $\delta_S : E^{[2]} \rightarrow E$. On the conormal domain its exact row is

$$\delta_S \iota_N(a_1, \dots, a_k) = \frac{1}{k} \sum_i h'(s_i) a_i.$$

Indeed differentiation of a lifted coefficient leaves a factor $h(s_i)$ and therefore belongs to I . In subsequent rank formulas δ_S denotes this restricted conormal map; its domain is $I/I^2 \cong E^k$, not all of $E^{[2]}$. For an actual original relation $P = \sum_i h(s_i) P_i$, differentiation of its Mellin transform yields

$$J^{(k)} \mathcal{L}_k \mathcal{J}_h^{(k)} P = \frac{U}{k} \sum_i h'(s_i) [P_i]_I.$$

The terms differentiating U or P_i still contain $h(s_i)$, so they vanish only at this displayed final quotient. In the split lift the quotient $G(\pi)$ sends a supported conormal vector to e_E , whereas the lifted coefficient derivative sends it to $(\delta_S n)^\bullet$. Both send external absence to τ . This is the original map through which the subsequent calculation observes a theta relation before it becomes its receiving supported zero.

For a reflection-stable packet the involution is $f^\dagger(s) = \overline{f(1 - \overline{s_1}, \dots, 1 - \overline{s_k})}$. The ordered arithmetic residue form used below is explicitly

$$\mathcal{R}_k(f, v) = \sum_{(\rho_1, \dots, \rho_k) \in Z^k} \text{Res}_{s_1=\rho_1} \cdots \text{Res}_{s_k=\rho_k} \frac{f^\dagger(s) v(s)}{\prod_{i=1}^k g(s_i)} ds_1 \cdots ds_k.$$

The order of the iterated residues is the displayed original order; there is no extra Koszul factor in this definition. To check independence of representatives, changing one input by $h(s_i)a$ cancels that factor against $g(s_i) = h(s_i)v_h(s_i)$; the resulting expression is holomorphic in the i -th variable at every corresponding centre and its residue is zero. The same statement for the first input uses $h^\dagger = (-1)^d h$.

Its nondegeneracy retains every local unit. At a centre ρ , write $z = s - \rho$ and $g(\rho + z) = z^{m_\rho} u_\rho(z)$, $u_\rho(0) \neq 0$. The local residue is the coefficient of $z^{m_\rho-1}$ in $f^\dagger(z)v(z)/u_\rho(z)$. Multiplication by u_ρ^{-1} is invertible in $\mathbb{C}[z]/(z^{m_\rho})$. The coefficient pairing of the basis $1, z, \dots, z^{m_\rho-1}$ with itself has entries 1 when the two exponents sum to $m_\rho - 1$, and 0 otherwise; its anti-diagonal determinant is nonzero. The reflected conjugate germ is an invertible conjugate-linear map and only pairs matched centres. The CRT decomposition and tensor product of these finite perfect pairings prove nondegeneracy of $\mathcal{R}_k$. For any finite coefficient algebra A , $M_a : A \rightarrow A$ denotes multiplication by a , and $\text{Tr}_A M_a$ is its ordinary complex-linear trace. The inertia (n_+, n_-, n_0) of a Hermitian form means the dimensions of its positive, negative and zero subspaces in a diagonalization by congruence; the zero dimension is the dimension of its radical.

The entire conormal image of an actual zero packet

Let Z be the finite actual packet in the transcript, with all its distinct centres and full positive multiplicities. For the residue and Hermitian trace calculations retain the transcript's reflection stability: $\iota(\rho) = 1 - \bar{\rho}$ permutes Z and $m_{\iota\rho} = m_\rho$. Its conjugate-linear involution is $f^\dagger(s) = \overline{f(1 - \bar{s})}$; since $h^\dagger = (-1)^d h$, it descends to E_h and componentwise to its tensor powers. The rank calculations alone do not need this involution. Write

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad d = \sum_{\rho \in Z} m_\rho, \quad r = |Z|, \quad E_h = \mathbb{C}[s]/(h).$$

No factor of h is removed. Denote by e_ρ the full Chinese-remainder idempotent of E_h . The exact CRT map sends a polynomial class to its classes modulo $(s - \rho)^{m_\rho}$; its inverse is multiplication by these idempotents followed by summation. At the rho component use the named local coordinate $z = s - \rho$. Let

$$h_\rho(s) = h(s)/(s - \rho)^{m_\rho}.$$

Its value $h_\rho(\rho)$ is nonzero. Before reduction the derivative is

$$h'(\rho + z) = m_\rho z^{m_\rho - 1} h_\rho(\rho + z) + z^{m_\rho} h'_\rho(\rho + z).$$

In $\mathbb{C}[z]/(z^{m_\rho})$, the second displayed term maps to zero and the first equals $m_\rho h_\rho(\rho) z^{m_\rho - 1}$. This explicitly records the killed term and every original scalar. Therefore the ideal (h') on this factor is the line generated by $z^{m_\rho - 1}$. For $m_\rho = 1$ this line is the whole one-dimensional factor. We have proved

$$\dim(h')E_h = r, \quad \dim E_h/(h') = d - r. \quad (C1)$$

For the ordered k -fold packet put $E = E_h^{\otimes k}$. The image of δ_S is exactly

$$\mathcal{J} = (h'(s_1), \dots, h'(s_k))E. \quad (C2)$$

Containment follows from its displayed row. For the reverse inclusion, a term $h'(s_i)b$ is the image of the vector with i th entry kb and all other entries zero. This retains the original $1/k$ instead of dropping it. There is the exact quotient morphism

$$E/\mathcal{J} \xrightarrow{\sim} (E_h/(h'))^{\otimes k}, \quad [a_1 \otimes \dots \otimes a_k] \mapsto [a_1] \otimes \dots \otimes [a_k]. \quad (C3)$$

To prove its kernel, choose any basis of $(h')E_h$ and extend it to a basis of E_h . The tensor basis decomposes into those tensors with at least one ideal-basis factor and those with none. The former span exactly the sum of the k ideals in (C2); the latter project bijectively onto a basis of the target. The basis is used only to prove the named quotient map, and it does not replace the arithmetic Gram or the original polynomial coordinates. Equations (C1)–(C3) imply

$$\text{rank } \delta_S = d^k - (d - r)^k, \quad \dim \ker \delta_S = (k - 1)d^k + (d - r)^k. \quad (C4)$$

The full local version, for an ordered tuple $(\rho_1, \dots, \rho_k)$, has image dimension

$$\prod_i m_{\rho_i} - \prod_i (m_{\rho_i} - 1) \quad (C5)$$

inside the unchanged local algebra $\mathbb{C}[z_1, \dots, z_k]/(z_i^{m_{\rho_i}} : 1 \leq i \leq k)$. Indeed the quotient in (C3) replaces each local vector-space dimension by $m_{\rho_i} - 1$. This includes every repeated-root direction rather than passing to residue values.

The actual coefficient U in A1959 is an invertible element of E . Multiplication by U is therefore an automorphism with explicit inverse multiplication by U^{-1} . Since an ideal is invariant under multiplication by a unit, the actual arithmetic observation $U\delta_S$ has precisely the same image (C2) and kernel as δ_S . These are rank calculations for the original arithmetic source relation map, including its unit, not merely ranks of an unrelated polynomial map.

The full mixed derivative and its arithmetic trace

Let

$$J_k = \prod_{i=1}^k g'(s_i) \in E, \quad g = 2\xi.$$

A1959 lines 7743–7764 gives the exact source identity

$$\left[\partial_{s_1} \dots \partial_{s_k} \left(UP \prod_i h(s_i) \right) \right]_I = J_k[P]_I. \quad (C6)$$

The product rule proves this without suppressing terms: every term in which some derivative misses its corresponding h retains an undifferentiated $h(s_i)$ and therefore belongs to I ; exactly the one term hitting every h survives. Its coefficient is $U \prod_i h'(s_i) = \prod_i g'(s_i)$ in E because $g' = h'(g/h) + h(g/h)'$. The term involving h has thus been recorded before the quotient.

The actual unit $v_h = g/h$ is nonzero at every centre. On the rho factor,

$$g'(\rho + z) = m_\rho u_\rho(0) z^{m_\rho - 1} \pmod{z^{m_\rho}}, \quad u_\rho(z) = g(\rho + z)/z^{m_\rho}, \quad u_\rho(0) \neq 0. \quad (C7)$$

Thus $M_{g'}$ has rank one at each centre, and the tensor-product multiplication map in (C6) has

$$\text{rank } M_{J_k} = r^k, \quad \dim \ker M_{J_k} = d^k - r^k. \quad (C8)$$

Its image on an ordered tuple is the line with the original coefficient

$$\left(\prod_i m_{\rho_i} u_{\rho_i}(0) \right) z_1^{m_{\rho_1} - 1} \dots z_k^{m_{\rho_k} - 1}. \quad (C9)$$

The map may land in the nilradical when a multiplicity exceeds one, but the original perfect residue pairing still detects (C9). Concretely, the ordered residue contraction is

$$\mathcal{R}_k(f, J_k u) = \sum_{\rho \in Z^k} \left(\prod_i m_{\rho_i} \right) \overline{f(1 - \bar{\rho})} u(\rho). \quad (C10)$$

Here $\mathcal{R}_k$ means the ordinary ordered iterated residue in the fixed order $s_1, \dots, s_k$, with no additional Koszul factor. Equation (C10) is repeated application of the actual simple-pole residue of g'/g , whose coefficient is the full m_ρ . It is also the ordinary multiplication trace on the full ordered algebra. Formula (C10) proves the exact map between the conormal/Jacobian observation and the trace; no assertion that a nilpotent is irrelevant is used. The cohomological supertrace has its separate factor $(-1)^k$. A differently signed tensor evaluation must be related by its explicit sign multiplication; no automatic cancellation is claimed here.

Restriction to the actual signed symmetric cochain image

The original idempotent in A1859 lines 6054–6078 is

$$S_k = \frac{1}{k!} \sum_{\pi \in S_k} \text{sgn}(\pi) T_\pi, \quad T_\pi = \text{sgn}(\pi) P_\pi \text{ in top degree.}$$

Its top-degree action is therefore the ordinary Reynolds projector $R_k = \frac{1}{k!} \sum_{\pi \in S_k} P_\pi$. We use the exact image $B_k = (E_h^{\otimes k})^{S_k}$, including full nilpotents. Simultaneous permutation of variables and conormal indices makes δ_S equivariant. Averaging a preimage proves that its invariant image is $\mathcal{J}^{S_k}$. This argument uses the same $1/k!$ as the actual chain idempotent. Taking invariants in (C3) is exact: if an invariant quotient vector has a lift, averaging that lift gives an invariant lift. It follows that

$$\text{rank}(\delta_S \text{ on invariant conormals}) = \binom{d+k-1}{k} - \binom{d-r+k-1}{k}. \quad (\text{C11})$$

Here $\binom{k-1}{k} = 0$ when $d-r=0$, since the k th symmetric power of the zero vector space is zero for $k \geq 1$. To verify the dimensions, a basis of a t -dimensional vector space gives an unscaled orbit-sum basis of its k th tensor invariants indexed by nonnegative t -tuples summing to k ; placing k identical marks among t slots gives $\binom{t+k-1}{k}$. This is a vector-space basis calculation, not a metric replacement.

The invariant conormal domain has dimension

$$d \binom{d+k-2}{k-1}. \quad (\text{C12})$$

Indeed an invariant tuple of k coefficients is determined by its k th coefficient, which must be invariant under the permutations of the first $k-1$ tensor positions. The other coefficients are recovered by permutation; independence of the chosen permutation is exactly this stabilizer invariance. This constructs the isomorphism to $E_h \otimes \text{Sym}^{k-1}(E_h)$ and proves (C12). Subtracting (C11) gives the complete invariant kernel dimension.

The mixed Jacobian is permutation invariant. Its image is the k th tensor power of the r -dimensional image $(g')E_h$; the same averaging proves surjectivity after taking invariants. Therefore

$$\text{rank}(M_{J_k}|_{B_k}) = \binom{r+k-1}{k}. \quad (\text{C13})$$

These are exact maps on the original signed symmetric cochain image, not on the unsigned cochain average that kills repeated degree-one vectors.

Intrinsic symmetric trace: the exact comparison of its multiplicities

It is necessary to keep both trace maps, with their actual comparison. Let an occupation vector be $n = (n_\rho)_{\rho \in Z}$, with every $n_\rho \geq 0$ and $\sum_{\rho \in Z} n_\rho = k$. Define e_n to be the unscaled sum of the full ordered CRT idempotents whose ordered tuples have those occupations. Products of distinct summands vanish, so $e_n^2 = e_n$, and these are the full orthogonal idempotents of B_k . The associated local factor is the invariant tensor product of the original local algebras. Its dimension is

$$D_n = \prod_{\rho \in Z} \binom{m_\rho + n_\rho - 1}{n_\rho}. \quad (\text{C14})$$

For proof, one chosen ordering has stabilizer $\prod_{\rho \in Z} S_{n_\rho}$. An invariant tuple of local components is determined by that component and its stabilizer invariance. The latter is $\bigotimes_{\rho \in Z} ((\mathbb{C}[z]/(z^{m_\rho}))^{\otimes n_\rho})^{S_{n_\rho}}$. Applying the orbit-sum dimension argument separately proves (C14), and also constructs the isomorphism without changing local variables.

The component has nilpotent ideal of zero constant terms: in the full ordered local algebra the ideal $(z_1, \dots, z_k)$ has a power equal to zero, hence so does its intersection with the invariant factor. Every element with nonzero constant is invertible by the finite geometric series in this ideal. Thus this is a local algebra, with residue field $\mathbb{C}$ and dimension (C14). Multiplication by $a = c + n$, n nilpotent, has trace $D_n c$: multiplication by n is nilpotent, hence has zero trace (triangularize its successive kernels). We obtain the intrinsic symmetric trace form

$$\mathcal{T}_k^{\text{sym}}(f, u) = \text{Tr}_{B_k} M_{f^\dagger u} = \sum_n D_n \overline{f(\iota n)} u(n), \quad \iota \rho = 1 - \bar{\rho}. \quad (\text{C15})$$

Here $\iota n = (n_{\iota \rho})_{\rho \in Z}$, the induced permutation of the original occupations. It is also, by the image/kernel decomposition of the Reynolds projector,

$$\mathcal{T}_k^{\text{sym}}(f, u) = \text{Tr}_{E_h^{\otimes k}} (R_k M_{f^\dagger u}). \quad (\text{C16})$$

Indeed $M_{f^\dagger u}$ commutes with R_k , and $R_k M$ equals M on its image B_k and zero on its kernel. This is the exact operator joining the ordered tensor trace and the actual symmetric cohomology trace.

The unprojected ordered trace in (C10), restricted to invariant inputs, has the different weight

$$O_n = \frac{k!}{\prod_\rho n_\rho!} \prod_\rho m_\rho^{n_\rho}.$$

The exact further map between the two arithmetic forms is multiplication by

$$B = \sum_n \frac{D_n}{O_n} e_n, \quad \mathcal{T}_k^{\text{sym}}(f, u) = \mathcal{R}_k(f, J_k B u). \quad (\text{C17})$$

Every ratio is a specified positive nonzero rational number, and the inverse replaces each ratio by its reciprocal. The equation follows by comparing the two literal finite sums. Thus the symmetric intrinsic trace is reached by the same actual theta mixed derivative, followed by its explicitly displayed coefficient map B . No orbit factor or local multiplicity has been normalized away.

For precision, this contraction has the exact induced isomorphism

$$B_k / \text{rad } \mathcal{T}_k^{\text{sym}} \xrightarrow{\sim} \text{im}(M_{J_k}|_{B_k}), \quad [u] \mapsto J_k B u. \quad (\text{C17a})$$

The local formula (C7) shows that multiplication by the full mixed Jacobian kills exactly those classes whose constant term is zero on every ordered tuple. Restricting to invariants gives exactly the nilpotent ideal of B_k , already computed from its local factors. Multiplication by B preserves that ideal and is invertible. Its kernel is therefore precisely the trace radical, and surjectivity onto the stated invariant image follows from invertibility of B . The image here has dimension (C13); it is the invariant part of the ordered image (C8), not the full ordered image. This proves (C17a), with its exact kernel, rather than identifying its two spaces without a map.

An explicit original source representative for its value is obtained as follows. Choose the unique full CRT polynomial representative P of Bu , of degree less than d in each of the unchanged variables; for invariant u this representative is invariant by uniqueness and the permutation invariance of h . Then

$$F_{u,k} := \left(\prod_{i=1}^k \log x_i \right) P(D_1, \dots, D_k) (\Theta \phi_*)^{\otimes k} \quad \text{satisfies} \quad J^{(k)} F_{u,k} = J_k B u. \quad (\text{C17b})$$

Indeed the undifferentiated source on the right has Mellin transform $P \prod_i g(s_i) = U P \prod_i h(s_i)$. Each log multiplication differentiates its specified Mellin variable, so (C6) gives precisely (C17b). These source functions and their logarithmic multiples remain in the given rapidly decreasing test spaces by the bounds defining those spaces. The undifferentiated expression is an original theta tensor boundary, for example using the first source leg ϕ_* with the other theta legs and the same polynomial operator. The mixed derivative is taken before that relation is quotiented. For $m > 1$ its observed jet may be trace-radical, even while its residue contraction in (C17) is nonzero; (C17a) and (C17b) retain exactly which map is being used.

Consequence for the actual RH counterfactual, with every multiplicity

Suppose the counterfactual supplies an actual quartet

$$Z = \left\{ \frac{1}{2} + \delta + i\gamma, \frac{1}{2} - \delta + i\gamma, \frac{1}{2} + \delta - i\gamma, \frac{1}{2} - \delta - i\gamma \right\}, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 0,$$

with the common full multiplicity m supplied by the original reflection and conjugation. The existence assumption in this paragraph is exactly the RH counterfactual. The statements below compute its consequences on the original objects. Put the first and second centres in one reflection pair, and the third and fourth in the other. Then $d=4m$ and $r=4$, so

$$\text{rank } \delta_S = (4m)^k - (4(m-1))^k, \quad \text{rank } M_{J_k} = 4^k, \quad (\text{C18})$$

and on the actual signed symmetric image,

$$\text{rank } \delta_S^{\text{sym}} = \binom{4m+k-1}{k} - \binom{4(m-1)+k-1}{k}, \quad \text{rank } M_{J_k}^{\text{sym}} = \binom{k+3}{3}. \quad (\text{C19})$$

The trace radical is exactly the nilpotent ideal. To prove this, a zero-constant element is orthogonal to everything by (C15). Conversely, if its value at occupation n is nonzero, pairing against $e_{\iota n}$ gives the nonzero scalar $D_{\iota n}$ times that value or its conjugate. The radical therefore has dimension

$$\binom{4m+k-1}{k} - \binom{k+3}{3}. \quad (\text{C20})$$

On a two-element reflection orbit of occupations n and ιn , the literal Gram in the idempotent basis is

$$\begin{pmatrix} 0 & D_n \\ D_n & 0 \end{pmatrix}, \quad D_n = \prod_{i=1}^4 \binom{m+n_i-1}{n_i} > 0,$$

since matched multiplicities make $D_n = D_{\iota n}$. The vectors $e_n + e_{\iota n}$ and $e_n - e_{\iota n}$ have values $+2D_n$ and $-2D_n$, respectively. The basis and both factors two have been retained. A fixed occupation gives the positive one-dimensional Gram D_n .

A fixed occupation must have $n_1 = n_2$ and $n_3 = n_4$. There are no such occupations for odd k . For $k = 2l$, choosing $n_1 = n_2 = a$ and $n_3 = n_4 = l - a$, with $0 \leq a \leq l$, gives exactly $l+1$. Let

$$b_k = \binom{k+3}{3}, \quad f_k = \begin{cases} 0 & k \text{ odd,} \\ k/2 + 1 & k \text{ even.} \end{cases}$$

Counting each nonfixed orbit twice proves the complete inertia, including the entire radical:

$$\boxed{\begin{aligned} n_+(\mathcal{T}_k^{\text{sym}}) &= (b_k + f_k)/2, \\ n_-(\mathcal{T}_k^{\text{sym}}) &= (b_k - f_k)/2, \\ n_0(\mathcal{T}_k^{\text{sym}}) &= \binom{4m + k - 1}{k} - b_k. \end{aligned}} \quad (\text{C21})$$

These negative directions are nonradical. Each of their negative trace values is the pairing with the explicit original theta mixed-derivative observation $J_k Bf$ in (C6) and (C17). This states the exact map: it does not assert that an idempotent f is itself in the Jacobian image, which would fail for repeated roots. The class f survives the nilradical quotient and the actual signed symmetric projector; its mixed-derivative observation remains available before the corresponding source relation is quotiented to its supported zero. Formula (C21) supplies a growing family of exact nonzero arithmetic directions that every claimed purity comparison for the RH counterfactual must control. It establishes no off-line zeta zero and no unconditional positivity contradiction: neither assertion occurs in the authoritative source, and neither was used in this calculation.

The relative fibres at the expected sum weight still retain negative directions

The sum pushforward in A1912 retains the occupation idempotents even when their sum eigenvalues coincide. On the quartet label ordering just specified, its eigenvalue at occupation n is exactly

$$\lambda_n = \frac{k}{2} + \delta(n_1 - n_2 + n_3 - n_4) + i\gamma(n_1 + n_2 - n_3 - n_4). \quad (\text{C22})$$

To verify this with nilpotents, restrict multiplication by $S = \sum_i s_i$ to one ordered CRT factor, where it is the displayed scalar plus multiplication by $\sum_i z_i$. The latter has a power equal to zero, since all monomials of total degree above $k(m-1)$ vanish. Passing to stabilizer invariants retains this action and its scalar; it does not set the nilpotent to zero. This proves the full generalized eigenvalue statement on the occupation factor.

For $k = 2l$, let B_{bal} be the direct summand defined by the sum of all full idempotents with $n_1 + n_3 = l$ and $n_2 + n_4 = l$. It is a specified algebra direct summand of B_k , invariant under both reflection and the sum operator. Its eigenvalues all have real part $k/2$ by (C22). There are exactly $(l+1)^2$ such occupations: independently distribute l among the two right centres and l among the two left centres. Exactly $l+1$ of them are reflection-fixed, as in (C21). Its trace form therefore has

$$n_-(\mathcal{T}_k^{\text{sym}}|_{B_{\text{bal}}}) = \frac{l(l+1)}{2}, \quad n_+(\mathcal{T}_k^{\text{sym}}|_{B_{\text{bal}}}) = \frac{(l+1)(l+2)}{2}, \quad (\text{C23})$$

and radical dimension $\sum_{n \text{ balanced}} D_n - (l+1)^2$, with the same literal weights (C14). This is proved by the same explicit two-idempotent Gram blocks, now on this exact invariant direct summand. It is not a new positivity assumption.

Already at $k=2$ the two distinct occupations $(1,0,0,1)$ and $(0,1,1,0)$ both have the actual sum eigenvalue 1. Their full idempotents remain different, and

$$\mathcal{T}_2^{\text{sym}}(e_{(1,0,0,1)} - e_{(0,1,1,0)}, e_{(1,0,0,1)} - e_{(0,1,1,0)}) = -2m^2. \quad (\text{C24})$$

For $m > 1$ the local nilpotent sum operators remain attached; no assertion that the idempotents are eigenvectors is needed. Equation (C24) is exactly why the original program's relative algebra cannot be replaced by the list of sums alone. It is an actual counterfactual consequence inside A1912's constructed pushforward. It does not say that an upper bound on the entire tensor image

would be insufficient: the same whole image also retains repeated extreme eigenlines of real part $k(1/2 + \delta)$. Rather, it specifies the additional negative relative directions that remain even in its expected-real-part summand, with their exact source and trace maps (C17a)–(C17b).

The negative count can be located on each exact sum fibre. For an integer q with $-l \leq q \leq l$, the fibre whose eigenvalue is $l + 2i\gamma q$ consists of the full occupation factors

$$n = (a, b, l - a, l - b), \quad 0 \leq a, b \leq l, \quad a + b = l + q.$$

There are $N_q = l + 1 - |q|$ such pairs: when $q \geq 0$, a runs from q to l ; when $q \leq 0$, a runs from 0 to $l + q$. Reflection exchanges a and b , so its fixed count is $f_q = 1$ if $l + q$ is even and zero otherwise. This also equals one precisely when N_q is odd. The exact nonradical inertia of that particular sum fibre is consequently

$$n_- = \left\lfloor \frac{l + 1 - |q|}{2} \right\rfloor, \quad n_+ = \left\lceil \frac{l + 1 - |q|}{2} \right\rceil, \quad (\text{C25})$$

with its unchanged nilpotent radical of dimension $\sum_{a+b=l+q} D_{(a,b,l-a,l-b)} - N_q$. The proof is the full Gram-block argument above restricted by the literal spectral projector which sums these occupation idempotents. Thus the persistent negative directions are established inside individual exact sum fibres, not only in a union of fibres with the same real part.

Chapter 11

The original boundary at every holonomy phase

The off-line counterfactual inside the original holonomy construction

This calculation uses the original theta source in the final analytic continuation of the programme. In particular, the scalar measure below is computed from g/h with $g = 2\xi$; it is not a Gamma reference measure. The source equations being continued are (H5)–(H17), (H24)–(H28), and (H53)–(H55) of *All-holonomy descent of the original theta quotient*. The complete source record and locators are recorded in the accompanying audit. The conclusions concern the stated counterfactual and do not assert the existence of an off-line zero.

The retained arithmetic objects and every phase

Suppose $1/2 + \delta + i\gamma$ is a zero of g of its full order m , where $0 < \delta < 1/2$ and $\gamma > 0$. Reflection and conjugation give the complete quartet, with the same order. Retain

$$h(s) = \prod_{\epsilon, \eta \in \{-1, 1\}} (s - 1/2 - \epsilon\delta - i\eta\gamma)^m, \quad v_h = g/h, \quad w_h(u) = \frac{|v_h(1/2 + iu)|^2}{2\pi}. \quad (\text{HT1})$$

The division is entire and its full jet $v_h = j_h(v_h)$ is a unit. For an integer $k \geq 2$, put

$$c = k/2, \quad \ell_k = 1 + k(m-1), \quad q = \ell_k(k+1)^2, \quad m_{h,k} = w_h^{*k},$$

$$\chi(S) = \prod_{a,b=0}^k (S - c - (2a-k)\delta - i(2b-k)\gamma)^{\ell_k}, \quad E = \mathbb{C}[S]/(\chi), \quad A = M_S. \quad (\text{HT2})$$

These are the original cyclic sum quotient, its full multiplicities, and its multiplication operator. For completeness, in each tensor local factor the nilpotent sum is $n_1 + \dots + n_k$ with $n_i^m = 0$. Its power $k(m-1)$ has nonzero coefficient $(k(m-1))!/((m-1)!)^k$ on $n_1^{m-1} \dots n_k^{m-1}$, whereas the next power vanishes. Thus its nilpotency index is exactly ℓ_k . Every pair (a, b) is realized by choosing independently the real and imaginary signs in the k tensor factors. Distinct pairs give distinct complex sums, so the annihilator in the sum variable is exactly χ .

The original analytic construction provides F_h with Mellin transform v_h , and the original source and jet maps are

$$\mathcal{V}P = P(D_1 + \dots + D_k)F_h^{\otimes k}, \quad J^{(k)}\mathcal{V} = \eta\pi_\chi, \quad \eta[P] = v_h^{\otimes k}P(A_k)1. \quad (\text{HT3})$$

Here η is an injective complex linear module map. Its unit factor is retained. The norm of this source is exactly $\int_{\mathbb{R}} |P(c + iu)|^2 m_{h,k}(u) du$. The half-density map is

$$(\mathcal{U}_k F)(r, z) = e^{(kr + \sum_{i < k} z_i)/2} F(e^{r+z_1}, \dots, e^{r+z_{k-1}}, e^r), \quad r = \log x_k, \quad z_i = \log x_i - \log x_k. \quad (\text{HT4})$$

It maps into $L^2(\mathbb{R}, dr; \mathcal{K})$ with $\mathcal{K} = L^2(\mathbb{R}^{k-1}, dz)$ and intertwines $D_1 + \dots + D_k$ with $-\partial_r + c$. No relative variable is removed.

Fix any $L > 0$ and $\theta \in [0, 2\pi)$. The source of phase θ is

$$(\mathcal{Z}_{L,\theta} \psi)(r) = \sum_{j \in \mathbb{Z}} e^{ij\theta} \psi(r + jL), \quad \psi(r + L) = e^{-i\theta} \psi(r) \quad \text{in the target domain.} \quad (\text{HT5})$$

On the original finite source, its Gram form is exactly the polynomial norm for the measure

$$\nu_{L,\theta} = \frac{2\pi}{L} \sum_{n \in \mathbb{Z}} m_{h,k}(u_n) \delta_{u_n}, \quad u_n = \frac{2\pi n + \theta}{L}, \quad S = c + iu. \quad (\text{HT6})$$

The factor $2\pi/L$ is unchanged. The measure has positive mass at every u_n . Indeed the nonzero entire function v_h has only isolated zeros on the line; hence $w_h > 0$ almost everywhere. For every u , the product $w_h(v)w_h(u - v)$ is positive almost everywhere in v , so its integral is positive. Iterated convolution preserves strict positivity. The original weighted Schwartz estimates give convergence and all moments in (HT6). A nonzero polynomial cannot vanish at every lattice point; consequently every finite polynomial Gram matrix is positive definite.

For $N \geq q - 1$, let $P_N = \mathbb{C}[S]_{\leq N}$ and $J_N = \pi_\chi|_{P_N}$. With the fixed coefficient bases define

$$\begin{aligned} M_{N,\theta} &= (\mathcal{Z}_{L,\theta} \mathcal{U}_k \mathcal{V}|_{P_N})^* (\mathcal{Z}_{L,\theta} \mathcal{U}_k \mathcal{V}|_{P_N}), \quad K = J_N M_{N,\theta}^{-1} J_N^*, \quad G = K^{-1}, \\ C_{N,\theta} &= M_{N,\theta}^{-1} J_N^* G, \quad R = \mathcal{Z}_{L,\theta} \mathcal{U}_k \mathcal{V} C_{N,\theta}. \end{aligned} \quad (\text{HT7})$$

Thus $J_N C_{N,\theta} = I_E$ and $R^* R = G$. All inverses in (HT7) exist: $M_{N,\theta}$ is positive definite and J_N is surjective. No inverse at degree $q - 2$ is used.

A uniform phasewise lower bound for the original control

Define

$$H = G^{-1/2} (A^* G + G A - k G) G^{-1/2}, \quad \mathcal{L}_{h,k} = 2\delta \ell_k (k + 1) \left\lfloor \frac{(k + 1)^2}{4} \right\rfloor. \quad (\text{HT8})$$

For every $k \geq 2$, $L > 0$, $\theta \in [0, 2\pi)$, and $N \geq q - 1$, the nonzero eigenvalues of H are $\varepsilon, -\varepsilon$, and

$$\boxed{\varepsilon \geq \mathcal{L}_{h,k} > 0, \quad \frac{\mathcal{L}_{h,k}}{q} \geq \frac{\delta k}{2}.} \quad (\text{HT9})$$

Proof. Let $p_j(S)$ be the monic orthogonal polynomials for (HT6), with squared norms $\omega_j > 0$, and let $b_j = [p_j] \in E$. In the real coordinate u , write $p_j(c + iu) = i^j r_j(u)$, where r_j is real monic. Orthogonality and the symmetry of multiplication by u show

$$u r_j = r_{j+1} + a_j r_j + \frac{\omega_j}{\omega_{j-1}} r_{j-1}, \quad a_j \in \mathbb{R},$$

with the last term omitted at $j = 0$. To verify that no earlier term occurs, pair $u r_j$ with r_i , $i \leq j - 2$, and move u to r_i ; its degree is at most $j - 1$. Pairing with r_{j-1} gives precisely ω_j / ω_{j-1} . In the S coordinate this is

$$S p_j = p_{j+1} + (c + i a_j) p_j - \frac{\omega_j}{\omega_{j-1}} p_{j-1}. \quad (\text{HT10})$$

Since the source Gram is diagonal in this basis, $K = \sum_{j=0}^N b_j b_j^* / \omega_j$. Apply A on the left and A^* on the right and use (HT10). The imaginary diagonal terms cancel, and each adjacent term except the last cancels its predecessor. This gives, including $N = q - 1$,

$$AK + KA^* - kK = \frac{b_{N+1}b_N^* + b_N b_{N+1}^*}{\omega_N}. \quad (\text{HT11})$$

Congruence by $G^{1/2}$ identifies the left side with H . Thus its rank is at most two. Also $\text{Tr } H = 2 \text{Re } \text{Tr } A - kq = 0$ by the complete grid (HT2).

Let E_+ be the direct sum of the complete primary spaces with $a > k/2$, and let P be the orthogonal projection onto $G^{1/2}E_+$ in the standard Hermitian space. The latter subspace is invariant under $G^{1/2}AG^{-1/2}$. Taking traces in an orthonormal basis of this subspace gives

$$\text{Tr}(PH) = 2 \sum_{a > k/2} \sum_{b=0}^k \ell_k(2a - k) \delta = 2\delta \ell_k(k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor = \mathcal{L}_{h,k}. \quad (\text{HT12})$$

The finite sum is r^2 if $k = 2r - 1$, and $r(r+1)$ if $k = 2r$. A Hermitian rank-at-most-two matrix of trace zero has eigenvalues $\varepsilon, -\varepsilon$ and zeroes. For every orthogonal projection, $\text{Tr}(PH) \leq \varepsilon$: in an eigenbasis the positive contribution is at most ε and the negative one is nonpositive. Equation (HT12) proves the first inequality of (HT9), and $4\lfloor (k+1)^2/4 \rfloor \geq k(k+1)$ proves the second. $\square$

The exact energy forced into the retained theta boundary

In $\mathcal{H}_{L,\theta} = L^2([0, L], dr; \mathcal{K})$ put $D_{L,\theta} = -\partial_r + c$, with domain $\{f \in H^1([0, L]; \mathcal{K}) : f(L) = e^{-i\theta} f(0)\}$. The domain of its square is $\{f \in H^2([0, L]; \mathcal{K}) : f(L) = e^{-i\theta} f(0), f'(L) = e^{-i\theta} f'(0)\}$. The original smooth periodized source columns belong to these domains. Monic division in the original coefficient source gives the unique polynomial map $Q_{N,\theta}$ with

$$SC_{N,\theta} - C_{N,\theta}A = \chi Q_{N,\theta}.$$

Indeed applying J_{N+1} gives zero. Define the actual boundary

$$\mathcal{B} = \mathcal{Z}_{L,\theta} \mathcal{U}_k \mathcal{V}(\chi Q_{N,\theta}) = D_{L,\theta} R - RA. \quad (\text{HT13})$$

This retains its theta primitive. Explicitly, tensor monic division gives

$$\chi(S)Q(S) = \sum_i h(s_i)Q_i(\mathbf{s}),$$

because its image in the full tensor jet algebra vanishes. In the original theta complex, ϕ_h has degree zero, F_h has degree one, and $h(D)F_h = \Theta\phi_h$. The degree- $(k-1)$ tensor primitive is exactly

$$\sum_{i=1}^k (-1)^{i-1} Q_i(D_1, \dots, D_k) (F_h^{\otimes(i-1)} \otimes \phi_h \otimes F_h^{\otimes(k-i)}).$$

The tensor differential contributes a further sign $(-1)^{i-1}$ when it acts on ϕ_h , because the preceding $i-1$ factors have degree one. The two signs cancel, so its differential is precisely $\mathcal{V}(\chi Q)$. Applying the same half-density and phase maps gives precisely (HT13).

Set

$$U = RG^{-1/2}, \quad T = \mathcal{B}G^{-1/2}, \quad Z = U^*T, \quad T_\perp = (I - UU^*)T.$$

Then the following exact energy identity, with no omitted source term, holds:

$$\text{Tr}(G^{-1}\mathcal{B}^*\mathcal{B}) = \frac{\varepsilon^2}{2} + \left\| \frac{Z - Z^*}{2} \right\|_{\text{HS}}^2 + \|T_\perp\|_{\text{HS}}^2 \geq \frac{\mathcal{L}_{h,k}^2}{2} \geq \frac{\delta^2 k^2 q^2}{8}. \quad (\text{HT14})$$

All quantities on the left are calculated from g/h and its original theta representatives. In particular the same estimate holds after integration with the original Fourier inversion measure:

$$\int_0^{2\pi} \text{Tr}(G_{N,\theta}^{-1} \mathcal{B}_{N,\theta}^* \mathcal{B}_{N,\theta}) \frac{d\theta}{2\pi} \geq \frac{\mathcal{L}_{h,k}^2}{2}. \quad (\text{HT15})$$

Proof. The two arguments in integration by parts have the same boundary multiplier $e^{-i\theta}$, so the endpoint products cancel. Hence $D_{L,\theta}^* + D_{L,\theta} = 2cI$ on the stated domain. Inserting $D_{L,\theta}R = RA + \mathcal{B}$ and $R^*R = G$ gives

$$A^*G + GA - kG = -(R^*\mathcal{B} + \mathcal{B}^*R), \quad H = -Z - Z^*.$$

Since $U^*U = I$, the decomposition $T = UZ + T_\perp$ is orthogonal in Hilbert–Schmidt norm. The Hermitian and skew-Hermitian parts of Z are orthogonal for the real Hilbert–Schmidt scalar product. Therefore

$$\|T\|_{\text{HS}}^2 = \frac{1}{4}\|H\|_{\text{HS}}^2 + \left\|\frac{Z - Z^*}{2}\right\|_{\text{HS}}^2 + \|T_\perp\|_{\text{HS}}^2.$$

By (HT9), $\|H\|_{\text{HS}}^2 = 2\varepsilon^2$, which proves (HT14). For fixed k, L, N the weighted source sums and their derivatives converge uniformly in θ . Thus $M_{N,\theta}$ is continuous. Its smallest eigenvalue is positive on the compact phase circle, so its inverse, $G_{N,\theta}$, and the boundary columns are continuous there. Integration of (HT14) proves (HT15). $\square$

The full lattice test for the original Laplacian boundary

Put $\mathcal{L}_{L,\theta} = D_{L,\theta}^2 - kD_{L,\theta} = \partial_r^2 - c^2$. Direct substitution of (HT13) gives the exact map

$$\mathcal{L}_{L,\theta}R - R(A^2 - kA) = (D_{L,\theta} - k)\mathcal{B} + \mathcal{B}A. \quad (\text{HT16})$$

If $Av = \lambda v$ with $v \neq 0$, write $\lambda = c + x + iy$. Then

$$\frac{\|((D_{L,\theta} - k)\mathcal{B} + \mathcal{B}A)v\|}{\|Rv\|} \geq \min_{n \in \mathbb{Z}} \sqrt{(u_n^2 + x^2 - y^2)^2 + 4x^2y^2}. \quad (\text{HT17})$$

In particular every phase and every admitted degree satisfy, on an eigenvector for $\lambda = k(1/2 + \delta + i\gamma)$,

$$\boxed{\frac{\|((D_{L,\theta} - k)\mathcal{B} + \mathcal{B}A)v\|}{\|Rv\|} \geq 2k^2\delta\gamma.} \quad (\text{HT18})$$

For an eigenvalue with $x \neq 0$ and $y = 0$, (HT17) also gives the positive lower bound $x^2 + \min_n u_n^2$, even though its imaginary part is zero.

Proof. The Fourier basis $L^{-1/2}e^{-iu_n r}$ diagonalizes the quasi-periodic Laplacian with eigenvalues $-u_n^2 - c^2$, each acting identically on $\mathcal{K}$. The source vectors lie in its domain by the original weighted differentiability estimates. Writing Rv in that basis and using Parseval gives

$$\|(\mathcal{L}_{L,\theta} - (\lambda^2 - k\lambda))Rv\|^2 = \sum_n ((u_n^2 + x^2 - y^2)^2 + 4x^2y^2) \|\widehat{Rv}(n)\|_{\mathcal{K}}^2.$$

The lattice is discrete and the expression tends to infinity as $|n| \rightarrow \infty$, so its infimum is a minimum. Also R is injective because $R^*R = G \succ 0$. Divide the displayed identity by $\|Rv\|^2$; (HT16) proves (HT17). The extreme primary factor in (HT2) is nonzero and has an eigenvector, so substituting $x = k\delta, y = k\gamma$ proves (HT18). If $y = 0$, the square root in (HT17) is $u_n^2 + x^2$. $\square$

What these computations decide

The original off-line counterfactual survives passage to every finite holonomy quotient only with the explicit boundary energy (HT14) and Laplacian residual (HT17). These quantities belong to the original arithmetic source. The statement is compatible with the exact all-phase identity $\int M_{N,\theta} d\theta / (2\pi) = M_N$ and with the original small gluing-norm estimate. That estimate concerns

$C_N - C_{N,\theta}$; (HT13) applies S and subtracts right multiplication by A , preserving a different, explicitly displayed source quantity. There is no claim here that averaging the $G_{N,\theta}$ preserves the rank-two identity (HT11): the lower bound (HT15) integrates the proved phasewise energies themselves.

The transcript's threshold-four volume lower bound was already present before its final Deligne comparison. Its last counterexamples retain many generic identities but do not satisfy the actual divisor condition $h \mid 2\xi$. Consequently they disprove the reduced generic inference stated there; they do not disprove the entire arithmetic programme or provide an RH counterexample. Equations (HT9), (HT14), and (HT17) give further quantified consequences within the original programme without replacing its theta metric or its represented relations.

Chapter 12

Exact arithmetic transport of all four determinants

This calculation retains the arithmetic coefficient complex over the absolute pointed base $\mathbf{F}_{1,\tau} = \{\tau, 1\}$. It computes the signed arithmetic/reference determinant term on the original finite theta source. The new formulas are a complete first and second variation, an exact four-window projection trace, and a quantitative bound in terms of the actual common source Gram. No arithmetic source multiplier is assumed bounded above or bounded away from zero.

Original objects and the fixed comparison

The retained tau-base construction has the four-point chart $+, - < \eta < \sigma$ and the coefficient sheaf $\mathcal{T}$ with $(\mathcal{T}_+, \mathcal{T}_-, \mathcal{T}_\eta, \mathcal{T}_\sigma) = (V, V, \mathcal{B}, 0)$. Its two nonzero restrictions are Θ and $J\Theta = \Theta\mathcal{F}$. The actual projective resolution gives

$$\mathbf{A}_\tau(\mathcal{T}) = [V \oplus V \xrightarrow{\Theta(\phi - \widehat{\psi})} \mathcal{B}], \quad H^1 \mathbf{A}_\tau(\mathcal{T}) = Q = \mathcal{B}/\Theta V. \quad (\text{AT1})$$

These are the fixed source objects of the original Tau Base note, equations (18)–(25), supplied in the source closure. The tensor resolution in its equations (46)–(48) retains the original differential signs and identifies its top cohomology with $Q^{\otimes k}$. The operation here uses that pushforward, with its arithmetic classes. Its separately computed restriction to σ remains an available comparison; it is not substituted for (AT1).

Fix an original actual packet h of full zero orders of $g = 2\xi$, and a positive integer k . Retain the full tensor-sum operator, its original cyclic annihilator $\chi = \chi_{h,k}$ of degree $q \geq 1$, and $E = \mathbb{C}[S]/(\chi)$ in the increasing-power remainder frame. The original construction supplies

$$\mathcal{V}P = P(D_1 + \cdots + D_k)F_h^{\otimes k}, \quad J^{(k)}\mathcal{V}P = \eta_{h,k}J_N P, \quad \eta_{h,k}[P] = v_h^{\otimes k}P(A_h^{(k)})1, \quad (\text{AT2})$$

where $v_h = j_h(g/h)$ is the full invertible Taylor unit, J_N is literal remainder modulo χ , and $\eta_{h,k} : E \hookrightarrow E_h^{\otimes k}$ is the original injective full-jet observation, $E_h = \mathbb{C}[s]/(h)$. The original section $\sigma_h : E_h \hookrightarrow Q$ gives the separate arithmetic cohomology injection and equality

$$q^{(k)}\mathcal{V}P = \sigma_h^{\otimes k}\eta_{h,k}J_N P, \quad \sigma_h^{\otimes k}\eta_{h,k} : E \hookrightarrow Q^{\otimes k}. \quad (\text{AT2a})$$

The supplied arithmetic-input and cyclic-sum proofs construct these maps; all operations below keep them fixed. In particular the sum coordinate is $S = k/2 + iu$, with no division by k .

For $N \geq q-1$, set $\mathcal{P}_N = \mathbb{C}[S]_{\leq N}$, $\mathcal{P}_{-1} = 0$, and let $B_N : \mathcal{P}_{N-q} \rightarrow \mathcal{P}_N$ be multiplication by the literal monic χ . Write $s_N : E \rightarrow \mathcal{P}_N$ for the degree-less-than- q remainder section. Polynomial division proves

$$0 \longrightarrow \mathcal{P}_{N-q} \xrightarrow{B_N} \mathcal{P}_N \xrightarrow{J_N} E \longrightarrow 0, \quad J_N s_N = I_E. \quad (\text{AT3})$$

The arithmetic norm has density $m_1(u) = m_{h,k}(u) = w_h^*(u)$, $w_h(u) = |(g/h)(1/2 + iu)|^2/(2\pi)$. The retained Gamma reference has density

$$m_0(u) = r_{1/4,k}(u) = \frac{(\sqrt{2\pi})^k}{c_{k/4}} \frac{|\Gamma(k/4 + iu/2)|^2}{2\pi}, \quad c_a = 2^{1-2a}\Gamma(2a). \quad (\text{AT4})$$

Its mass is $(2\pi)^{k/2}$; the arithmetic mass is the original μ_h^k . The source proofs included with this calculation establish finite moments of every degree and positivity almost everywhere for both densities. The calculation below only integrates polynomials against these two actual measures and their sum. It needs no estimate on the pointwise ratio m_1/m_0 .

For $0 \leq t \leq 1$ define, in the unchanged coefficient frame,

$$\begin{aligned} m_t &= (1-t)m_0 + tm_1, \quad \Delta m = m_1 - m_0, \quad v_N(u) = (1, S(u), \dots, S(u)^N), \\ M_N(t) &= \int v_N(u)^* v_N(u) m_t(u) du, \quad \Delta M_N = M_N(1) - M_N(0), \\ H_N(t) &= B_N^* M_N(t) B_N. \end{aligned} \quad (\text{AT5})$$

Every $M_N(t)$ is positive definite: a nonzero polynomial has finitely many real zeros, and m_t is positive almost everywhere. B_N is injective, so $H_N(t)$ is positive definite on its actual relation domain. When $N = q - 1$, that domain is zero; its determinant is one, its unique endomorphism is its inverse, and all compositions through that domain are zero. The real parameter t interpolates norms on (AT3); it changes neither the absolute base nor a cochain map.

Canonical sections, determinant lines and full variations

Theorem 12.1. *On the fixed exact sequence (AT3), its least-norm section and quotient Gram are*

$$\begin{aligned} R_N(t) &= s_N - B_N H_N(t)^{-1} B_N^* M_N(t) s_N, \\ J_N R_N(t) &= I_E, \quad B_N^* M_N(t) R_N(t) = 0, \\ G_N(t) &= R_N(t)^* M_N(t) R_N(t) = (J_N M_N(t)^{-1} J_N^*)^{-1}, \\ V_N(t) &:= \det G_N(t) = \frac{\det M_N(t)}{\det H_N(t)}. \end{aligned} \quad (\text{AT6})$$

In the original polynomial frame the exact first and second variations are

$$\begin{aligned} \dot{R}_N &= -B_N H_N^{-1} B_N^* \Delta M_N R_N, \quad \dot{G}_N = R_N^* \Delta M_N R_N, \\ \ddot{G}_N &= -2R_N^* \Delta M_N B_N H_N^{-1} B_N^* \Delta M_N R_N \preceq 0. \end{aligned} \quad (\text{AT7})$$

The exact logarithmic variation, with both trace terms retained, is

$$\begin{aligned} \frac{d}{dt} \log V_N &= \text{Tr}(M_N^{-1} \Delta M_N) - \text{Tr}(H_N^{-1} B_N^* \Delta M_N B_N) \\ &= \text{Tr}(G_N^{-1} R_N^* \Delta M_N R_N). \end{aligned} \quad (\text{AT8})$$

Its second derivative is

$$\begin{aligned} \frac{d^2}{dt^2} \log V_N &= -2 \text{Tr}(G_N^{-1} R_N^* \Delta M_N B_N H_N^{-1} B_N^* \Delta M_N R_N) \\ &\quad - \text{Tr}(G_N^{-1} \dot{G}_N G_N^{-1} \dot{G}_N) \leq 0. \end{aligned} \quad (\text{AT9})$$

Proof. Every lift of $x \in E$ is $s_N x + B_N y$. Expanding its squared norm and completing its finite quadratic expression gives the unique minimizer $y = -H_N^{-1} B_N^* M_N s_N$. This proves the first two lines of (AT6). For the inverse expression, every polynomial vector p decomposes as $B_N y + R_N J_N p$.

Orthogonality therefore gives $R_N^* M_N p = G_N J_N p$ for every p . Taking adjoints yields $M_N R_N = J_N^* G_N$. Substituting this into $J_N R_N = I_E$ gives $(J_N M_N^{-1} J_N^*) G_N = I_E$, which proves the inverse expression.

The determinant calculation preserves orientation. Put $r = N - q + 1$ and $L_N = [B_N \ s_N]$. In the ordered monomial bases, $[s_N \ B_N]$ has determinant one, since χ is monic and the lower diagonal block is triangular with diagonal one. Thus $\det L_N = (-1)^{qr}$. The upper-left block of $L_N^* M_N L_N$ is H_N , and its Schur complement is G_N . Consequently $\det H_N \det G_N = |(-1)^{qr}|^2 \det M_N = \det M_N$. Equivalently the map

$$\det \mathcal{P}_{N-q} \otimes \det E \longrightarrow \det \mathcal{P}_N, \quad (y_1 \wedge \cdots \wedge y_r) \otimes (x_1 \wedge \cdots \wedge x_q) \longmapsto B_N y_1 \wedge \cdots \wedge B_N y_r \wedge s_N x_1 \wedge \cdots \wedge s_N x_q \quad (\text{AT10})$$

is the fixed determinant-line isomorphism. Replacing any lift by a relation does not change its wedge. Orthogonal lifts give the product metric just computed. This proves the determinant identity including the empty relation space, without changing any scalar mass.

Differentiate $J_N R_N = I$; then $\dot{R}_N$ lies in $\text{im } B_N$. Differentiating $B_N^* M_N R_N = 0$ determines its unique coefficient and proves the first identity of (AT7). On differentiating $R_N^* M_N R_N$, the two terms involving $\dot{R}_N$ vanish by that orthogonality. A second differentiation gives the two equal negative terms in (AT7), since ΔM_N is Hermitian. This proves the claimed second derivative and its sign. Applying the finite determinant derivative to (AT6) gives the first equality of (AT8); applying it to G_N gives the second. One more derivative gives (AT9). The first trace there is the squared Hilbert–Schmidt norm of $H_N^{-1/2} B_N^* \Delta M_N R_N G_N^{-1/2}$; the second is the squared Hilbert–Schmidt norm of $G_N^{-1/2} \dot{G}_N G_N^{-1/2}$. These identities establish their signs and retain the original forms. All inverse matrices are smooth on a neighborhood of the compact interval $[0, 1]$, by positivity and continuity. No infinite-dimensional inverse has been used. $\square$

The derivative $\dot{R}_N$ in (AT7) is a literal polynomial relation. Applying the original $\mathcal{V}$ in (AT2) therefore retains its original theta primitive and sends its arithmetic class to zero; $J^{(k)} \mathcal{V} R_N(t) = \eta_{h,k}$ for every t . At each nonempty original support label A , the cochain identities lift by $(A, P) \mapsto (A, fP)$, while $\tau \mapsto \tau$. Thus a relation-valued derivative has a supported zero class at label A . Its source vector and its full nonnegative energy in (AT7)–(AT9) remain present. These formulas compute their contribution rather than deleting it after taking cohomology.

The signed four-endpoint term on one retained source

Let $N_* = 2q$. Every space in (AT3) for the four endpoints sits in $\mathcal{P}_{N_*}$ by its actual coefficient inclusion. Give this common space the form $M_*(t) = M_{N_*}(t)$. Write $P_N(t)$ for its orthogonal projection onto $\mathcal{P}_N$, and $Q_N(t)$ for its orthogonal projection onto the relation subspace $\mathcal{R}_N = \chi \mathcal{P}_{N-q}$. The latter is zero at $N = q - 1$. For the coefficient inclusion $I_N : \mathcal{P}_N \hookrightarrow \mathcal{P}_{N_*}$, the formulas, with their domains retained, are

$$P_N = I_N M_N^{-1} I_N^* M_*, \quad Q_N = I_N B_N H_N^{-1} B_N^* I_N^* M_*, \quad C(t) = M_*(t)^{-1} \Delta M_*. \quad (\text{AT11})$$

Both P_N and Q_N are $M_*(t)$ -self-adjoint idempotents with their displayed images, verified by substitution. Since $\mathcal{R}_N \subset \mathcal{P}_N$, $P_N Q_N = Q_N P_N = Q_N$. Their difference is the orthogonal projection onto the finite canonical quotient section and has trace q . Cyclicity of finite matrix trace in (AT11) proves the exact common-source version of (AT8):

$$\frac{d}{dt} \log V_N(t) = \text{Tr}((P_N(t) - Q_N(t))C(t)). \quad (\text{AT12})$$

Put $i_0 = q - 1$, $i_1 = q$, $j_0 = 2q - 1$, and $j_1 = 2q$. Define the two positive operators

$$\begin{aligned} U(t) &= (Q_{j_0} - Q_{i_0}) + (Q_{j_1} - Q_{i_1}), \\ W(t) &= (P_{j_0} - P_{i_0}) + (P_{j_1} - P_{i_1}). \end{aligned} \quad (\text{AT13})$$

Each difference in this display is an orthogonal projection, because the two spaces in its pair are nested in the same Hilbert space. Each has rank q . The sums need not be projections and are not asserted to be orthogonal sums; they are positive operators with $\text{Tr } U = \text{Tr } W = 2q$.

Write

$$\begin{aligned}\mathcal{B}(t) &= \log \frac{V_{i_0}(t)V_{i_1}(t)}{V_{j_0}(t)V_{j_1}(t)}, \quad T_N = V_N(1)/V_N(0), \\ \Delta\mathcal{B} &= \mathcal{B}(1) - \mathcal{B}(0) = \log \frac{T_{q-1}T_q}{T_{2q-1}T_{2q}}.\end{aligned}\tag{AT14}$$

Subtracting the four exact formulas (AT12), with their original signs, and integrating gives

$$\boxed{\Delta\mathcal{B} = \int_0^1 \text{Tr}((U(t) - W(t))C(t)) dt.}\tag{AT15}$$

This is the actual arithmetic correction in the audit's formula (54). It compares the two added relation windows with the two added polynomial windows. Concavity of each individual $\log V_N(t)$ in (AT9) does not assign a sign to their signed combination in (AT15); that combination has now been computed.

There is also an exact pointwise integral, using no division by the arithmetic density. Define

$$\begin{aligned}k_N(t, u) &= v_N(u)M_N(t)^{-1}v_N(u)^*, \\ r_N(t, u) &= |\chi(S(u))|^2 v_{N-q}(u)H_N(t)^{-1}v_{N-q}(u)^*, \\ d_N(t, u) &= k_N(t, u) - r_N(t, u), \quad r_{q-1}(t, u) = 0, \\ d_N(t, u) &= v_N(u)R_N(t)G_N(t)^{-1}R_N(t)^*v_N(u)^* \geq 0.\end{aligned}\tag{AT16}$$

The last equality follows by writing the orthogonal splitting of M_N^{-1} as $B_N H_N^{-1} B_N^* + R_N G_N^{-1} R_N^*$; multiplying by M_N and evaluating on $\text{im } B_N \oplus \text{im } R_N$ proves that inverse identity. In particular $\int d_N(t, u)m_t(u) du = q$. Consequently (AT8) also reads

$$\frac{d}{dt} \log V_N(t) = \int d_N(t, u) \Delta m(u) du, \quad \Delta\mathcal{B} = \int_0^1 \int (d_{i_0} + d_{i_1} - d_{j_0} - d_{j_1})(t, u) \Delta m(u) du dt. \tag{AT17}$$

All coefficients of these fixed-degree polynomials in u and $\bar{S}(u)$ are bounded for $t \in [0, 1]$. Since $|\Delta m| \leq m_0 + m_1$ and these measures have the required finite moments, absolute integrability proves the exchange of integrals. This remains valid at every zero of m_1 .

An exact finite-source bound for the correction

Let $b_{\min}$ and $b_{\max}$ be the extreme eigenvalues of the actual positive $M_*(0)$ -self-adjoint operator $D_* = M_*(0)^{-1}M_*(1)$, and set $\kappa_* = b_{\max}/b_{\min}$. These are finite original source data at the common cutoff $N_* = 2q$. They are not presumed uniform in either the packet or k .

Theorem 12.2. *The exact correction on the original finite source satisfies*

$$\boxed{|\Delta\mathcal{B}| \leq 2q \log \kappa_*}.\tag{AT18}$$

Equivalently, retaining the original Gamma and arithmetic endpoint quantities,

$$\mathcal{B}_{h,k}^\Gamma - 2q \log \kappa_* \leq \mathcal{B}_{h,k}^{\text{ar}} \leq \mathcal{B}_{h,k}^\Gamma + 2q \log \kappa_*.\tag{AT19}$$

Proof. The operator $C(t)$ is $M_*(t)$ -self-adjoint. If its extreme eigenvalues are $c_{\min}(t), c_{\max}(t)$, positivity of U, W and their exact traces give

$$|\mathrm{Tr}((U - W)C)| \leq 2q(c_{\max}(t) - c_{\min}(t)). \quad (\text{AT20})$$

For example $\mathrm{Tr}(UC) \leq c_{\max} \mathrm{Tr} U$ follows by taking the trace of the product of U and the positive operator $c_{\max}I - C$ in the original Hilbert space. The lower bound is identical, and subtraction proves (AT20), without assuming that U, W, C commute.

In the fixed coefficient space,

$$C(t) = ((1 - t)I + tD_*)^{-1}(D_* - I).$$

Thus each eigenvalue is $c_b(t) = (b - 1)/(1 - t + tb)$ for an eigenvalue $b > 0$ of D_* . Its derivative with respect to b is $(1 - t + tb)^{-2} > 0$. The extreme values therefore correspond to the same $b_{\min}, b_{\max}$ throughout the interval, including repeated eigenvalues. Direct integration gives

$$\int_0^1 (c_{\max}(t) - c_{\min}(t)) dt = \log b_{\max} - \log b_{\min} = \log \kappa_*. \quad (\text{AT21})$$

Integrating (AT20) in the exact identity (AT15) proves (AT18). No pointwise multiplier bound or asymptotic estimate has entered. $\square$

When $M_*(1) = cM_*(0)$ with its literal $c > 0$, the exact source form at time t is $(1 - t + tc)M_*(0)$. Each quotient determinant is $(1 - t + tc)^q V_N(0)$, so all four factors in (AT14) cancel and $\Delta\mathcal{B} = 0$. This is also forced by $\kappa_* = 1$ in (AT18). The individual masses and volume factors remain as written; cancellation occurs only in this specified ratio. An arbitrary fixed change of the coefficient frame conjugates D_* , transports every subspace and its projection, and multiplies each quotient determinant by the same squared frame determinant. Hence (AT15)–(AT21) preserve their original values under that explicitly stated map.

For the reflection-stable quartet calculation of the supplied Gamma audit, let its actual displacement be $\delta > 0$ and retain $q = [1 + k(m - 1)](k + 1)^2$. Its fully written Gamma theorem gives $\mathcal{B}_{h,k}^\Gamma \geq 4q \log(\delta k / (2\sqrt{5}))$. Substitution in (AT19) gives the new quantitative comparison on the same source:

$$\mathcal{B}_{h,k}^{\mathrm{ar}} \geq 4q \log\left(\frac{\delta k}{2\sqrt{5}}\right) - 2q \log \kappa_*, \quad \log \kappa_* \geq 2 \log\left(\frac{\delta k}{2\sqrt{5}}\right) - \frac{\mathcal{B}_{h,k}^{\mathrm{ar}}}{2q}. \quad (\text{AT22})$$

The quartet and actual packet restriction are retained in this application; a reference test polynomial has not been declared to divide 2ξ . Equations (AT15) and (AT17) give the full signed quantity behind this bound. In particular they provide the exact integrand that an arithmetic cancellation calculation must evaluate, with both relation windows and both polynomial windows present. No subthreshold arithmetic bound or weight purity is asserted by this source comparison alone.

Source dependencies and precise status

The complete retained Tau Base note supplies (AT1), its tensor pushforward, right adjoint and supported maps. The original arithmetic-input and cyclic-sum proofs supply (AT2), full packet units and the theta primitives. The Gamma/source-density notes supply the actual measures and finite moments in (AT4)–(AT5). The new Deligne-bound note supplies the full quartet Gamma lower bound used only in (AT22). Each is preserved in the accompanying source closure, with its actual edition and hash.

The new proved calculations here are (AT6)–(AT21) on those unchanged input objects, and their substitution (AT22). They give an analytic calculation on the canonical finite complex mapping to $\mathbf{A}_{\tau,k}(\mathcal{T}^{\boxtimes k})$. They neither replace that pushforward by a generic stalk nor deduce a verdict about the entire tau-based program from an auxiliary model.

Chapter 13

The full relation-window spectrum and its bound

This calculation continues arithmetic determinant transport on its original finite source over the absolute tau base. It computes the full spectra of the two retained window operators, constructs an isometry intertwining them in the actual source metric, and strengthens the signed determinant bound. The source measures, their masses, the cyclic polynomial, the Taylor unit, and the cohomology observation remain fixed.

The original source and its four windows

Use a full actual packet h for $g = 2\xi$, an integer $k \geq 1$, and the original cyclic polynomial $\chi = \chi_{h,k}$ of degree $q \geq 1$. Let $E = \mathbb{C}[S]/(\chi)$, $E_h = \mathbb{C}[s]/(h)$, and $v_h = j_h(g/h) \in E_h^\times$. The original tensor-sum observation is

$$\eta : E \hookrightarrow E_h^{\otimes k}, \quad \eta[P] = v_h^{\otimes k} P(A_h^{(k)}) 1. \quad (\text{AW1})$$

The original section $\sigma_h : E_h \hookrightarrow Q = \mathcal{B}/\Theta V$ gives the separate injection $\sigma_h^{\otimes k} \eta : E \hookrightarrow Q^{\otimes k}$. Here $Q^{\otimes k}$ is the top cohomology of the original tensor pushforward over $\mathbf{F}_{1,\tau} = \{\tau, 1\}$; the one-factor complex is $[V \oplus V \xrightarrow{\Theta(\phi - \psi)} \mathcal{B}]$. The complete Tau Base and cyclic source proofs accompany the preceding arithmetic determinant transport calculation (AT).

For $N \geq q - 1$, retain the increasing-power spaces and maps

$$0 \longrightarrow \mathcal{P}_{N-q} \xrightarrow{B_N = \times \chi} \mathcal{P}_N \xrightarrow{J_N} E \longrightarrow 0, \quad \mathcal{P}_{-1} = 0. \quad (\text{AW2})$$

The common source is $H = \mathcal{P}_{2q}$, of dimension $n = 2q + 1$. Its original coordinate on the integration line is $S = k/2 + iu$. Set $v_N(u) = (1, S(u), \dots, S(u)^N)$ and retain the two densities

$$\begin{aligned} m_1(u) &= w_h^{*k}(u), & w_h(u) &= \frac{|(g/h)(1/2 + iu)|^2}{2\pi}, \\ m_0(u) &= \frac{(\sqrt{2\pi})^k |\Gamma(k/4 + iu/2)|^2}{c_{k/4} 2\pi}, & c_a &= 2^{1-2a} \Gamma(2a). \end{aligned} \quad (\text{AW3})$$

Their actual masses are μ_h^k and $(2\pi)^{k/2}$. For the real parameter $0 \leq t \leq 1$, put $m_t = (1-t)m_0 + tm_1$, $M_N(t) = \int v_N^* v_N m_t$, and $M(t) = M_{2q}(t)$. The source density proofs give all finite moments and positivity almost everywhere, so all these finite Grams are positive definite. We use the original form $\langle x, y \rangle_t = x^* M(t) y$ on H .

Let $P_N(t)$ be the orthogonal projection onto $\mathcal{P}_N \subset H$, and $Q_N(t)$ the projection onto $\chi\mathcal{P}_{N-q} \subset H$. Set $Q_{q-1} = 0$. For the coefficient inclusion $I_N : \mathcal{P}_N \rightarrow H$, their fully typed coefficient formulas are

$$\begin{aligned} P_N &= I_N M_N^{-1} I_N^* M, \\ Q_N &= I_N B_N (B_N^* M_N B_N)^{-1} B_N^* I_N^* M. \end{aligned} \quad (\text{AW4})$$

The zero relation domain at $N = q-1$ contributes the zero operator. Let $G_N(t)$ be the minimum-lift Gram on E from (AW2) and $V_N(t) = \det G_N(t)$. Retain

$$\begin{aligned} \mathcal{B}(t) &= \log \frac{V_{q-1}(t) V_q(t)}{V_{2q-1}(t) V_{2q}(t)}, \\ C(t) &= M(t)^{-1} (M(1) - M(0)), \\ U(t) &= (Q_{2q-1} - Q_{q-1}) + (Q_{2q} - Q_q), \\ W(t) &= (P_{2q-1} - P_{q-1}) + (P_{2q} - P_q). \end{aligned} \quad (\text{AW5})$$

The AT proof, included in full, establishes $V_N = \det M_N / \det(B_N^* M_N B_N)$ with the original orientation $(-1)^{q(N-q+1)}$, and

$$\Delta \mathcal{B} = \mathcal{B}(1) - \mathcal{B}(0) = \int_0^1 \text{Tr}((U(t) - W(t))C(t)) dt. \quad (\text{AW6})$$

Thus the following computation evaluates the operators in the already proved signed arithmetic correction.

Full spectra and an explicit source isometry

Theorem 13.1. *In the original $M(t)$ metric, both $U(t)$ and $W(t)$ have spectrum*

$$0 \text{ with multiplicity } q, \quad 1 \text{ with multiplicity } 2, \quad 2 \text{ with multiplicity } q-1. \quad (\text{AW7})$$

The last multiplicity is zero when $q = 1$. There is an explicitly constructed smooth $M(t)$ -unitary $T(t) : H \rightarrow H$ such that $U(t) = T(t)W(t)T(t)^{-1}$.

Proof. Every subspace in either of the following flags is the literal displayed polynomial subspace, with its original coefficient inclusion. For the relation flag use

$$F_0 = \chi\mathcal{P}_0 \subset F_1 = \chi\mathcal{P}_{q-1} \subset F_2 = \chi\mathcal{P}_q \subset H. \quad (\text{AW8})$$

Their dimensions are $1, q, q+1$, since multiplication by χ is injective. When $q = 1$, $F_0 = F_1$ is retained as that equality. Projections of nested subspaces commute and their products equal the projection of the smaller subspace: each larger space is the orthogonal direct sum of the smaller space and its orthogonal complement within the larger space. On the original orthogonal decomposition $F_0 \oplus (F_1 \cap F_0^\perp) \oplus (F_2 \cap F_1^\perp) \oplus F_2^\perp$, the operator $U = Q_{F_1} + Q_{F_2} - Q_{F_0}$ acts respectively by $1, 2, 1, 0$. The dimensions are respectively $1, q-1, 1, q$.

For W use $\mathcal{P}_{q-1} \subset \mathcal{P}_q \subset \mathcal{P}_{2q-1} \subset \mathcal{P}_{2q} = H$. On the four corresponding orthogonal pieces, W acts by $0, 1, 2, 1$, with dimensions $q, 1, q-1, 1$. This proves (AW7), including the repeated middle subspaces at $q = 1$.

For an explicit intertwiner retain every Gram factor as follows. For any specified ordered independent list $z_0, \dots, z_d$, apply the recursion

$$y_j = z_j - \sum_{i < j} e_i \langle e_i, z_j \rangle_t, \quad a_j = (y_j^* M(t) y_j)^{1/2} > 0, \quad e_j = y_j / a_j. \quad (\text{AW9})$$

The positive real square root fixes its phase. No source density or mass is changed; each a_j is the norm in the original form. Independence proves $y_j \neq 0$ at every step, so this recursion is defined and smooth on the entire parameter interval.

First use $z_j = S^j$, $0 \leq j \leq 2q$, obtaining $p_0, \dots, p_{2q}$. Next use $z_j = \chi S^j$, $0 \leq j \leq q$, obtaining $r_0, \dots, r_q$. These lie in the successive relation spaces in (AW8). Finally let $R_{2q}(t) : E \rightarrow H$ be the original minimum section from AT6 and apply (AW9) to $R_{2q}(t)[1], \dots, R_{2q}(t)[S^{q-1}]$, obtaining $c_0, \dots, c_{q-1}$. The relation orthogonality in AT6 puts this entire list in $F_2^\perp$. Since $J_{2q}R_{2q} = I_E$, the list is independent and spans that q -dimensional complement.

Define T on the first orthonormal basis by

$$\begin{aligned} Tp_j &= c_j & (0 \leq j \leq q-1), \\ Tp_q &= r_0, & Tp_{2q} = r_q, \\ Tp_{q+j} &= r_j & (1 \leq j \leq q-1). \end{aligned} \tag{AW10}$$

The middle range is empty for $q = 1$, with p_q and p_{2q} still distinct. The target list is an orthonormal basis. Thus this linear map is invertible and $T^*MT = M$. The spectral actions already calculated agree on each indicated basis vector, proving $UT = TW$. The recursion, the smooth minimum section and the positive norms prove the asserted smoothness. This completes the construction. $\square$

The exact isometry gives a commutator form of the signed integrand:

$$\text{Tr}((U - W)C) = \text{Tr}(WT^{-1}[C, T]), \quad [C, T] = CT - TC. \tag{AW11}$$

Indeed cyclicity moves T in $\text{Tr}(TWT^{-1}C)$ to the right, and subtraction then gives the displayed product. This T is an isometry of the original source; its cohomology observation is computed, rather than silently assigned an intertwining property. The unchanged arithmetic observation $\mathcal{O} = \sigma_h^{\otimes k} \eta J_{2q} : H \rightarrow Q^{\otimes k}$ satisfies

$$\mathcal{O}T = \sigma_h^{\otimes k} \eta J_{2q} T, \quad \mathcal{O}T - \mathcal{O} = \sigma_h^{\otimes k} \eta J_{2q} (T - I), \quad \ker(\mathcal{O}T) = T^{-1}F_2. \tag{AW12}$$

The last identity follows from both injections in (AW1) and $\ker J_{2q} = F_2$. By (AW10), $T^{-1}F_2$ is exactly $\text{span}\{p_q, \dots, p_{2q}\} = \mathcal{P}_{q-1}^\perp$ in H . At any fixed nonempty support label A , lift every displayed linear map by $(A, x) \mapsto (A, fx)$ and send τ to τ . Consequently the supported-zero inverse image of the lifted $\mathcal{O}T$ at A is the full set $\{A\} \times \mathcal{P}_{q-1}^\perp$. The Taylor unit in (AW1) and the defect in (AW12) remain present.

The spectral bound with every eigenvalue retained

Let $b_1 \leq \dots \leq b_n$ be all eigenvalues, counted with multiplicity, of $D = M(0)^{-1}M(1)$ on its original $M(0)$ Hilbert space. All are positive. Define

$$\mathcal{L}_q(D) = \log \frac{b_{q+2}}{b_q} + 2 \sum_{j=1}^{q-1} \log \frac{b_{2q+2-j}}{b_j}, \quad \kappa(D) = b_n/b_1. \tag{AW13}$$

For $q = 1$ the sum is empty and $\mathcal{L}_1(D) = \log(b_3/b_1)$.

Theorem 13.2. *On the unchanged arithmetic source,*

$$|\Delta \mathcal{B}| \leq \mathcal{L}_q(D) \leq (2q - 1) \log \kappa(D). \tag{AW14}$$

In particular the coefficient $2q$ in AT18 can be improved to $2q - 1$, while the first bound retains the full relative spectrum.

Proof. We first prove the needed finite spectral trace inequality. Let C be self-adjoint in a finite Hilbert space with eigenvalues $c_1 \leq \dots \leq c_n$. If P is an orthogonal projection of rank r , let e_j be an orthonormal eigenbasis of C and $d_j = \langle e_j, P e_j \rangle$. Then $0 \leq d_j \leq 1$ and $\sum_j d_j = r$, since $d_j = \|P e_j\|^2$ and $\text{Tr } P = r$. Thus $\text{Tr}(PC) = \sum_j c_j d_j$. Its upper bound by $\sum_{j=n-r+1}^n c_j$ follows by subtracting the two expressions: the mass $\sum_{j \leq n-r} d_j$ equals $\sum_{j > n-r} (1 - d_j)$, and every coefficient in the latter group is at least every coefficient in the former. Pairing the same nonnegative mass, or bounding both groups by c_{n-r+1} , proves the inequality. Applying this argument to $-C$ proves the lower bound $\sum_{j=1}^r c_j$. This proof includes repeated eigenvalues and the ranks zero and n .

For an operator Z with spectrum (AW7), its spectral projection A_Z onto eigenvalues 1, 2 has rank $q + 1$, and its spectral projection B_Z onto eigenvalue 2 has rank $q - 1$. The exact decomposition is $Z = A_Z + B_Z$. Applying the proved projection bounds gives

$$\sum_{j=1}^{q+1} c_j + \sum_{j=1}^{q-1} c_j \leq \text{Tr}(ZC) \leq \sum_{j=q+1}^{2q+1} c_j + \sum_{j=q+3}^{2q+1} c_j. \quad (\text{AW15})$$

Each bound is attained for an abstract operator of this spectrum by assigning its eigenspaces to the appropriate eigenvectors of C . This attainment statement is only about the displayed spectral constraint; it assigns no freely chosen orientation to the actual polynomial relation spaces. Both U and W satisfy (AW15). Subtract their bounds and cancel the common c_{q+1} term only after writing both complete sums:

$$|\text{Tr}((U - W)C)| \leq c_{q+2} - c_q + 2 \sum_{j=1}^{q-1} (c_{2q+2-j} - c_j). \quad (\text{AW16})$$

No mutual commutation of U, W, C is used.

For the actual interpolated metric, $C(t) = ((1 - t)I + tD)^{-1}(D - I)$, so its eigenvalues in increasing order are

$$c_j(t) = \frac{b_j - 1}{1 - t + t b_j}. \quad (\text{AW17})$$

The denominator is positive and the derivative with respect to b_j is $(1 - t + t b_j)^{-2} > 0$. Thus the ordering is valid for the entire closed interval. Each $c_j(t)$ is the derivative of $\log(1 - t + t b_j)$ and has integral $\log b_j$. Integrating (AW16) in the exact signed identity (AW6) proves the first inequality of (AW14). Every ratio in (AW13) lies between 1 and b_n/b_1 , so retaining its coefficient gives $1 + 2(q - 1) = 2q - 1$ times $\log \kappa(D)$ for the second bound. This proves the theorem for all $q \geq 1$. $\square$

All bounds retain the literal source masses. If $M(1) = cM(0)$, every $b_j = c$, so (AW13) is zero. Directly each original quotient determinant is multiplied at time t by $(1 - t + t c)^q$, and the four displayed factors in (AW5) cancel. For a fixed invertible source-coordinate arrow F , the transported forms are $M' = F^* M F$, all subspaces have their F^{-1} images, and $D' = F^{-1} D F$. Hence the eigenvalues, (AW6) and (AW13) are unchanged under this explicit map.

The original quartet Gamma theorem, when applied to the actual full quartet packet with displacement $\delta > 0$ and common order m , retains $q = [1 + k(m - 1)](k + 1)^2$ and gives

$$\mathcal{B}_{h,k}^{\text{ar}} \geq 4q \log \left(\frac{\delta k}{2\sqrt{5}} \right) - \mathcal{L}_q(D). \quad (\text{AW18})$$

This is substitution of the complete supplied Gamma theorem into the proved first inequality of (AW14). It proves no uniform estimate for the eigenvalues b_j , and asserts no existence of an off-line packet. The new computed objects are the exact window spectra, their source isometry, its arithmetic observation and commutator, and the strengthened finite bound.

The one-dimensional quotient: exact angle and nonlinear transport

When $q = 1$, the same four endpoints are 0, 1, 1, 2. Their middle factors cancel in the specified ratio, leaving $\mathcal{B}(t) = \log(V_0(t)/V_2(t))$ with both original masses still present. Write $\chi = S - a$. The actual operators now are $U = Q_2$ and $W = I - P_0$. If $\mathcal{N} = (\chi\mathcal{P}_1)^\perp$ and $P_{\mathcal{N}} = I - Q_2$, then $U - W = P_0 - P_{\mathcal{N}}$. Both projections have rank one.

Let $e = 1 \in H$, $z = R_2(t)[1]$, and retain their actual norms $\langle e, e \rangle_t = V_0(t)$ and $\langle z, z \rangle_t = V_2(t)$. The source splitting gives $z = P_{\mathcal{N}}e$, since $J_2e = 1$. Thus $\langle e, z \rangle_t = V_2(t)$, and the angle between the two lines has the exact value

$$\cos^2 \theta(t) = \frac{|\langle e, z \rangle_t|^2}{\langle e, e \rangle_t \langle z, z \rangle_t} = \frac{V_2(t)}{V_0(t)} = e^{-\mathcal{B}(t)}. \quad (\text{AW19})$$

Both Grams are positive, so the cosine is positive. It is strictly less than one for the actual measure (AW3). Otherwise $e \in \mathcal{N}$, implying $\int \chi(S)m_t = 0$ and $\int S\chi(S)m_t = 0$. On the unchanged line $\bar{S} = k - S$, and consequently $|\chi(S)|^2 = ((k - \bar{a}) - S)\chi(S)$. The two equalities would force $\int |\chi(S)|^2 m_t = 0$, contrary to the positive Gram of this nonzero polynomial. Hence $0 < \theta(t) < \pi/2$ and $\mathcal{B}(t) > 0$ on the whole closed interval.

For completeness, take the unit vector $p = e/\sqrt{V_0}$ and write $z/\sqrt{V_2} = \cos \theta p + \sin \theta r$, where r is the resulting unit vector orthogonal to p . The phase is already fixed since $\langle e, z \rangle_t = V_2 > 0$. In this orthonormal pair the exact difference of projections is

$$(P_0 - P_{\mathcal{N}})|_{\text{span}(p,r)} = \begin{pmatrix} \sin^2 \theta & -\sin \theta \cos \theta \\ -\sin \theta \cos \theta & -\sin^2 \theta \end{pmatrix}. \quad (\text{AW20})$$

Its trace is zero and its determinant is $-\sin^2 \theta$. It vanishes on the remaining one-dimensional orthogonal complement. Thus its full eigenvalues are $\sin \theta, 0, -\sin \theta$. Applying the proved rank-one spectral trace bounds to its positive and negative eigenprojections yields

$$|\dot{\mathcal{B}}(t)| \leq \sqrt{1 - e^{-\mathcal{B}(t)}} (c_3(t) - c_1(t)). \quad (\text{AW21})$$

This retains the actual alignment of the two lines.

The function $\Phi(B) = 2 \operatorname{arcosh}(e^{B/2})$ for $B > 0$ has derivative $\Phi'(B) = 1/\sqrt{1 - e^{-B}}$, by differentiating the inverse of the hyperbolic cosine on its positive branch. The preceding strict positivity on the compact interval justifies its smooth composition with $\mathcal{B}(t)$. Divide (AW21) by its positive square-root factor and integrate (AW17). The exact nonlinear endpoint bound is

$$|2 \operatorname{arcosh}(e^{\mathcal{B}(1)/2}) - 2 \operatorname{arcosh}(e^{\mathcal{B}(0)/2})| \leq \log(b_3/b_1). \quad (\text{AW22})$$

Equivalently, with the original values $A_0 = \operatorname{arcosh}(e^{\mathcal{B}(0)/2})$ and $L = \frac{1}{2} \log(b_3/b_1)$, monotonicity of $2 \log \cosh A$ for $A \geq 0$ gives

$$2 \log \cosh(\max\{0, A_0 - L\}) \leq \mathcal{B}(1) \leq 2 \log \cosh(A_0 + L). \quad (\text{AW23})$$

This last lower bound can be zero although the actual $\mathcal{B}(1)$ is strictly positive, as proved above. Every step concerns the original $q = 1$ quotient and its degree-two source. No extension of that dimension restriction to a quartet quotient is asserted.

Complete proof dependencies

The complete AT calculation accompanies this note, together with its 29 complete source dependencies and independent proof review. Those sources prove the original pushforward, tensor signs, full Taylor-unit injections, theta primitive, positive arithmetic and Gamma moments, and the quartet application. Every new assertion (AW7)–(AW23) is proved above on those unchanged finite objects. A separate independent review supplies the spectral calculation and its edge-case checks.

Chapter 14

The actual four-window counterfactual interval

Actual bounds for the retained four-window costs

Continue with (HT1)–(HT18), now with $k \geq 3$. All unadorned source norms and quotient volumes in this section are those of the original $m_{h,k}(u) du$, with $g = 2\xi$ and the complete hypothetical quartet. Write $\mathcal{L} = \mathcal{L}_{h,k}$ and retain $q = \ell_k(k+1)^2$. The threshold-four lower bound is inherited from *Consecutive identities and the first window*, (CJ.9)–(CJ.19); it is already explicitly present in the recovered transcript at lines 4202–4228. It is not a new claim of the present continuation.

Every term of the inherited lower bound

Put $d_j = V_j/V_{j-1}$ for $j \geq q$. For the two original windows $I_- = (q-1, 2q-1)$ and $I_+ = (q, 2q)$, retain the contraction penalties

$$\begin{aligned} P_- &= -\log(1 - d_{2q-1}) - 2 \sum_{j=q}^{2q-2} \log(1 - d_j), \\ P_+ &= -\log(1 - d_q) - \log(1 - d_{2q}) - 2 \sum_{j=q+1}^{2q-1} \log(1 - d_j). \end{aligned} \tag{HC1}$$

The source identities prove $0 < d_j < 1$ in these ranges. The real phase ϕ_N is defined by the retained source pairing $b_N^* G_N b_{N+1} = i\omega_N \phi_N$. The first window uses its original boundary identity; no inverse at degree $q-2$ occurs. Define weights

$$w_{q-1} = w_{2q-1} = 1, \quad w_N = 2 \quad (q \leq N \leq 2q-2), \quad \sum_{N=q-1}^{2q-1} w_N = 2q.$$

Adding the two exact source products gives

$$\mathcal{B}_k := \log \frac{V_{q-1} V_q}{V_{2q-1} V_{2q}} = \sum_{N=q-1}^{2q-1} w_N \log(\varepsilon_N^2 + \phi_N^2) - \log \frac{\omega_{2q-1} \omega_{2q}}{\omega_{q-1} \omega_q} + P_- + P_+. \tag{HC2}$$

Here every $\varepsilon_N \geq \mathcal{L}$, including the first degree.

For clarity, retain all constants in the inherited comparable-window bound. With $\alpha = \pi/2$, $B =$

$42 + 8m$, and $\vartheta_h = \int_{-1}^1 w_h$, put

$$\begin{aligned} t_h &= \min(1, \vartheta_h), \quad c_0 = \min(1, 2c_h/3), \quad \ell_h = \frac{1}{2}e^{-\alpha}\sqrt{c_0 t_h} 2^{-B/2}, \\ X &= \max(1, M_h(b), M_h(-b)), \quad U_h = 4\sqrt{X}/b, \\ C_h^{\text{win}} &= 27 \max(1, U_h)^3 \max(1, \ell_h^{-1})^2, \quad D_h = \frac{\delta}{2C_h^{\text{win}}}. \end{aligned} \tag{HC3}$$

Here $0 < b < \pi/2$, and $c_h > 0$ is the original proved convolution constant. The source proves $(\omega_{n+r}/\omega_n)^{1/(2r)} \leq C_h^{\text{win}} n$ for $n \geq k \geq 3$, $n/2 \leq r \leq 2n$. Thus it applies to both literal windows, and $0 < D_h < 1/108$.

The exact residual in (HC2) has the following five nonnegative pieces:

$$\begin{aligned} \mathcal{R}_k &= P_- + P_+ + \sum_N w_N \log(1 + \phi_N^2/\mathcal{L}^2) \\ &\quad + \sum_N w_N \log \frac{\varepsilon_N^2 + \phi_N^2}{\mathcal{L}^2 + \phi_N^2} \\ &\quad + 4q \log(C_h^{\text{win}} q) - \log \frac{\omega_{2q-1} \omega_{2q}}{\omega_{q-1} \omega_q} + 4q \log \frac{2\mathcal{L}}{\delta k q}. \end{aligned} \tag{HC4}$$

Both sums use exactly $q-1 \leq N \leq 2q-1$. Nonnegativity follows, respectively, from $0 < d_j < 1$, the squared phases, (HT9), the two comparable-window estimates, and $\mathcal{L}/q \geq \delta k/2$. Expanding each logarithm in (HC4), with $D_h = \delta/(2C_h^{\text{win}})$, proves the exact identity

$$\boxed{\mathcal{B}_k = 4q \log(D_h k) + \mathcal{R}_k, \quad \mathcal{R}_k \geq 0.} \tag{HC5}$$

This retains, rather than discards, all the contraction, phase, norm, and trace savings in the original lower bound.

A proved upper bound for these same four original volumes

The original convolution estimate and Laplace moments are

$$m_{h,k}(u) \geq c_h \vartheta_h^{k-3} e^{-\alpha(|u|+k-3)} (1 + |u| + k - 3)^{-B}, \quad M_h(\pm b) = \int_{\mathbb{R}} e^{\pm bu} w_h(u) du.$$

Write $\mu_h = \int_{\mathbb{R}} w_h(u) du$ for the original source mass. Put

$$\begin{aligned} R_0 &= \sqrt{\delta^2 + \gamma^2}, \quad D = \max(1, 2/b, 2R_0), \quad T = Dq, \quad r = kR_0/T \leq 1/2, \\ a_k(T) &= c_h \vartheta_h^{k-3} e^{-\alpha(T+k-3)} (1 + T + k - 3)^{-B}, \\ M_k &= M_h(b)^k + M_h(-b)^k, \quad U_k(T) = \mu_h^k + \frac{(2q-2)!}{(bT)^{2q-2}} M_k, \\ A_N &= \sqrt{\frac{2N+1}{2}} \frac{4^{N+1}-1}{3}, \quad B_q(r) = \left(\frac{1+r}{1-r}\right)^q, \quad C_N = \frac{U_k(T) A_N^2 B_q(r)^2}{T a_k(T)}. \end{aligned} \tag{HC6}$$

The capital D in the interval scale is distinct from the inherited lower constant D_h and from the source differential. All masses, coordinates, and root multiplicities remain as in (HT1)–(HT7). One has the actual finite Hermitian comparisons

$$G_{q-1} \preceq C_{2q-1} G_{2q-1}, \quad G_q \preceq G_{q-1} \preceq C_{2q} G_{2q}.$$

Consequently, with the exact ratio

$$\rho_q = \frac{4q+1}{4q-1} \left(\frac{4^{2q+1}-1}{4^{2q}-1} \right)^2,$$

$$\boxed{0 \leq \mathcal{B}_k \leq q \log C_{2q-1} + q \log C_{2q} = 2q \log C_{2q-1} + q \log \rho_q.} \tag{HC7}$$

Proof of the degree extension and its maps. Use the exact substitution $f(x) = P(c + iTx)$ with inverse $P(S) = f((S - c)/(iT))$. Its source measure is $Tm_{h,k}(Tx) dx$, with mass μ_h^k , and its monic relation is $\chi_T(x) = (iT)^{-q}\chi(c + iTx)$. The roots $z_1, \dots, z_q$, listed with their repetitions, have modulus at most r . The raw jet of order d acquires exactly $(iT)^d$; the quotient determinant acquires $T^{-q(q-1)}$. This common factor cancels in each displayed quotient-volume ratio.

The Legendre expansion on $[-1, 1]$ gives $\|f\|_{\text{coeff},1} \leq A_N \|f\|_{L^2([-1,1])}$ for $\deg f \leq N$. Indeed Rodrigues' polynomial has coefficient norm at most $\binom{2j}{j} \leq 4^j$ and squared integral $2/(2j+1)$. Pairing the expansion with this polynomial bounds its coefficient by $\sqrt{(2j+1)/2} \|f\|_2$; summing the geometric series gives exactly A_N in (HC6).

Write $\prod_a (1 - z_a t)^{-1} = \sum_{j \geq 0} h_j t^j$. Geometric-series multiplication gives $|h_j| \leq \binom{q+j-1}{j} r^j$ and $\|\chi_T\|_{\text{coeff},1} \leq (1+r)^q$. For every integer $s \geq 0$, monic division of x^{q+s} has quotient $\sum_{j=0}^s h_j x^{s-j}$. The inverse-series identity cancels all other terms of degree at least q . Removing the leading coefficient one therefore gives

$$\|\text{rem}_{\chi_T} x^{q+s}\|_{\text{coeff},1} \leq (1+r)^q \sum_{j=0}^s \binom{q+j-1}{j} r^j - 1 \leq B_q(r).$$

The bound is valid also for $s = q$, the extra degree required here. At degrees below q the remainder is the original monomial, whose coefficient norm is one. Linearity gives the same remainder-map bound for every degree at most $2q$.

For a remainder v of degree below q ,

$$\int |v(u/T)|^2 m_{h,k}(u) du \leq U_k(T) \|v\|_{\text{coeff},1}^2.$$

This follows by bounding each monomial by $\max(1, |u/T|^{q-1})$ and using $\int u^{2q-2} m_{h,k} \leq (2q-2)! b^{-(2q-2)} M_k$. Restriction of the original norm to $[-T, T]$ gives the lower bound $T a_k(T) \|f\|_{L^2([-1,1])}^2$. Combining these three estimates proves

$$\|\text{rem}_\chi P\|_{H_{q-1}}^2 \leq C_N \|P\|_{H_N}^2 \quad (N = 2q - 1, 2q).$$

The map is the literal remainder representative of the original jet. Apply it to the canonical least lift of $x \in E$ at degree N . Its norm at degree $q-1$ is at least $x^* G_{q-1} x$; this proves the Hermitian comparisons preceding (HC7). Inclusion of polynomial degrees gives $G_q \preceq G_{q-1}$ and monotonicity of all volumes. Taking determinants proves (HC7). Applying the same remainder inequality to any nonzero polynomial below degree q proves $C_N \geq 1$ without an additional assumption. $\square$

For an entirely scalar upper bound retain the original constants

$$K = \max(1, \vartheta_h^{-1}), \quad C_{\text{env}} = \max(1, 2/(3c_h D)), \quad a = \alpha D + \log 256,$$

and the original expression (AU.31)

$$\mathcal{U}_k = a q^2 + \left(\log(XK) + \alpha + \frac{16R_0}{3D} \right) k q + B q \log(1 + (D+1)q) + q \log C_{\text{env}}.$$

Then the proved four-window upper bound is

$$\boxed{\mathcal{B}_k \leq \widehat{\mathcal{U}}_k := 2\mathcal{U}_k + q \log 20.} \quad (\text{HC8})$$

To verify the constants, $bT \geq 2q$ implies the factorial ratio in (HC6) is at most one; also $\mu_h \leq X$ and $M_k \leq 2X^k$, so $U_k(T) \leq 3X^k$. Directly from (HC6),

$$A_{2q-1}^2 \leq (2q/9) 256^q, \quad A_{2q}^2 \leq (40q/9) 256^q.$$

The logarithmic derivative $2/(1-r^2) \leq 8/3$ on $[0, 1/2]$ gives $\log B_q(r)^2 \leq 16kR_0/(3D)$. Substitution into C_N yields $q \log C_{2q-1} \leq \mathcal{U}_k$ and $q \log C_{2q} \leq \mathcal{U}_k + q \log 20$. Adding proves (HC8). This argument does not use the false inequality $\rho_2 < 20$; in the actual quartet domain $q \geq 16$ anyway.

The exact compatible interval for the signed arithmetic correction

Let V_N^Γ denote the inherited Gamma-reference quotient volume for the *same* original polynomial χ , and define

$$T_N = V_N/V_N^\Gamma, \quad \mathcal{B}_k^\Gamma = \log \frac{V_{q-1}^\Gamma V_q^\Gamma}{V_{2q-1}^\Gamma V_{2q}^\Gamma}, \quad \Delta_k^\Gamma = \log \frac{T_{q-1} T_q}{T_{2q-1} T_{2q}}.$$

Cancellation of the four literal volume factors gives $\mathcal{B}_k = \mathcal{B}_k^\Gamma + \Delta_k^\Gamma$. No auxiliary polynomial is introduced and no sign is imposed on Δ_k^Γ . From (HC5) and (HC8) the actual retained residuals must belong to the following explicit intervals:

$$\boxed{\max\{0, -4q \log(D_h k)\} \leq \mathcal{R}_k \leq \widehat{\mathcal{U}}_k - 4q \log(D_h k),} \quad (\text{HC9})$$

$$\boxed{\max\{0, 4q \log(D_h k)\} - \mathcal{B}_k^\Gamma \leq \Delta_k^\Gamma \leq \widehat{\mathcal{U}}_k - \mathcal{B}_k^\Gamma.} \quad (\text{HC10})$$

These intervals have strictly positive width for every $k \geq 3$. Indeed all nonquadratic terms of $\mathcal{U}_k$ are nonnegative, $a > 1$, $q \geq (k+1)^2 > 2k$, and $D_h < 1$. Thus

$$\widehat{\mathcal{U}}_k \geq 2aq^2 > 4qk \geq \max\{0, 4q \log(D_h k)\}.$$

Moreover, keeping the fixed packet constants, each lower-order term divided by q^2 tends to zero, so

$$\lim_{k \rightarrow \infty} \frac{\widehat{\mathcal{U}}_k - \max\{0, 4q \log(D_h k)\}}{q^2} = 2(\alpha D + \log 256) > 0. \quad (\text{HC11})$$

This proves numerical compatibility of the stated actual bound functions. It does not construct a packet realizing an arbitrarily chosen point of either interval, and it does not assert the existence of the hypothetical zero. In particular the interval may lie entirely at negative values of Δ_k^Γ ; determinant-ratio positivity does not imply positivity of its signed four-window logarithm.

The holonomy energy is tied to the same endpoint contractions

Fix an actual holonomy θ and period L . In this paragraph all metrics, lifts and volumes carry that same phase. Put

$$e_N = \log(V_N/V_{N+1}), \quad \beta_N = \lambda_{\max}(G_{N+1}^{-1/2} G_N G_{N+1}^{-1/2}), \\ J_N^D = \|D_{L,\theta} R_N G_N^{-1/2}\|_{\text{HS}}^2, \quad \mathcal{E}_N = \|\mathcal{B}_N G_N^{-1/2}\|_{\text{HS}}^2.$$

These use original source moments through degree $2N+2$. They obey

$$\boxed{\frac{\mathcal{L}^2}{2} \leq \mathcal{E}_N \leq (1 + \sqrt{\beta_N})^2 J_N^D, \quad e_N \geq 2 \log \max \left\{ 1, \frac{\mathcal{L}}{\sqrt{2J_N^D}} - 1 \right\}.} \quad (\text{HC12})$$

The denominator is positive: integration by parts on the stated twisted domain gives $\|D_{L,\theta} f\|^2 = c^2 \|f\|^2 + \|\partial_r f\|^2$, whence $J_N^D \geq c^2 q > 0$.

To prove the upper inequality, $D_{L,\theta} R_N v$ is an original degree- $(N+1)$ representative of Av . The same quotient minimum therefore gives

$$A^* G_{N+1} A \preceq R_N^* D_{L,\theta}^* D_{L,\theta} R_N.$$

Use $G_N \preceq \beta_N G_{N+1}$, sandwich by $G_N^{-1/2}$, and take traces. This gives $\|R_N A G_N^{-1/2}\|_{\text{HS}}^2 \leq \beta_N J_N^D$. The triangle inequality in $\mathcal{B}_N = D_{L,\theta} R_N - R_N A$ proves the upper inequality in (HC12); (HT14) proves

the lower. All eigenvalues in the definition of β_N are at least one by degree monotonicity. In fact the original rank-one kernel update proves the exact identity

$$e_N = \log \beta_N, \quad \beta_N = 1 + \frac{b_{N+1}^* G_N b_{N+1}}{\omega_{N+1}}. \quad (\text{HC12a})$$

Here b_{N+1} and ω_{N+1} are the remainder column and monic squared norm for this same original phase measure. Indeed $K_{N+1} = K_N + b_{N+1} b_{N+1}^* / \omega_{N+1}$, so

$$K_N^{-1/2} K_{N+1} K_N^{-1/2} = I + vv^*, \quad v = K_N^{-1/2} b_{N+1} / \sqrt{\omega_{N+1}}.$$

The relative matrix defining β_N is $K_{N+1}^{1/2} K_N^{-1} K_{N+1}^{1/2} = X^* X$ for the invertible $X = K_N^{-1/2} K_{N+1}^{1/2}$. The displayed matrix $I + vv^*$ is XX^* , and $X(X^* X)X^{-1} = XX^*$ proves they have identical eigenvalues. These are 1 with multiplicity at least $q-1$ and $1 + \|v\|^2$. Taking the determinant gives $V_N/V_{N+1} = \det K_{N+1} / \det K_N = 1 + \|v\|^2 = \beta_N$, which proves (HC12a), including a zero update if one occurs. Rearranging the two energy bounds now proves the second inequality of (HC12) with this exact endpoint identity.

The same weights as in (HC2) reconstruct the actual phase volume:

$$\mathcal{B}_{k,\theta} = \sum_{N=q-1}^{2q-1} w_N e_N \geq 2 \sum_{N=q-1}^{2q-1} w_N \log \max \left\{ 1, \frac{\mathcal{L}}{\sqrt{2J_N^D}} - 1 \right\}. \quad (\text{HC13})$$

Thus a small original derivative-source energy at a degree has the explicit endpoint contraction cost (HC12); that cost enters the same four-window budget, with its actual multiplicity.

Finally the supplied holonomy construction allows any fixed $0 < \eta < 1$ by choosing the original period from its computed weighted source comparison at degree $2q+2$. It proves simultaneously $(1-\eta)G_N \preceq G_{N,\theta} \preceq (1+\eta)G_N$ at all these degrees. Taking determinants at the four original endpoints gives

$$|\mathcal{B}_{k,\theta} - \mathcal{B}_k| \leq 2q \log \frac{1+\eta}{1-\eta}. \quad (\text{HC14})$$

Consequently (HC13)'s right side is at most $\widehat{\mathcal{U}}_k + 2q \log((1+\eta)/(1-\eta))$. For the averaged original phase metrics the already proved stronger comparison $(1-\eta^2)G_N \preceq \overline{G}_N \preceq G_N$ gives instead $|\mathcal{B}_k - \mathcal{B}_k| \leq 2q \log(1/(1-\eta^2))$. These changes are of order q for a fixed η , whereas the compatible width (HC11) has positive quadratic order. The exact Laplacian residual (HT17) remains present at every phase. No displayed upper estimate for its norm, or for the original derivative moments in (HC12), contradicts these lower bounds.

The sharper finite comparison retains its exact error

The original lower-envelope measure is

$$d\nu_k(u) = c_h \vartheta_h^{k-3} e^{-\alpha(k-3)} e^{-\alpha|u|} (k-2+|u|)^{-B} du.$$

Let V_N^ν be its quotient volume for the identical full remainder map. The actual finite upper expression for the four-window volume is

$$\mathcal{U}_{\nu,k} = \log(V_{q-1} V_q) - \log(V_{2q-1}^\nu V_{2q}^\nu),$$

with the exact nonnegative error

$$\mathcal{U}_{\nu,k} - \mathcal{B}_k = \log \frac{V_{2q-1}}{V_{2q-1}^\nu} + \log \frac{V_{2q}}{V_{2q}^\nu} \geq 0. \quad (\text{HC15})$$

Indeed $m_{h,k}(u) du \geq d\nu_k$ bounds every source polynomial quadratic form. Minimizing on each identical affine remainder fibre gives $G_N \succeq G_N^\nu \succ 0$. Conjugation by $(G_N^\nu)^{-1/2}$ shows that each volume ratio in (HC15) is at least one. Cancellation proves the equality. The low-degree factors require only original moments through $2q$, and the comparison moments in the last factors go through $4q$. These are well-defined finite expressions, with the actual full jet map and source mass. The supplied proof records do not evaluate their errors at the hypothetical arithmetic quartet or bound them at a rate opposing (HC5). Retaining this sharper finite expression therefore supplies no extra contradictory inequality.

Chapter 15

The combined restriction on the actual counterfactual

The same arithmetic source in every bound

Retain the actual quartet, $g = 2\xi$, h , $v_h = g/h$, δ , γ , and full order m from (HT1). For $k \geq 3$, retain $q = [1 + k(m-1)](k+1)^2$ and the exact cyclic sum polynomial of (HT2). All determinant factors below are taken in its original increasing-power remainder frame. They are positive because the original common source Grams are positive definite and the remainder maps are onto.

Use the original arithmetic measure $m_{h,k} = w_h^{*k}$ and its fixed Gamma reference from (AT4), with their unchanged masses. Set

$$B_{\text{ar}} = \log \frac{V_{q-1}^{\text{ar}} V_q^{\text{ar}}}{V_{2q-1}^{\text{ar}} V_{2q}^{\text{ar}}}, \quad B_{\Gamma} = \log \frac{V_{q-1}^{\Gamma} V_q^{\Gamma}}{V_{2q-1}^{\Gamma} V_{2q}^{\Gamma}}, \quad \Delta = B_{\text{ar}} - B_{\Gamma}. \quad (\text{CR1})$$

In the fixed degree- $2q$ source define the positive relative operator $D_* = M_{2q}(0)^{-1} M_{2q}(1)$. This symbol is not the Euler differential. Let $b_1 \leq \dots \leq b_{2q+1}$ be all its eigenvalues with their full multiplicities, and put

$$L_q = \log \frac{b_{q+2}}{b_q} + 2 \sum_{j=1}^{q-1} \log \frac{b_{2q+2-j}}{b_j}. \quad (\text{CR2})$$

Every eigenvalue here is computed from the two actual moment matrices on their common retained polynomial source. There is no free replacement by a diagonal arithmetic metric. The diagonalization used in the AW proof is the spectral theorem applied to that specified positive relative operator, and its proof transports the original form.

Let D_h and $\widehat{\mathcal{U}}_k$ have their full definitions in (HC3), (HC6) and (HC8). Let $\mathcal{U}_{\nu,k}$ be the sharper finite lower-envelope comparison in (HC15), including its two actual low-degree determinants and two envelope determinants. Then the following combined restriction follows on the original objects:

$$\max\{0, 4q \log(D_h k), B_{\Gamma} - L_q\} \leq B_{\text{ar}} \leq \min\{\widehat{\mathcal{U}}_k, \mathcal{U}_{\nu,k}, B_{\Gamma} + L_q\}. \quad (\text{CR3})$$

For the signed arithmetic correction itself this is

$$\max\{-B_{\Gamma}, 4q \log(D_h k) - B_{\Gamma}, -L_q\} \leq \Delta \leq \min\{\widehat{\mathcal{U}}_k - B_{\Gamma}, \mathcal{U}_{\nu,k} - B_{\Gamma}, L_q\}. \quad (\text{CR4})$$

Proof. Polynomial-degree inclusion decreases the minimum quotient Gram in the positive-semidefinite order, so each original volume ratio in (CR1) is at least one. This proves $B_{\text{ar}} \geq 0$. The full inherited two-window identity and the new retained residual calculation give $B_{\text{ar}} \geq 4q \log(D_h k)$

in (HC5). The actual remainder-map estimates prove $B_{\text{ar}} \leq \widehat{\mathcal{U}}_k$ in (HC8), while comparison on the identical affine remainder fibres proves $B_{\text{ar}} \leq \mathcal{U}_{\nu,k}$ in (HC15). The computed full window spectrum and the actual relative moment eigenvalues give $|\Delta| \leq L_q$ in (AW14). Substituting (CR1) into this last inequality gives the remaining lower and upper entries of (CR3). Taking the maximum of the three proved lower bounds and minimum of the three proved upper bounds proves (CR3), with no independent choice of their source data. Subtracting the same literal B_{Γ} everywhere gives (CR4). $\square$

The signed value appearing in these inequalities has also been calculated as an exact integral, not only bounded:

$$\begin{aligned} \Delta &= \int_0^1 \text{Tr}((U(t) - W(t))C(t)) dt \\ &= \int_0^1 \text{Tr}(W(t)T(t)^{-1}[C(t), T(t)]) dt. \end{aligned} \tag{CR5}$$

Here U, W are the two original relation and polynomial window operators (AT13); $C = M_{2q}(t)^{-1}(M_{2q}(1) - M_{2q}(0))$; and T is the actual smooth source isometry constructed in (AW9)–(AW10). The first equality is (AT15), whose proof retains all four determinant orientations and both relation spaces. The second is (AW11), proved by inserting $U = TWT^{-1}$ and cyclically moving T in the finite trace. The source isometry does not preserve the arithmetic observation: its exact observation defect and full kernel are (AW12), and a nonzero witness is proved below. Thus (CR5) has not silently replaced the arithmetic map by an isometry that preserves different cohomology classes.

In particular every putative actual quartet must obey the additional explicit spectral compatibility inequality

$$L_q \geq \max\{0, B_{\Gamma} - \widehat{\mathcal{U}}_k, B_{\Gamma} - \mathcal{U}_{\nu,k}, \max(0, 4q \log(D_h k)) - B_{\Gamma}\}. \tag{CR6}$$

Indeed the first two nonzero entries follow by combining $B_{\Gamma} - L_q \leq B_{\text{ar}}$ with the two upper estimates. The last entry follows from $\max(0, 4q \log(D_h k)) \leq B_{\text{ar}} \leq B_{\Gamma} + L_q$. All logarithmic ratios defining L_q are nonnegative, proving its lower bound by zero as well. This proves (CR6) with the actual relative spectrum, not a freely chosen condition number.

An explicit original relation that the source isometry does not preserve

Here the arithmetic observation is the exact map $\mathcal{O} = \sigma_h^{\otimes k} \eta J_{2q} : H \rightarrow Q^{\otimes k}$ in (AW12), and $T(t) : H \rightarrow H$ is the constructed source isometry. Write the original monic polynomial as $\chi(S) = \sum_{j=0}^q a_j S^j$ and define its reflected conjugate on this fixed sum line by $\chi^{\dagger}(S) = \sum_{j=0}^q \overline{a_j}(k - S)^j$. The actual vector

$$v(S) = \chi(S)\chi^{\dagger}(S) \in \mathcal{P}_{2q} \quad \text{satisfies} \quad \mathcal{O}v = 0, \quad \mathcal{O}T(t)v \neq 0 \quad (0 \leq t \leq 1). \tag{CR7}$$

To prove this, $v \in \chi \mathcal{P}_q = \ker J_{2q}$ proves the first equality using the original observation map. On the unchanged line $S = k/2 + iu$ one has $k - S = \overline{S}$, so

$$\langle 1, v \rangle_t = \int_{\mathbb{R}} |\chi(k/2 + iu)|^2 m_t(u) du > 0.$$

The integral is finite by the original moment bounds, and positive because χ is a nonzero polynomial and m_t is positive almost everywhere. Therefore $v \notin \mathcal{P}_{q-1}^{\perp}$. Equation (AW12), proved by the explicit isometry on the two full flags, gives $\ker(\mathcal{O}T) = \mathcal{P}_{q-1}^{\perp}$, establishing the second assertion of (CR7). On a supported fibre the first value is its supported zero, whereas the second is a nonzero arithmetic

coefficient at that same label. Neither value is external τ . Thus the exact source isometry and its calculated commutator are available, but that particular isometry does not descend through the original relation quotient. This obstruction is proved on the same actual polynomial and measure used in (CR1)–(CR6).

The exact status after these calculations

The coarse actual upper and lower bound functions have strictly positive separation for every $k \geq 3$, as proved in (HC9)–(HC11). Keeping the packet constants fixed, their separation divided by q^2 tends to $2((\pi/2)D + \log 256) > 0$, with D exactly the original interval scale in (HC6). Thus those proved estimates give no contradiction. The more informative interval (CR3) retains actual unevaluated relative moment eigenvalues and the sharper envelope determinants. This calculation does not claim to have verified a nonempty instance of that refined interval for any off-line zeta packet: no such packet has been certified.

The strongest original-object restrictions proved here are simultaneous. The actual packet survives the full balanced finite comparison, the mixed-support coefficient retract, and the original tensor action. Its proper-label kernel contribution has the exact transition of DP and PL. Its full relative symmetric trace has the calculated negative directions and original derivative representatives in C. Its canonical source at every phase has the boundary energy and Laplacian residual in HT. Its phase derivative energy contributes to the same endpoint contraction costs by (HC12)–(HC14). Its original arithmetic correction is the computed common-source integral (CR5), bounded by the actual relative spectrum in (CR2)–(CR6).

These conclusions eliminate particular ways an off-line packet might have disappeared from the programme, and quantify the cost of its remaining original source representatives. They do not eliminate all such packets. Establishing the requested contradiction would require an opposing estimate on these original arithmetic quantities, proved from the same g/h and theta relations. Establishing RH false would require an actual off-critical zero of g , which has not been obtained. Neither outcome is asserted by this manuscript. The full calculations, source evidence and corrected comparisons are retained for continuation.

Chapter 16

Proof provenance and reproducibility

The source repository contains `proof_inputs/`, the exact authored TeX inputs; `tex/modules/`, copies with cross-reference identifiers namespaced for this reader; and `provenance/PROOF_INPUT_MANIFEST.json`, which records their cryptographic hashes and that mechanical transformation. Equation tokens, constants, original symbols and mathematical bodies are preserved by that transformation.

The tau-restriction and scaling chapters are attributed companion calculations from the concurrent complete-transcript audit. AT and AW are attributed companion calculations from the original programme’s coordinating calculation task; their full sources and reviews are retained in `companion_sources/`. The balanced kernel, finite derived comparison, proper-label example, coefficient retract, conormal trace and holonomy calculations are the present counterfactual continuation, with their independent review corrections recorded. Claims already proved in the shared conversation are explicitly identified as inherited. Finite symbolic checks support particular algebraic identities; the written proofs, rather than those checks, establish the quantified statements.

The source reconstruction reports are complete at `audit/`. Full visible and raw transcripts, source coverage, original record identifiers, the two supplied attachments, and verbatim current-session user provenance are retained at `provenance/`. The public transcript’s redacted payloads and absent attachment bodies are not reconstructed by inference.

The original proof dependency closure is retained in `original_programme/`, a fixed 821-page source cut. It includes the actual arithmetic-input proof, theta/Gamma source, source topology, tensor, conormal, period and volume constructions cited here. Later holonomy source bodies are retained in the complete transcript. The classical Hardy input used only for the explicit nonzero balanced-kernel cycle is retained with its original citation and provenance; it is not claimed as a newly proved result. No RH assumption is used in that input.

To rebuild this reader from its included TeX, run `python scripts/build_reader.py`. XeLaTeX, Latin Modern fonts and the packages declared in `tex/main.tex` are required; the script uses PyMuPDF only to record the resulting page count. The separate assembly script records how the present files were collected from the workspace; rebuilding the included manuscript does not require that workspace or any network connection. No Lean session, external publication, or certified search for an off-critical zero is performed by these scripts.